

PROPERTIES OF BLEND OPERATIONS

Mode name	Inverse	Transpose	Complement	Amount range	Relative exc time	Neutral Locus wrt FG	Neutral Locus wrt BG	Gradient Continuity	Local FG Inversion	Local BG Inversion	Saturated Fraction	Relative Contrast	Mean Difference	Darkened Fraction	Lightened Fraction
	Note 1	Note 1	Note 1	Note 2	Note 3	Note 4	Note 4	Note 5	Note 6	Note 6	Note 7	Note 8	Note 9	Note 9	Note 9
Opacity & Composition															
normal			normal	N/A		1.000 FG=BG	BG=FG	yes	none	none		0.01	1.00	0.00	0.50 0.50
Bidirectional Modes															
General Contrast & Mixing															
soft light			soft light	N/A		2.125 FG=0.5	BG=0; BG=1	yes	none	none		0.01	1.06	0.00	0.50 0.50
soft light ps				N/A		3.180 FG=0.5	BG=0; BG=1	no	none	none		0.01	1.02	0.00	0.50 0.50
soft light svq				N/A		3.894 FG=0.5	BG=0; BG=1	no	none	none		0.01	1.02	0.00	0.50 0.50
soft light eb				N/A		8.588 FG=0.5	BG=0; BG=1	yes	none	none		0.01	1.04	0.00	0.50 0.50
soft light eb2				[0 to +inf]		14.902 FG=0.5	BG=0; BG=1	yes	none	none		0.01	1.03	0.00	0.50 0.50
overlay		hard light	overlay	[0 to +inf]		2.129 FG=0.5	BG=0; BG=1	no	none	none		0.01	1.15	0.00	0.50 0.50
hard overlay				N/A		2.855 FG=0.5	BG=0	no	none	none		0.26	1.29	0.00	0.50 0.50
hard light		overlay	hard light	[0 to +inf]		2.102 FG=0.5	none	no	none	none		0.01	1.15	0.00	0.50 0.50
linear light			linear light	[0 1]		1.855 FG=0.5	none	no	none	none		0.50	1.41	0.00	0.50 0.50
vivid light			flat light	[0 to +inf]		4.980 FG=0.5	BG=0; BG=1	no	none	none		0.51	1.41	0.00	0.50 0.50
easy light			easy light	[0 to +inf]		15.216 FG=0.5	BG=0; BG=1	no	none	none		0.02	1.13	0.00	0.50 0.50
flat light		flat light	flat light	[0 to +inf]		3.373 FG=0.5	BG=0; BG=1	no	none	none		0.02	1.17	0.00	0.50 0.50
soft flatlight		soft flatlight	soft flatlight	[0 to +inf]		5.216 none	BG=0; BG=1	no	none	none		0.02	1.10	0.00	0.50 0.50
softer flatlight		softer flatlight	softer flatlight	[0 to +inf]		2.957 FG=BG	BG=FG; BG=0; BG=1	no	none	none		0.02	1.00	0.00	0.50 0.50
meanlight			meanlight	[0 to +inf]		3.090 FG=0.5	BG=0; BG=1	no	none	none		0.01	1.00	0.00	0.50 0.50
soft mean light			soft mean light	[0 to +inf]		3.031 FG=BG	BG=FG; BG=0; BG=1	no	none	none		0.01	0.83	0.00	0.50 0.50
star light			star light	[0 2]		4.000 FG=BG	BG=FG	yes	none	none		0.01	0.82	0.00	0.50 0.50
moon light			moon light	[0 2]		5.922 FG=BG	BG=FG	yes	none	none		0.00	0.91	0.00	0.50 0.50
moon light 2			moon light 2	[0 2]		5.686 FG=BG	BG=FG	yes	none	none		0.00	0.76	0.00	0.50 0.50
pin light			pin light	[0 2]		2.478 FG=0.5, FG=BG	BG=FG	no	none	none		0.01	1.00	0.00	0.25 0.25
super light		super light	super light	[1 to +inf]		1.953 FG=0.5	none	no	none	none		0.50	1.41	0.00	0.50 0.50
fog lighten			fog darken	N/A		2.267 FG=BG, FG=0	BG=FG	yes	1	none		0.00	0.81	0.08	0.50 0.50
fog darken			fog lighten	N/A		2.329 FG=BG, FG=1	BG=FG	yes	1	none		0.00	0.81	-0.08	0.50 0.50
Hard Mix															
hard mix ps		hard mix ps	hard mix ps	[0 2]		2.094 none	BG=0; BG=1	no	none	none		1.00	1.73	0.00	0.49 0.50
hard mix ib		hard mix ib	hard mix ib	[0 2]		2.278 none	BG=0; BG=1	no	(1 2)	none		1.00	1.73	0.00	0.50 0.49
hard mix kr			hard mix kr	[0 2]		2.349 none	BG=0; BG=1	no	[0 2]	none		0.76	1.60	0.00	0.49 0.50
Quadratics															
glow		reflect	heat	(-inf to +inf)		2.349 FG=1-BG	BG=0; BG=1; BG=1-FG	no	(-1 0)	(-inf -1)		0.34	1.37	0.05	0.50 0.49
heat		freeze	glow	(-inf to +inf)		2.557 FG=1-BG	BG=0; BG=1; BG=1-FG	no	(-1 0)	(-inf -1)		0.34	1.37	-0.05	0.50 0.50
reflect		glow	freeze	(-inf to +inf)		2.537 none	none	no	none	(-inf -1)		0.34	1.37	0.05	0.39 0.60
freeze		heat	reflect	(-inf to +inf)		2.494 none	none	no	none	(-inf -1)		0.34	1.37	-0.05	0.60 0.39
helow		frect	helow	(-inf to +inf)		3.196 FG=1-BG	BG=0; BG=1; BG=1-FG	yes	(-1 0)	(-inf -1)		0.01	1.24	0.00	0.49 0.50
gleat		reeze	gleat	(-inf to +inf)		3.020 FG=1-BG	BG=0; BG=1; BG=1-FG	no	(-1 0)	(-inf -1)		0.67	1.52	0.00	0.50 0.50
frect		helow	frect	(-inf to +inf)		2.929 none	none	yes	none	(-inf -1)		0.01	1.24	0.00	0.50 0.50
reeze		gleat	reeze	(-inf to +inf)		2.949 none	none	no	none	(-inf -1)		0.67	1.52	0.00	0.50 0.50
Penumbra (Soft Dodge/Burn)															
penumbra a1		penumbra b1	penumbra b1	[0 to +inf]		2.247 none	BG=0	yes	none	none		0.01	1.00	0.00	0.58 0.41
penumbra b1		penumbra a1	penumbra a1	[0 to +inf]		2.349 none	BG=1	yes	none	none		0.01	1.00	0.00	0.41 0.58
penumbra a2		penumbra b2	penumbra b2	[0 to +inf]		3.898 none	BG=0	yes	none	none		0.01	0.92	0.00	0.56 0.44
penumbra b2		penumbra a2	penumbra a2	[0 to +inf]		3.961 none	BG=1	yes	none	none		0.01	0.92	0.00	0.44 0.56
penumbra a3		penumbra b3	penumbra b3	[0 to +inf]		2.878 FG=BG	BG=FG; BG=0	yes	none	none		0.01	0.83	0.00	0.50 0.49
penumbra b3		penumbra a3	penumbra a3	[0 to +inf]		3.000 FG=BG	BG=FG; BG=1	yes	none	none		0.01	0.83	0.00	0.49 0.50
Mean-Scaled															
scale add				(-inf to +inf)*		35.294 content-dependent	none	no	(-inf 0)	none		0.25	1.17	0.00	0.50 0.50
scale mult				[0 to +inf]*		31.255 content-dependent	BG=0	no	none	none		0.16	1.20	-0.05	0.50 0.50
contrast				[0 to +inf]*		32.824 content-dependent	BG=0; BG=1	no	none	none		0.26	1.22	0.00	0.50 0.50

Unidirectional Pairs															
Dodges & Burns															
color dodge			color burn	[0 10]		1.804 FG=0	BG=0; BG=1	no	none	none		0.51	1.12	0.25	0.00 0.99
color burn			color dodge	[0 10]		2.039 FG=1	BG=0; BG=1	no	none	none		0.51	1.12	-0.25	0.99 0.00
poly dodge			poly burn	[0 to +inf]		2.655 FG=0	BG=0; BG=1	no	none	none		0.50	1.12	0.25	0.00 0.99
poly burn			poly dodge	[0 to +inf]		2.655 FG=1	BG=0; BG=1	no	none	none		0.50	1.12	-0.25	0.99 0.00
linear dodge		linear dodge	linear burn	[0 1]		1.937 FG=0	BG=1	no	none	none		0.50	0.82	0.33	0.00 0.99
linear burn		linear burn	linear dodge	[0 1]		1.855 FG=1	BG=0	no	none	none		0.50	0.82	-0.33	0.99 0.00
easy dodge			easy burn	[0 to +inf]		8.353 FG=0	BG=0; BG=1	yes	none	none		0.01	0.91	0.19	0.00 0.99
easy burn			easy dodge	[0 to +inf]		8.824 FG=1	BG=0; BG=1	yes	none	none		0.01	0.91	-0.19	0.99 0.00
gamma dodge			gamma burn	[0 to +inf]		8.549 FG=0	BG=0; BG=1	yes	none	none		0.04	1.07	0.19	0.00 0.99
gamma burn			gamma dodge	[0 to +inf]		8.510 FG=1	BG=0; BG=1	yes	none	none		0.04	1.07	-0.19	0.99 0.00
maleki dodge		tint	maleki burn	[0 1]		3.439 FG=0	BG=1	no	none	none		0.39	0.58	0.39	0.00 0.99
maleki burn		shade	maleki dodge	[0 1]		3.376 FG=1	BG=0	no	none	none		0.39	0.58	-0.39	0.99 0.00
suau dodge				[0 to +inf]		15.176 FG=0	BG=0; BG=1	yes	none	none		0.08	0.94	0.09	0.00 0.99
suau burn				[0 to +inf]		15.020 FG=1	BG=0; BG=1	yes	none	none		0.08	1.04	-0.09	0.99 0.00
Hybrid Glow & Shadow															
flat glow			flat shadow	[0 2]		3.224 FG<=BG	BG=0; BG>=FG	no	none	none		0.01	0.96	-0.08	0.00 0.49
flat shadow			flat glow	[0 2]		3.286 FG>=BG	BG=1; BG<=FG	no	none	none		0.01	0.96	-0.08	0.49 0.00
mean glow			mean shadow	[0 2]		3.647 FG<=BG	BG>=FG	no	none	none		0.00	0.82	0.08	0.00 0.50
mean shadow			mean glow	[0 2]		3.600 FG>=BG	BG<=FG	no	none	none		0.00	0.82	-0.08	0.50 0.00
star glow			star shadow	[0 2]		3.612 FG<=BG	BG>=FG	no	none	none		0.01	0.84	0.13	0.00 0.50
star shadow			star glow	[0 2]		3.741 FG>=BG	BG<=FG	no	none	none		0.01	0.84	-0.13	0.50 0.00
moon glow			moon shadow	[0 2]		5.020 FG<=BG	BG>=FG	no	none	none		0.00	0.82	0.13	0.00 0.50
moon shadow			moon glow	[0 2]		5.255 FG>=BG	BG<=FG	no	none	none		0.00	0.82	-0.13	0.50 0.00
moon glow 2			moon shadow 2	[0 2]		4.902 FG<=BG	BG>=FG	yes	none	none		0.00	0.83	0.09	0.00 0.50
moon shadow 2			moon glow 2	[0 2]		5.373 FG>=BG	BG<=FG	yes	none	none		0.00	0.83	-0.09	0.50 0.00
Krita/EB Assortment															
tint		maleki dodge	shade	N/A		3.439 none	BG=1	no	none	none		0.39	0.58	0.39	0.00 1.00
shade		maleki burn	tint	N/A		3.388 none	BG=0	no	none	none		0.39	0.58	-0.39	1.00 0.00
lighteneb			darkeneb	N/A		4.980 none	BG=0	no	none	none		0.07	0.93	0.24	0.15 0.84
darkeneb			lighteneb	N/A		4.431 none	BG=1	no	none	none		0.07	0.93	-0.24	0.84 0.15

Relational															
lighten RGB	lighten RGB	darken RGB	[0 1]		2.067 FG<=BG	BG>=FG	no	none	none		0.01	0.82	0.17	0.00	0.50
darken RGB	darken RGB	lighten RGB	[0 1]		2.020 FG>=BG	BG<=FG	no	none	none		0.01	0.82	-0.17	0.50	0.00
lighten Y	lighten Y	darken Y	[0 1]		5.176 FG_y<=BG_y	BG_y>=FG_y	no	none	none		0.01	0.82	0.17	0.00	0.50
darken Y	darken Y	lighten Y	[0 1]		5.137 FG_y>=BG_y	BG_y<=FG_y	no	none	none		0.01	0.82	-0.17	0.50	0.00
near	far	near	[0 1]		7.961 FG=BG; FG≠ BG	BG=FG; BG≠FG	no	(0 1)	none		0.01	1.00	0.00	0.50	0.50
far	near	far	[0 1]		8.039 FG=BG	BG=FG	no	none	(0 1)		0.01	1.00	0.00	0.00	0.00
replace bg	replace fg	replace bg	[0 1]*		6.039 FG=BG	BG=FG, BG=AMOUNT	no	none	[1 0]		0.00	0.99	-0.01	0.01	0.00
replace fg	replace bg	replace fg	[0 1]*		6.000 FG=BG; FG=AMOUNT	BG=FG	no	[1 0]	none		0.00	0.99	-0.01	0.50	0.49
preserve bg	preserve fg	preserve bg	[0 1]*		5.922 FG=BG	BG=FG; BG≠AMOUNT	no	none	[1 0]		0.01	1.01	0.01	0.49	0.50
preserve fg	preserve bg	preserve fg	[0 1]*		5.882 FG=BG; FG≠AMOUNT	BG=FG	no	[1 0]	none		0.01	1.01	0.01	0.00	0.01

Plots for selected blend modes

This PDF file also has an index for easier navigation.

Overview Maps

- Intensity map overview
- R-BG Contrast map overview
- R vs BG (meridian) map overview
- R vs FG map overview
- Loci of Neutral response (conditions in which BG remains unchanged)

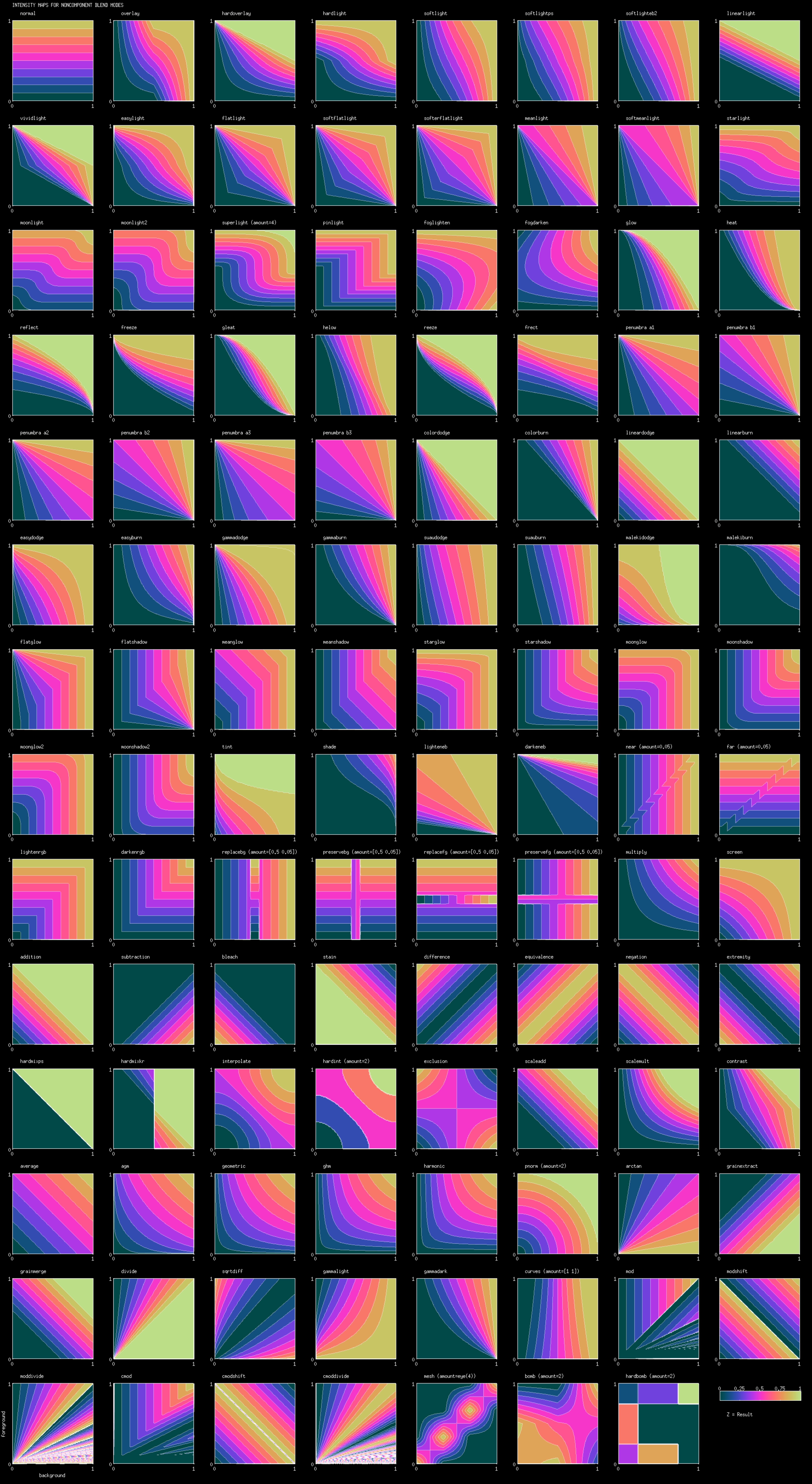
Soft Light Properties

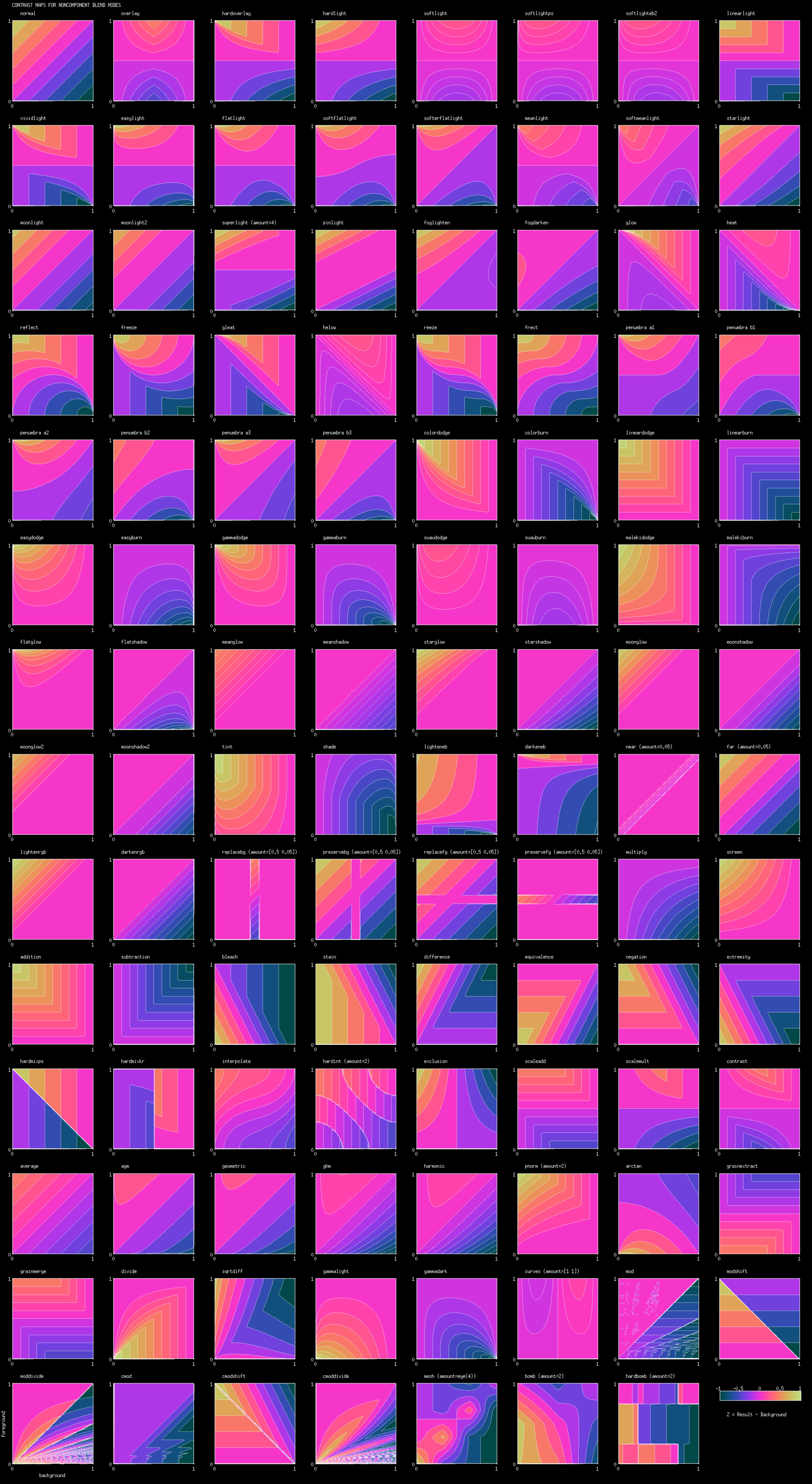
- Softlight comparisons
- Soft Light EB2 (meridian maps)

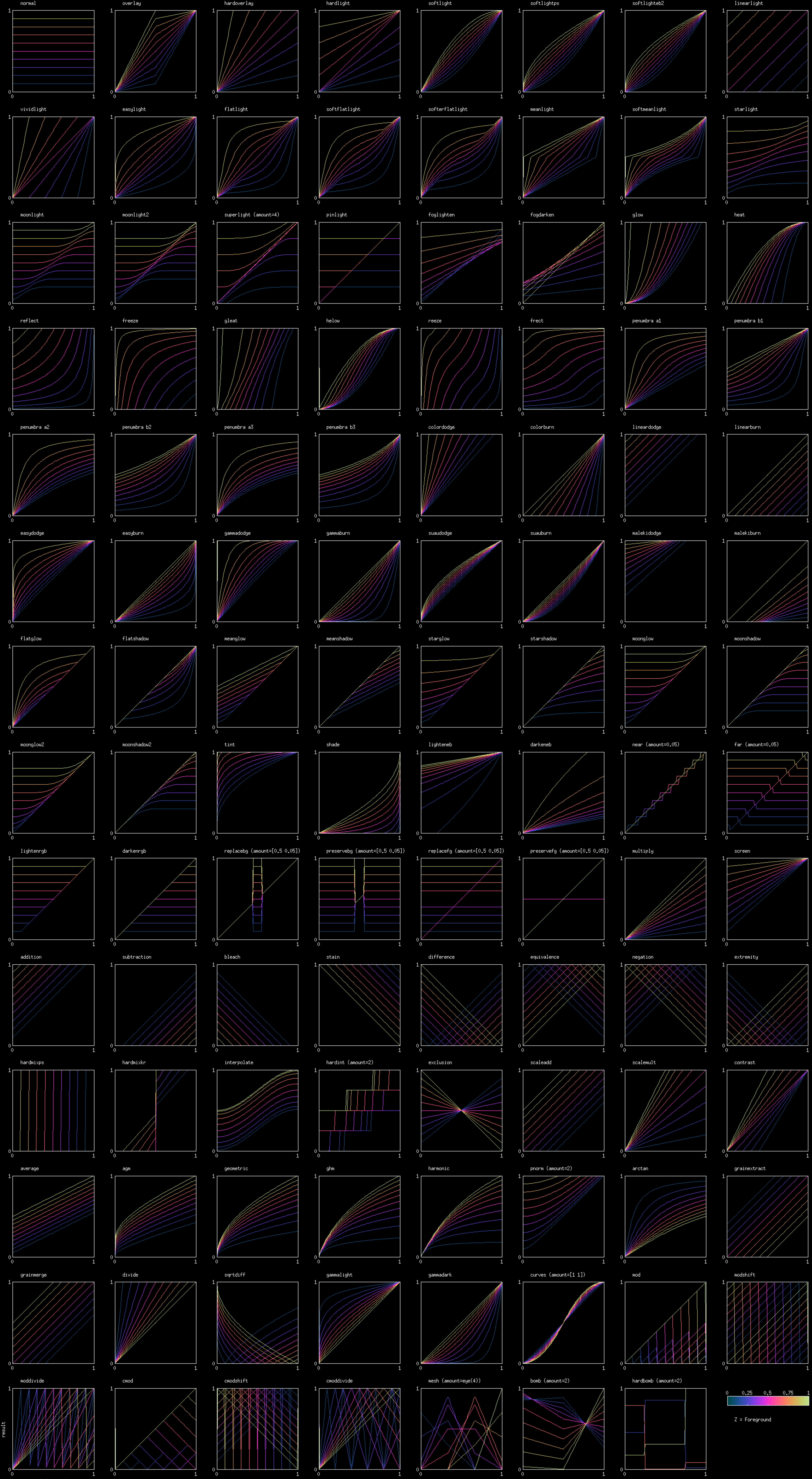
Parameter Sweeps

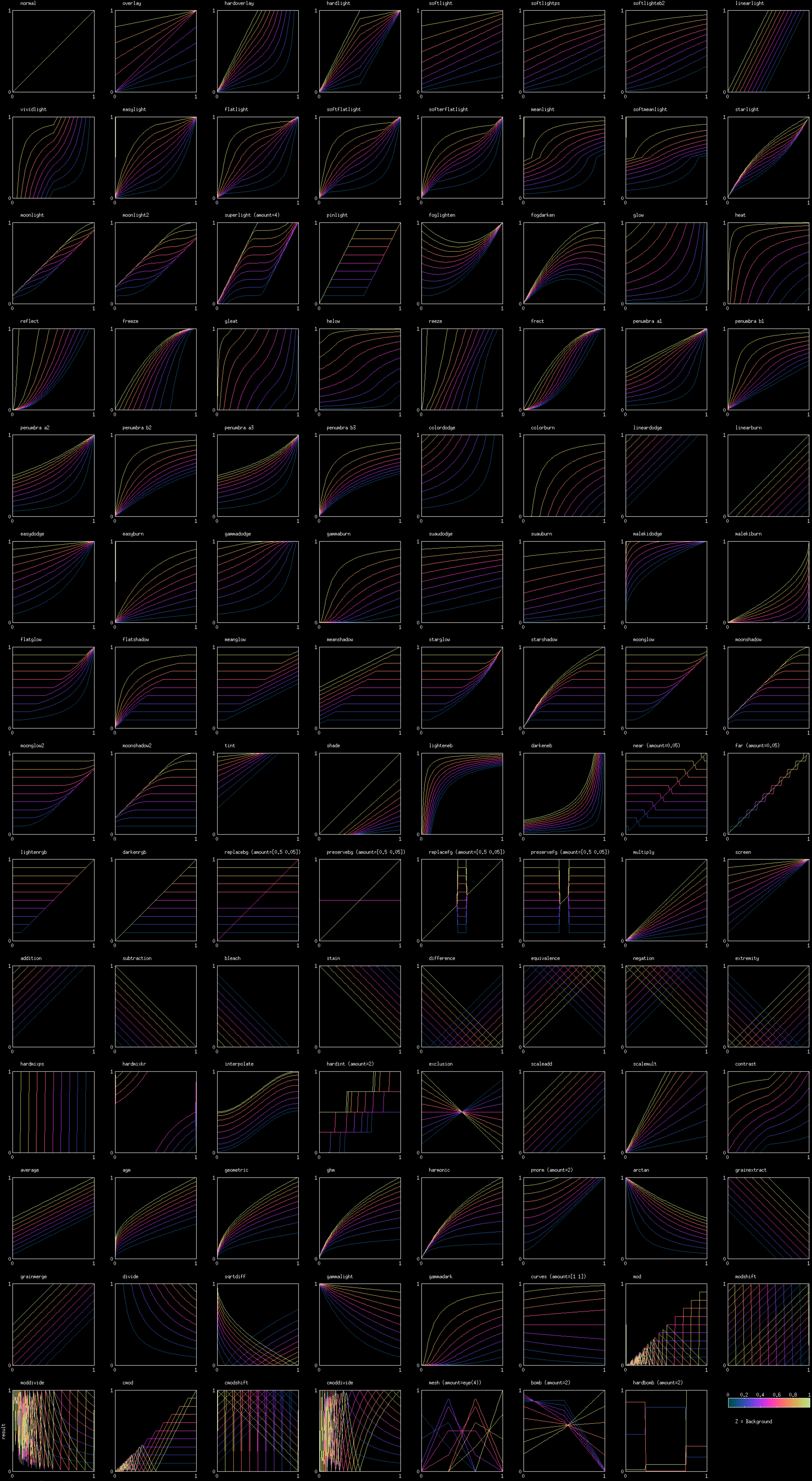
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|----------------|--------------|---------------|
| Soft Light EB2 | Color Burn | Moon Glow 2 |
| Overlay | Poly Dodge | Moon Shadow 2 |
| Linear Light | Poly Burn | Penumbra A1 |
| Vivid Light | Linear Dodge | Penumbra B1 |
| Easy Light | Linear Burn | Glow |
| Flat Light | Easy Dodge | Heat |
| Mean Light | Easy Burn | Gleat |
| Star Light | Suau Dodge | Helow |
| Moon Light | Suau Burn | Difference |
| Moon Light 2 | Maleki Dodge | Extremity |
| Super Light | Maleki Burn | Sqrtdiff |
| Pin Light | Flat Glow | Interpolate |
| Hard Mix PS | Flat Shadow | Average |
| Hard Mix IB | Mean Glow | Geometric |
| Hard Mix KR | Mean Shadow | Harmonic |
| Scale Add | Star Glow | Pnorm |
| Scale Mult | Star Shadow | Lighten RGB |
| Contrast | Moon Glow | Curves |
| Color Dodge | Moon Shadow | |

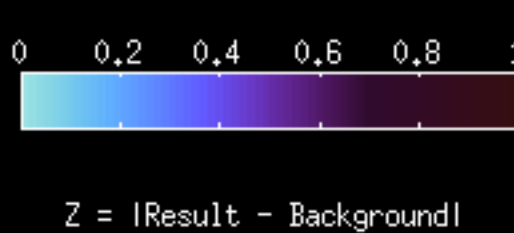
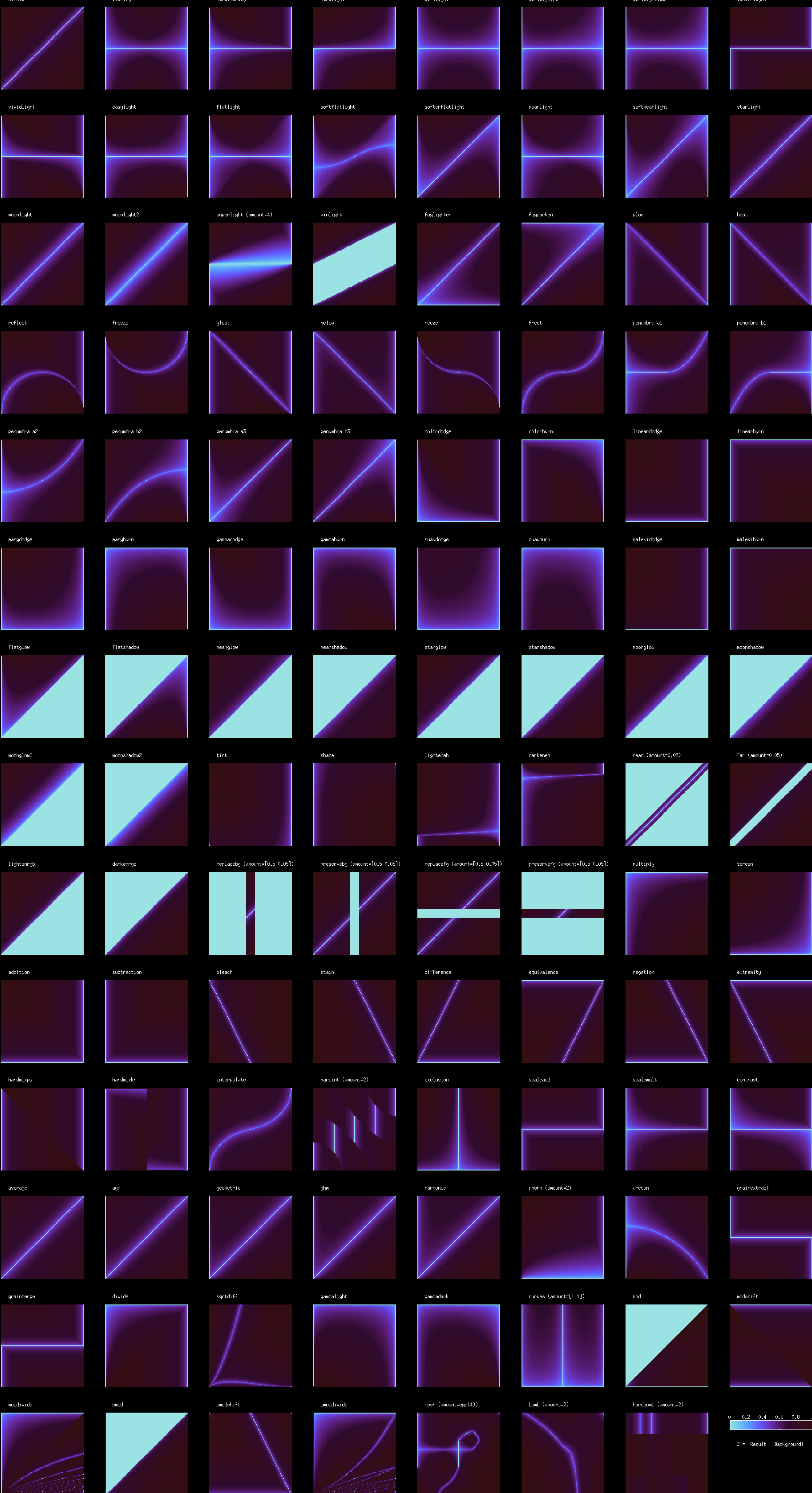
Composition Mode Examples









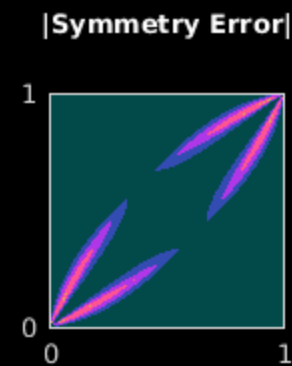
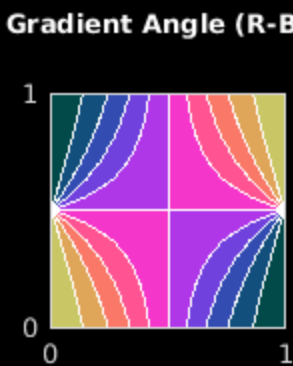
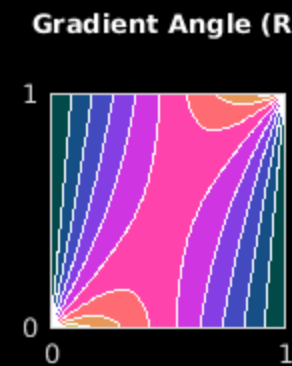
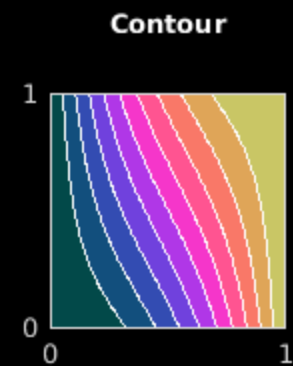
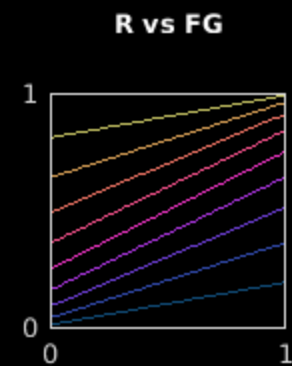
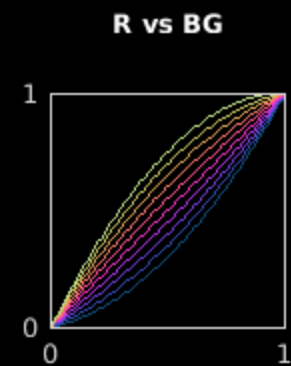


foreground

background

softlight

max(dR/dBG): 1.9900
relative speed: 12.8806
RMS value err: 0.0268
RMS sym err: 0.0752

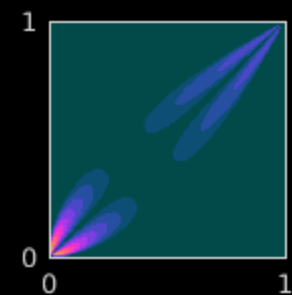
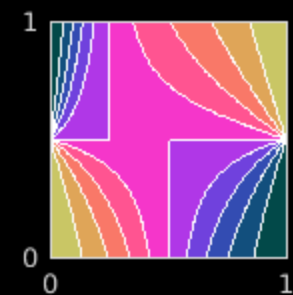
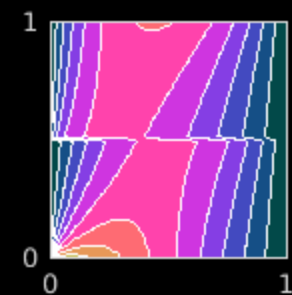
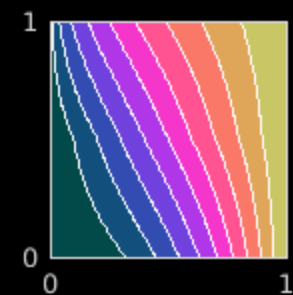
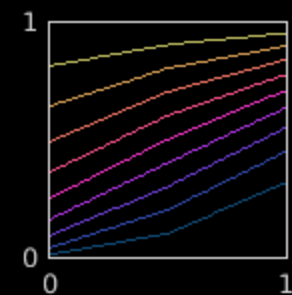
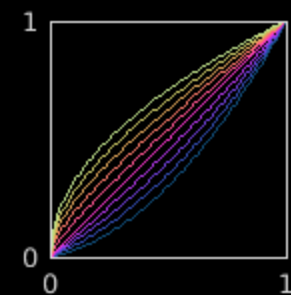


Formulae

$$R = I.^2 + 2 * M .* I .* (1 - I);$$

softlightps

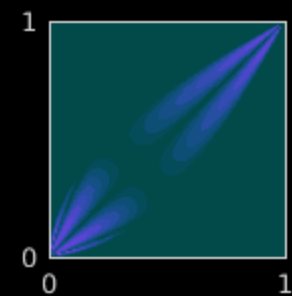
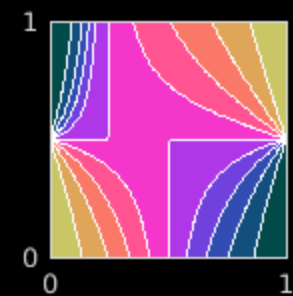
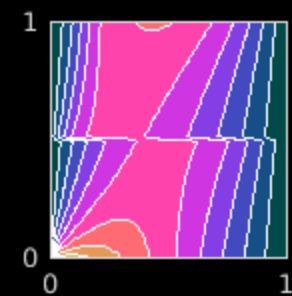
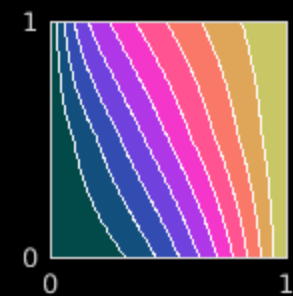
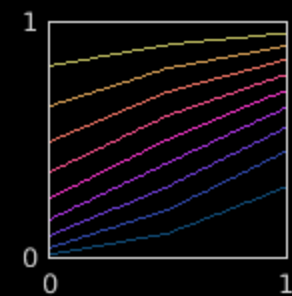
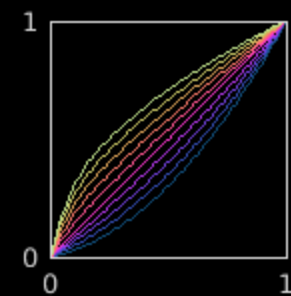
max(dR/dBG): 10.0000
relative speed: 6.9040
RMS value err: 0.0068
RMS sym err: 0.0435



$$hi = M > 0.5;$$
$$R = (I + (2 * M - 1) .* (\sqrt{I} - I)) .* hi ...$$
$$+ (I - (1 - 2 * M) .* I .* (1 - I)) .* \sim hi;$$

softlightsvg

max(dR/dBG): 3.8816
relative speed: 4.5183
RMS value err: 0.0079
RMS sym err: 0.0339

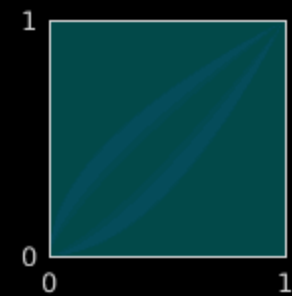
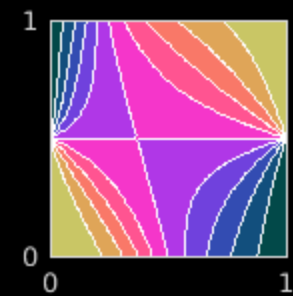
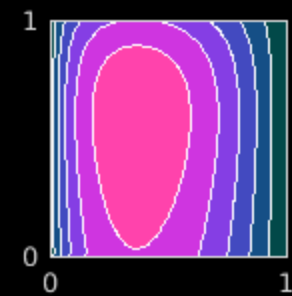
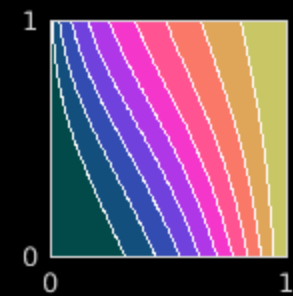
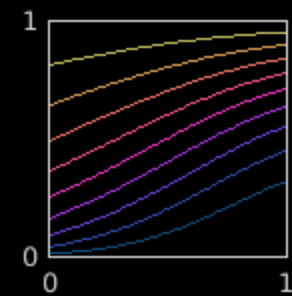
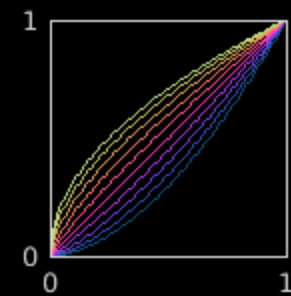


$$m1 = M \leq 0.50;$$
$$m2 = I \leq 0.25;$$
$$m3 = \sim m1 \& m2;$$
$$m4 = \sim m1 \& \sim m2;$$

$$R = (I - (1 - 2 * M) .* I .* (1 - I)) .* m1 ...$$
$$+ (I + (2 * M - 1) .* (4 * I .* (4 * I + 1) .* (I - 1) + 7 * I)) .* m3 ...$$
$$+ (I + (2 * M - 1) .* (I.^{0.5} - I)) .* m4;$$

softlighteb

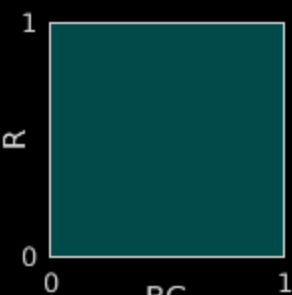
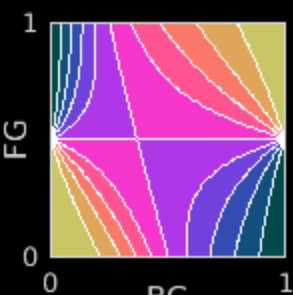
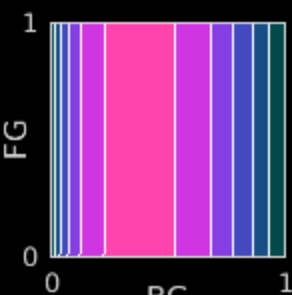
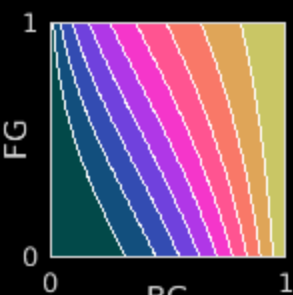
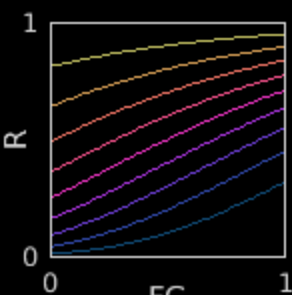
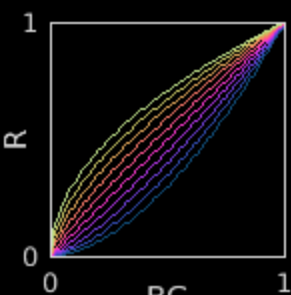
max(dR/dBG): 10.0000
relative speed: 1.8323
RMS value err: 0.0046
RMS sym err: 0.0074



$$R = I.^{(M.^2 - 2.5 * M + 2)};$$

softlighteb2

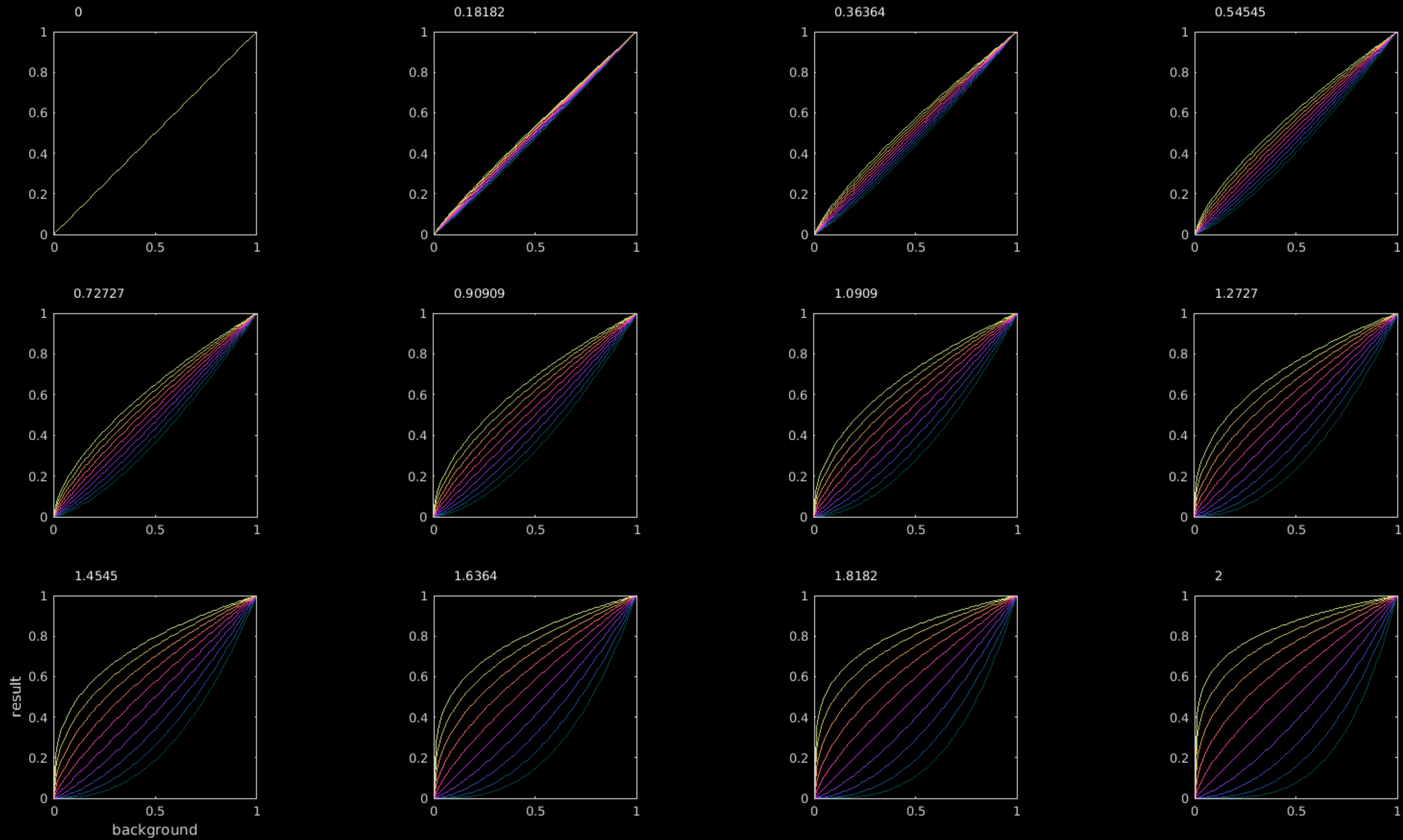
max(dR/dBG): 10.0000
relative speed: 1.0000
RMS value err: 0.0000
RMS sym err: 0.0000



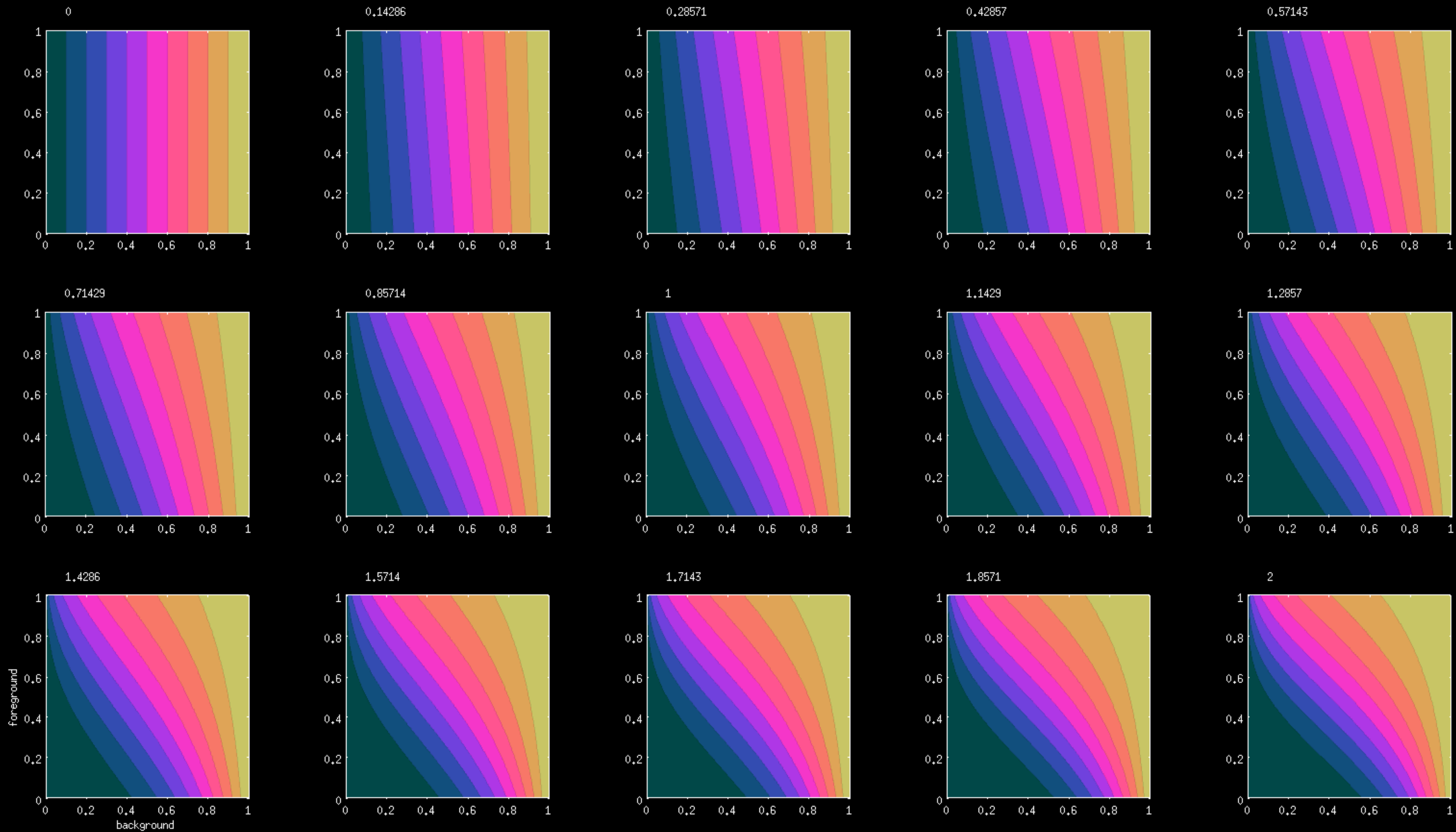
$$\text{amount} = \max(\text{amount}, 0);$$
$$R = I.^{(2.^{(\text{amount} * (1 - 2 * M))}};$$

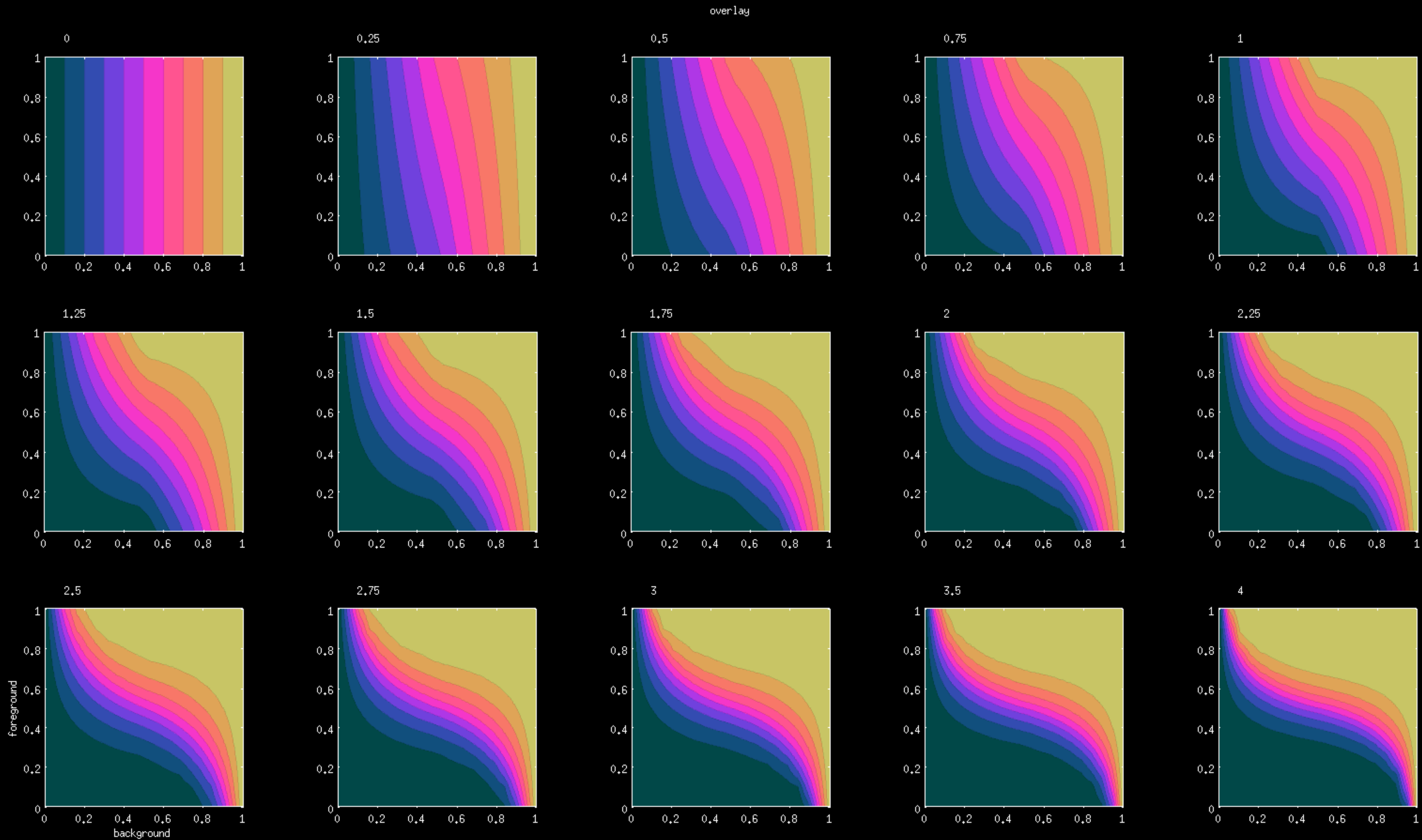


softlighteb2

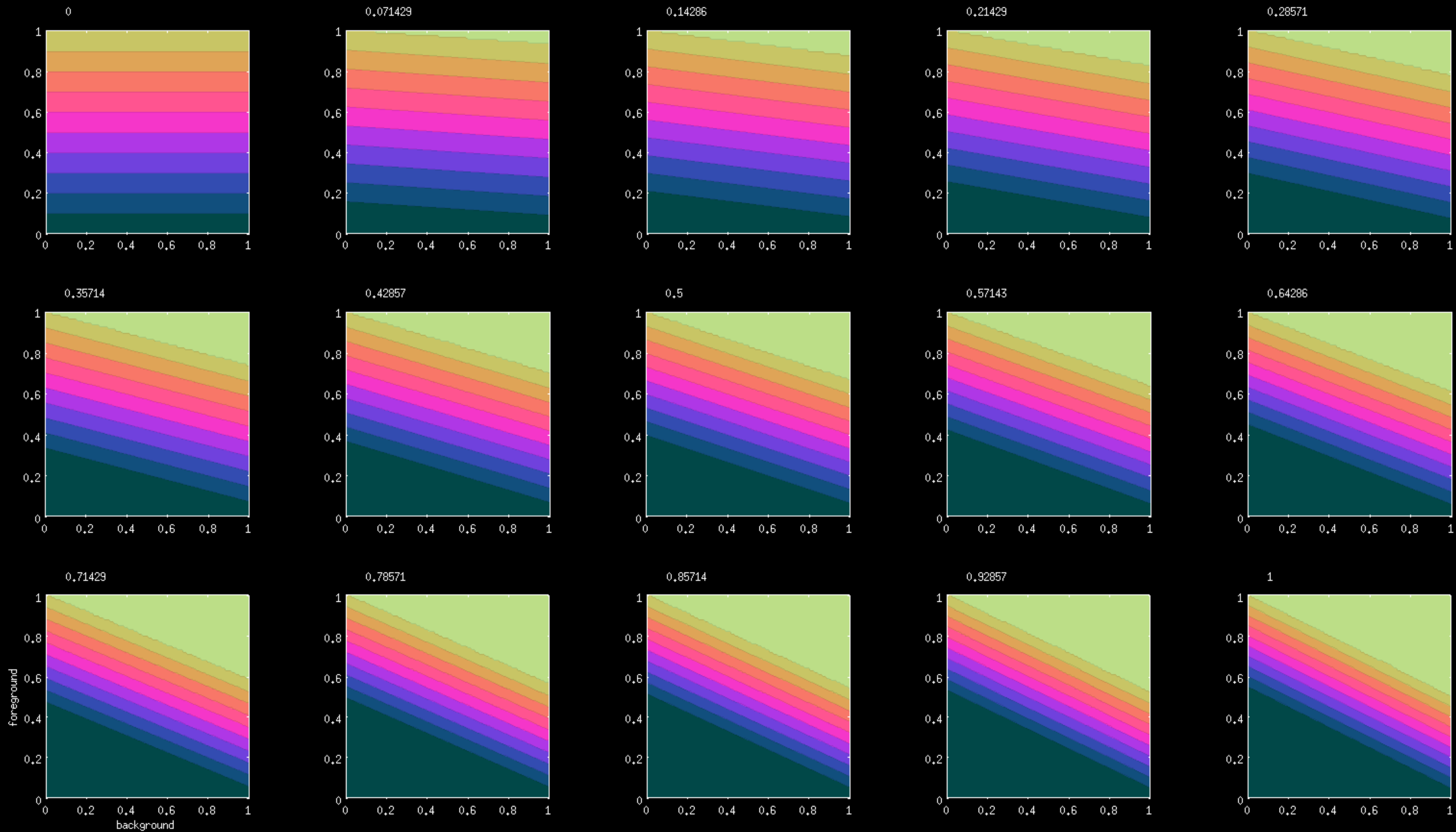


softlighteb2

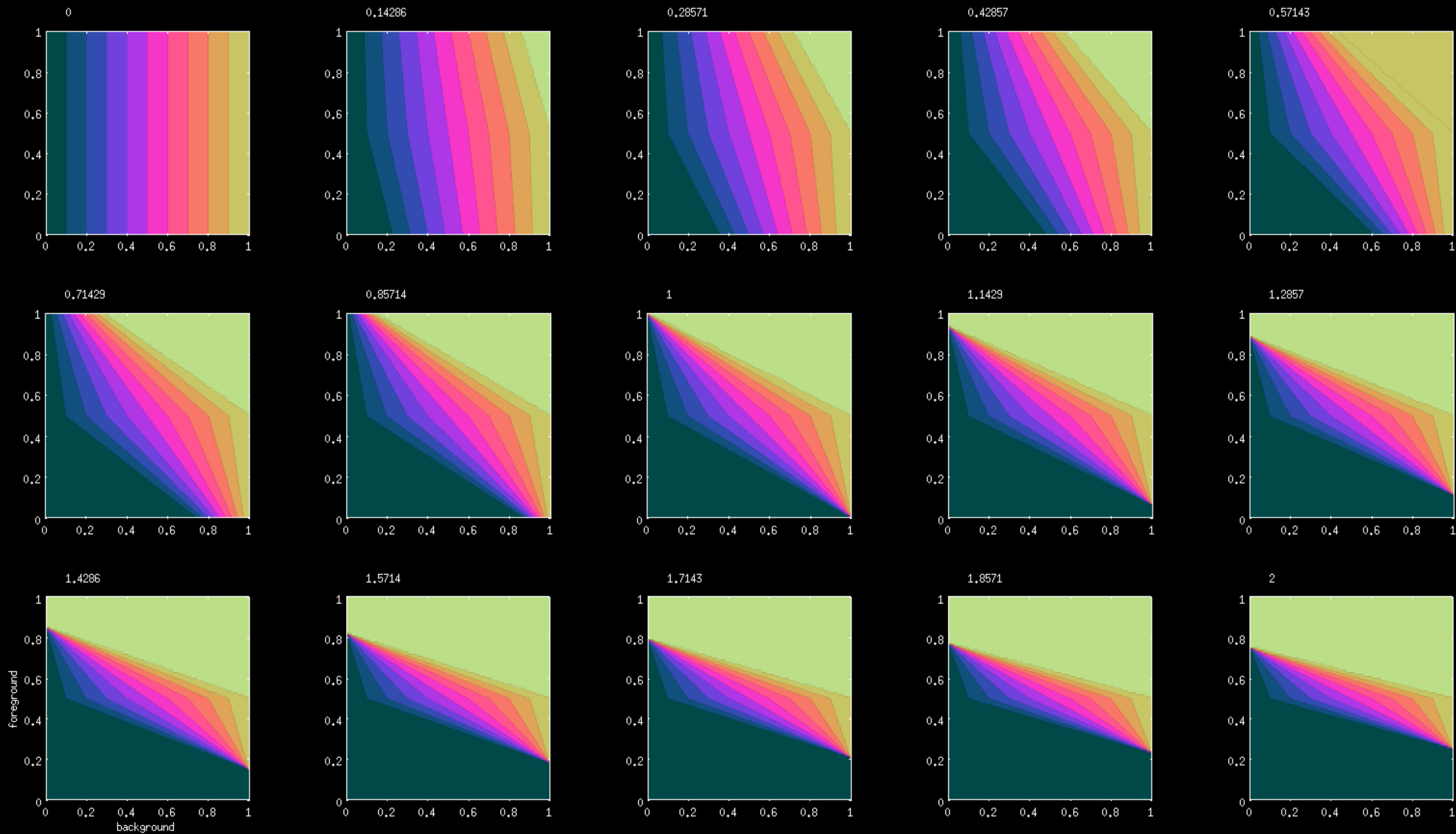




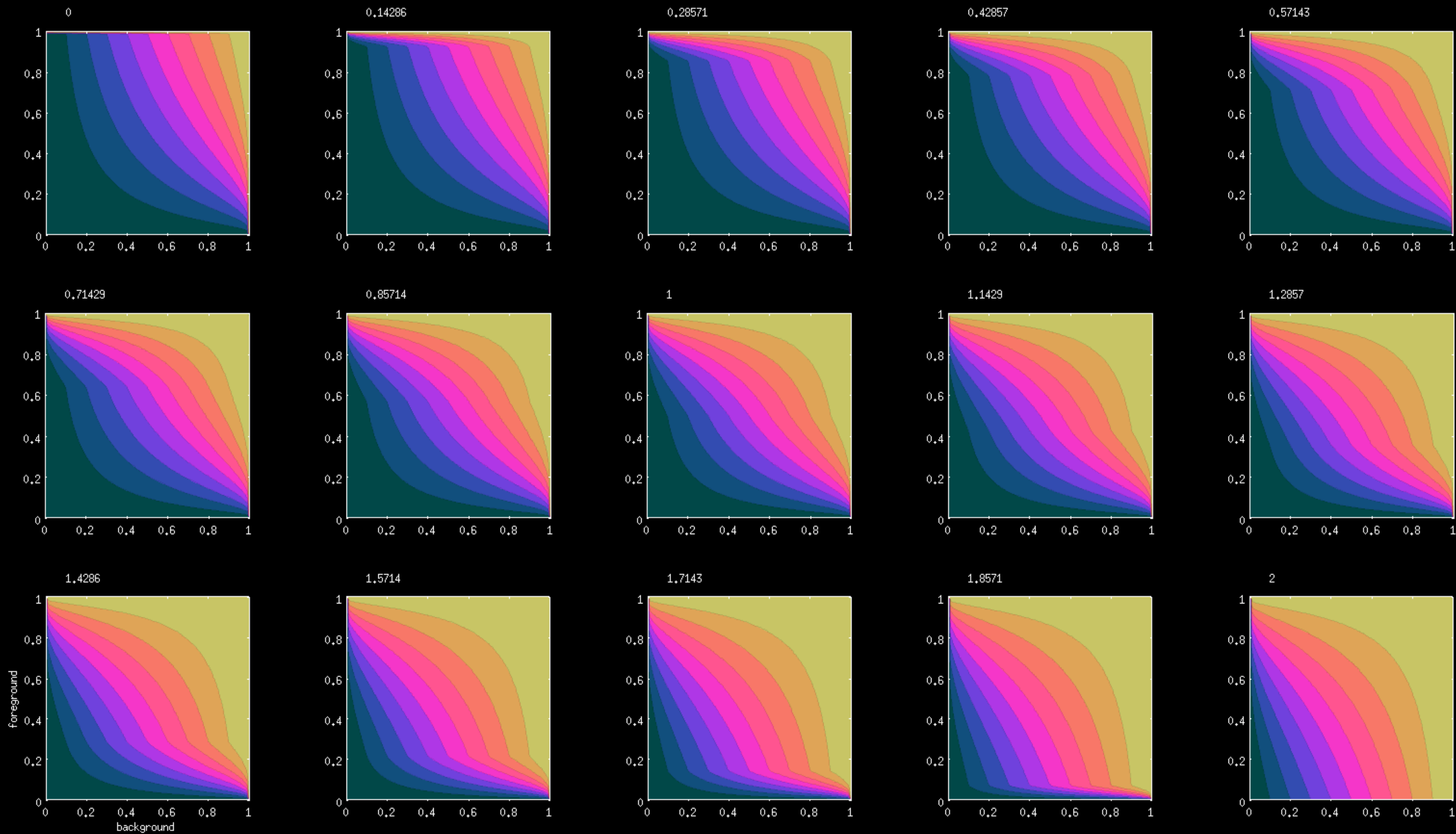
linearlight



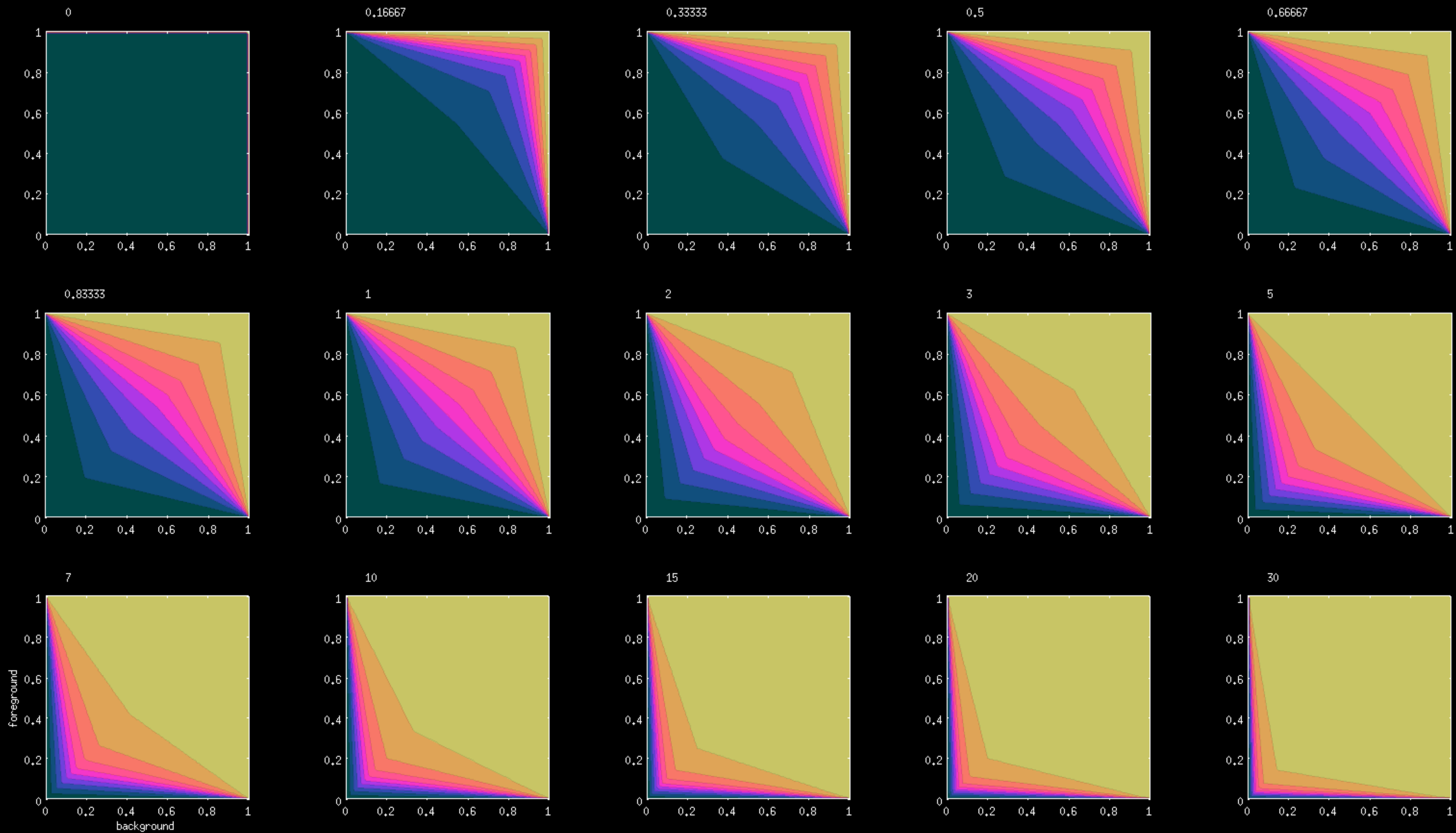
vividlight



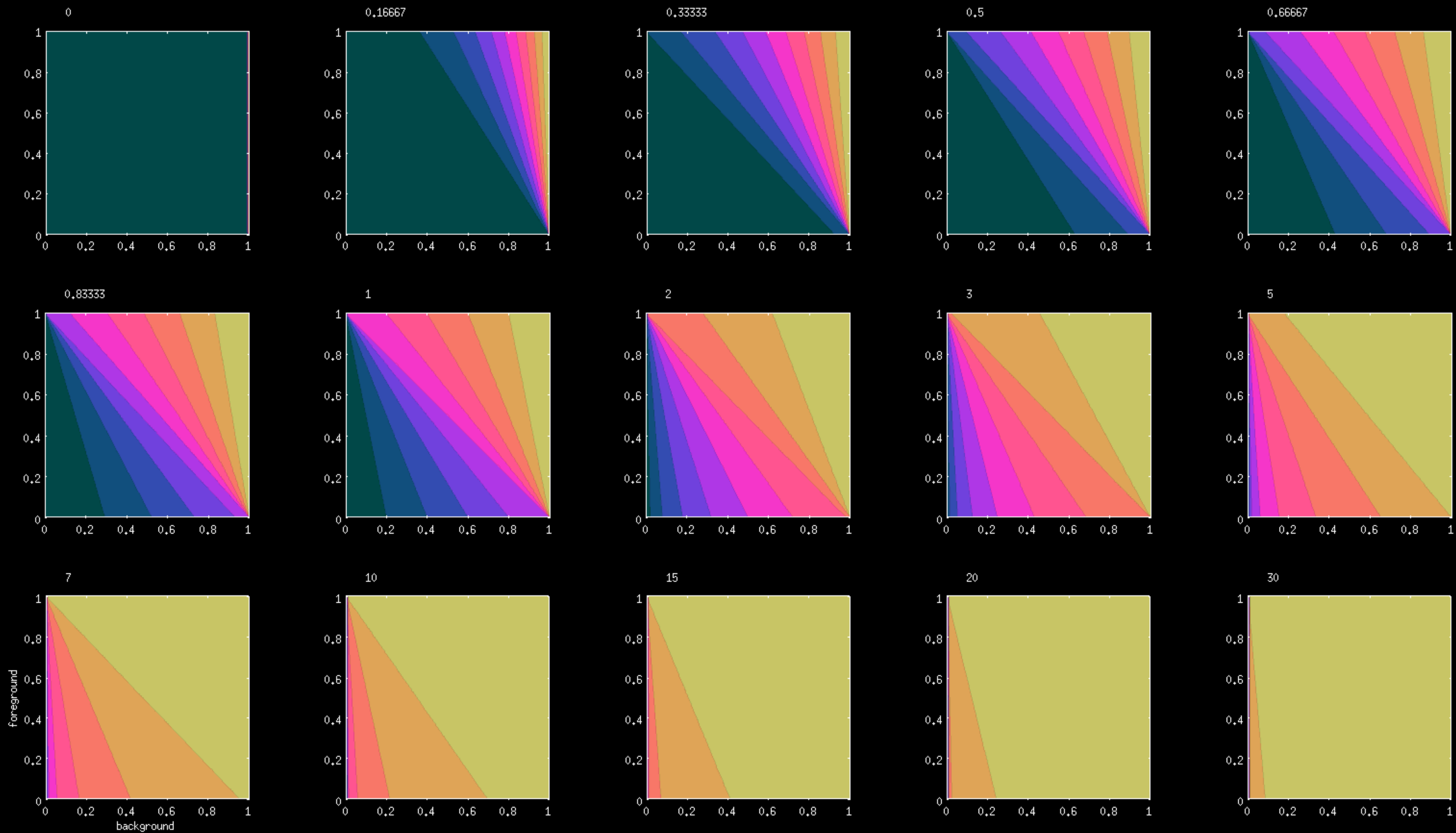
easylight



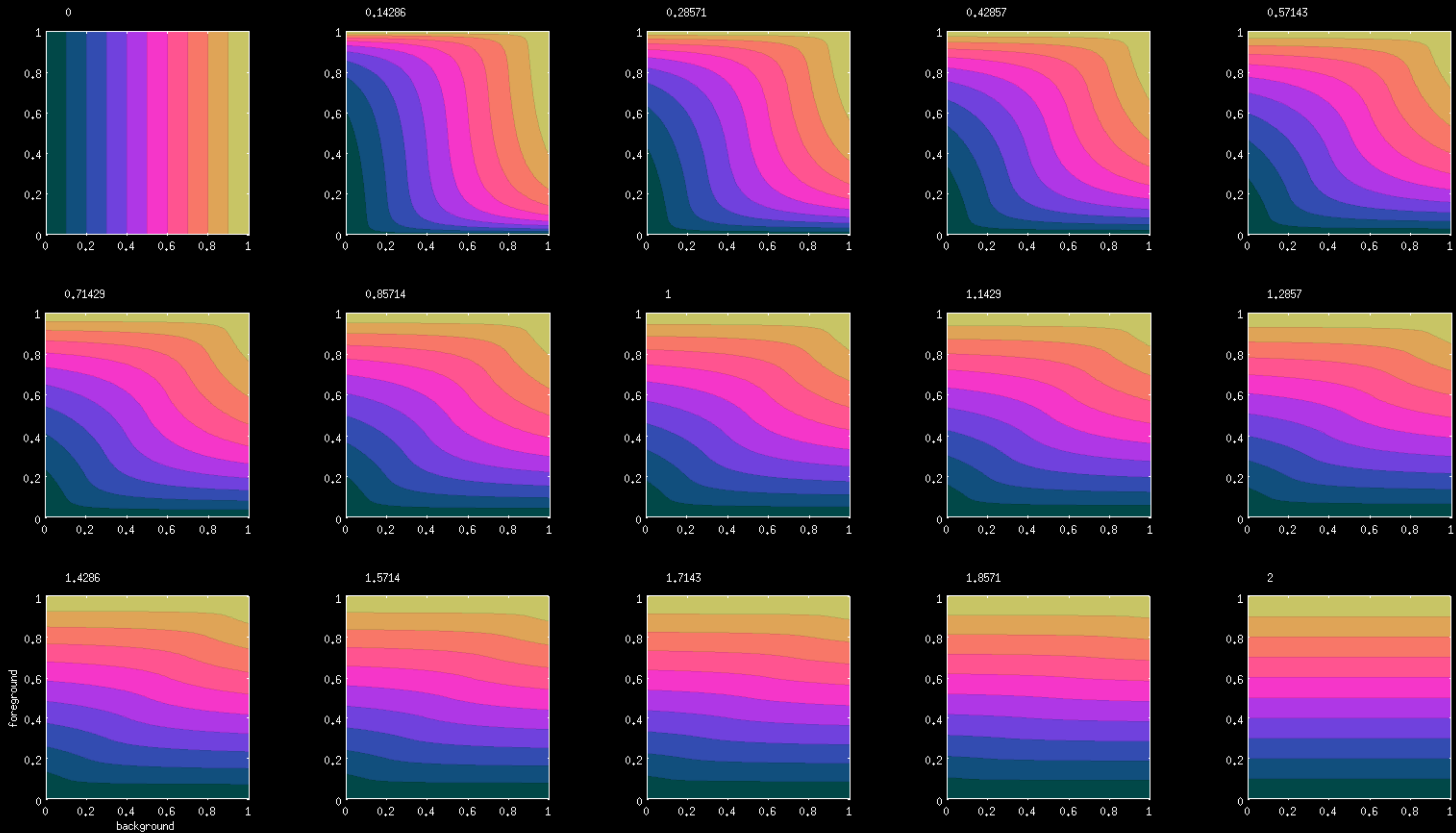
flatlight



meanlight

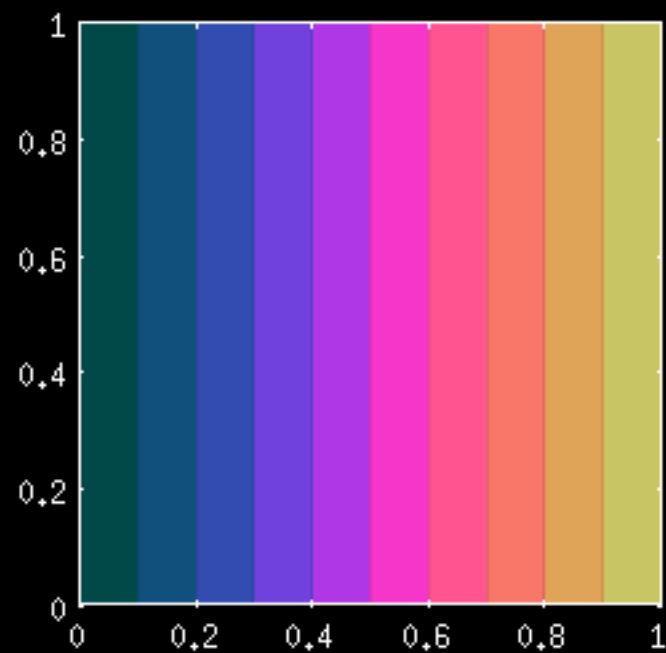


starlight

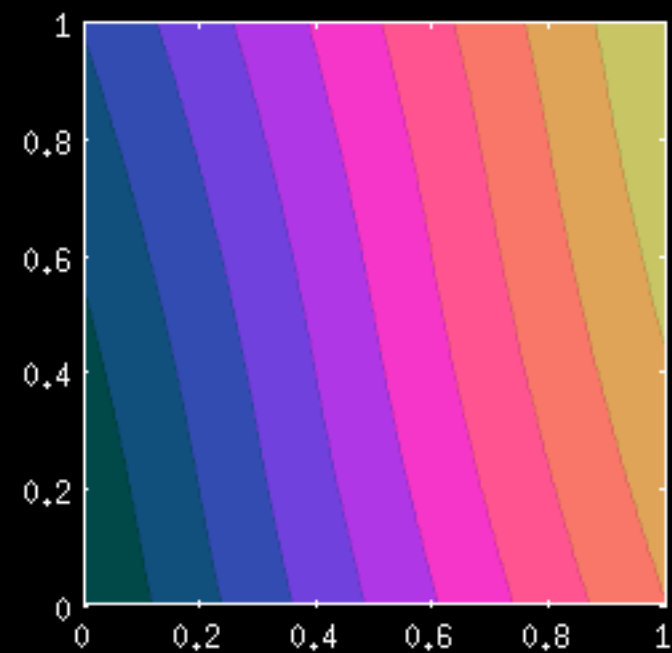


moonlight

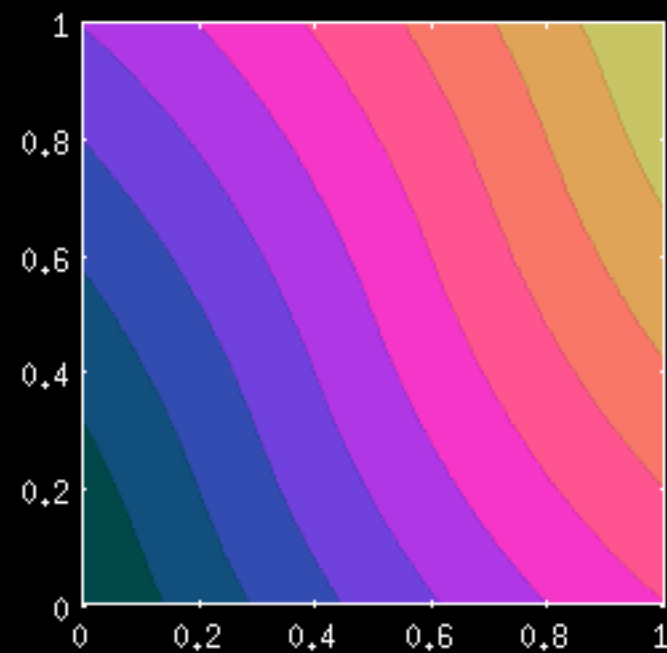
0



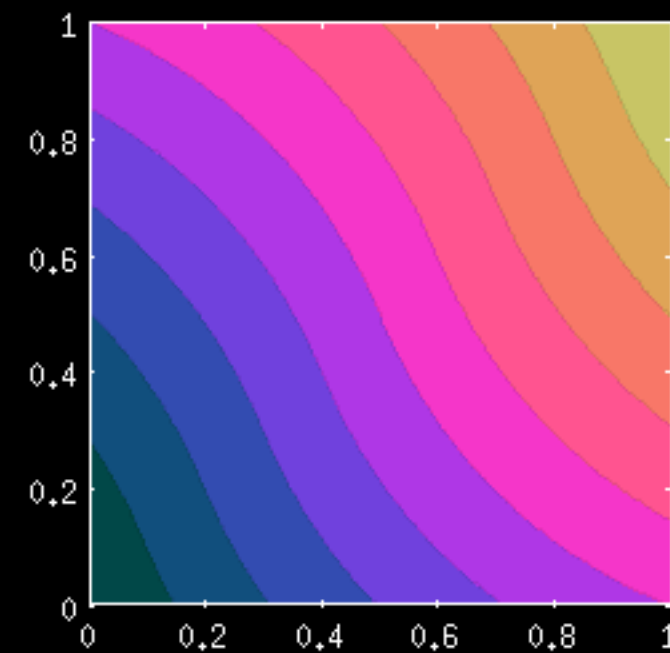
0.14286



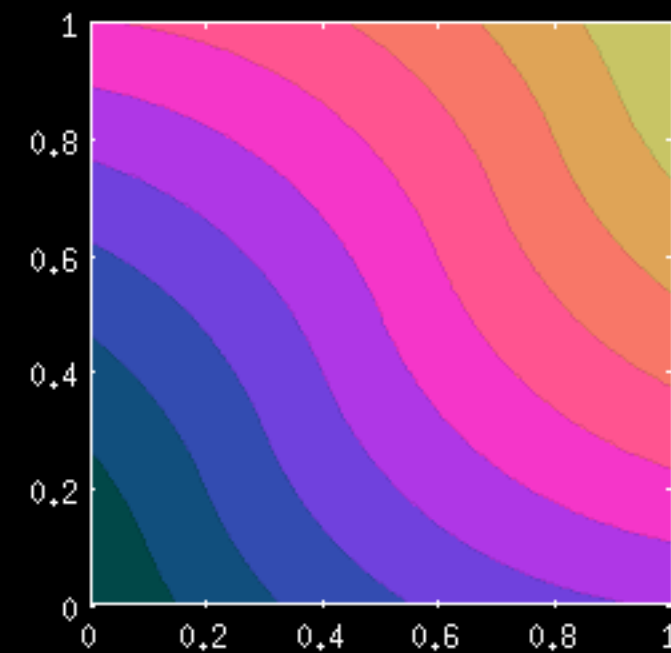
0.28571



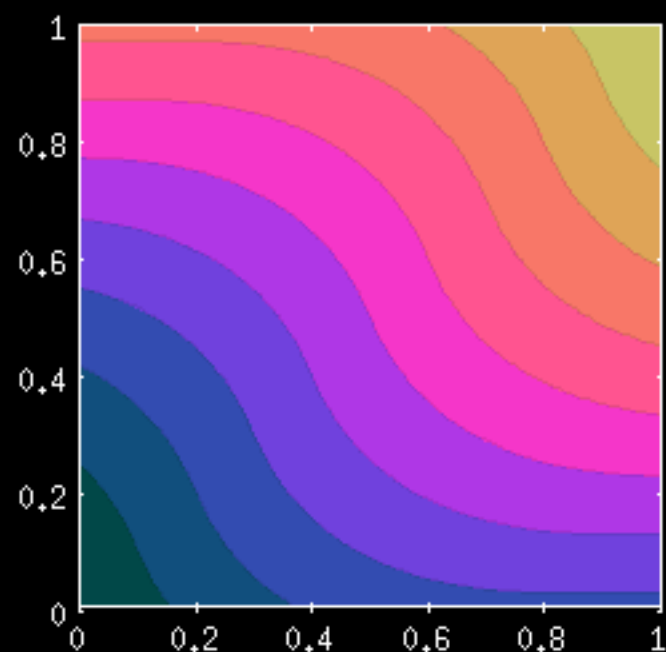
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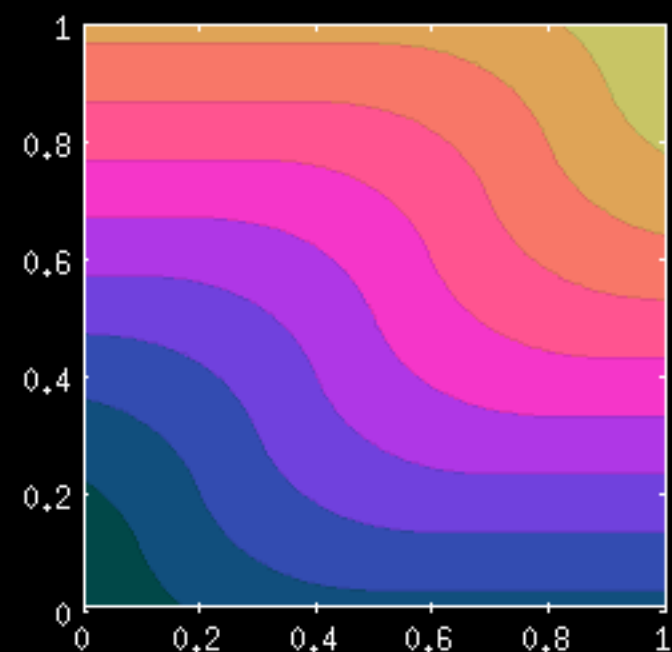
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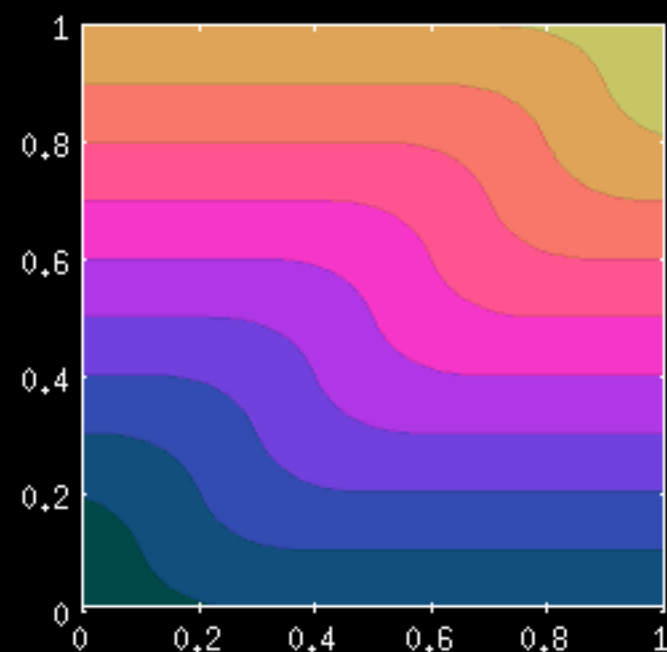
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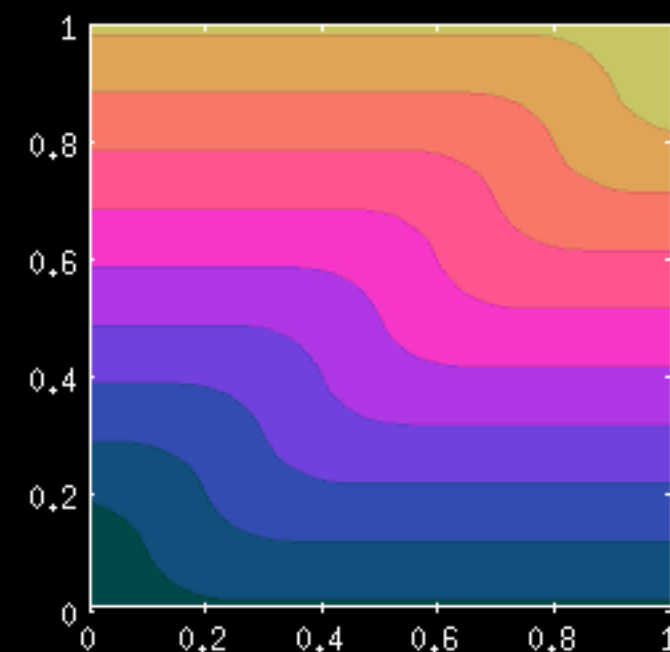
0.85714



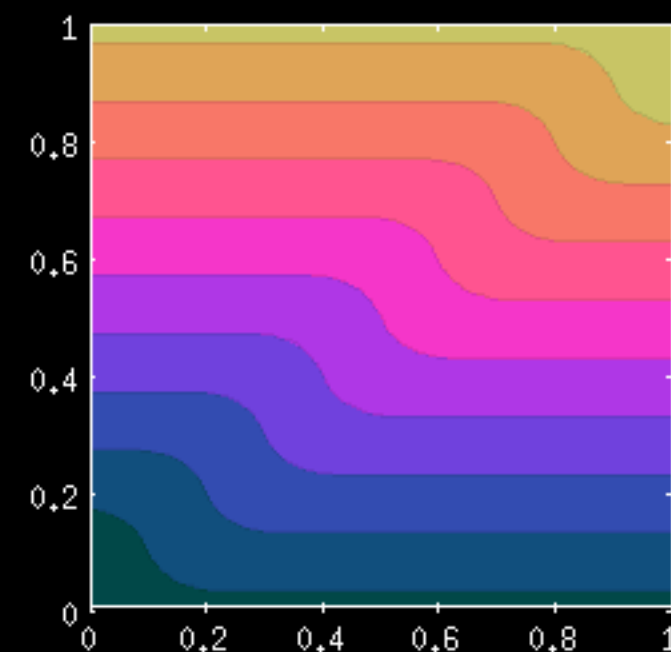
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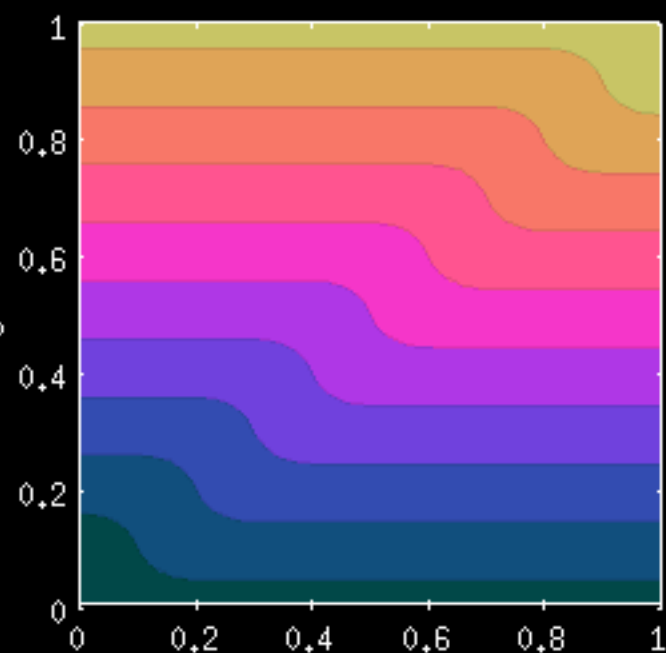
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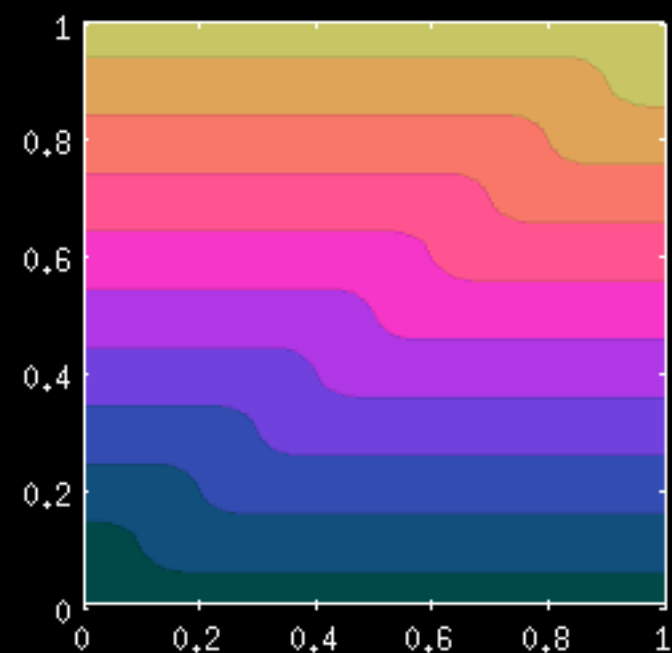
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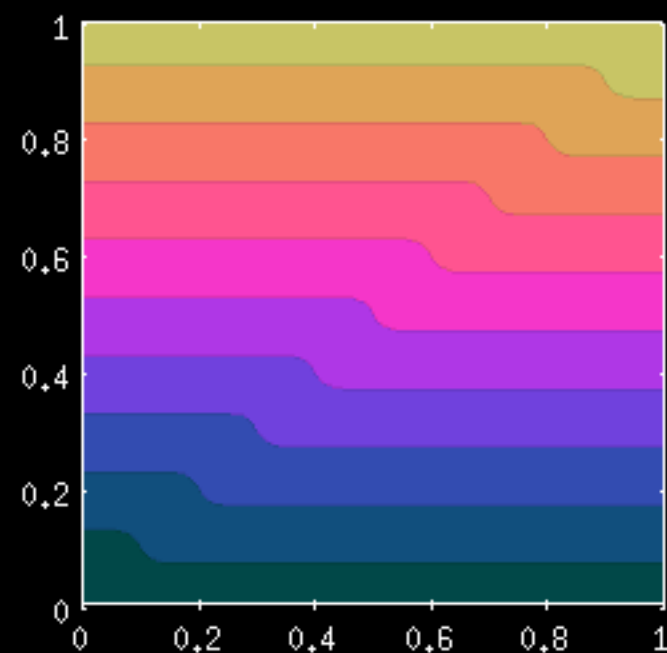
1.4286



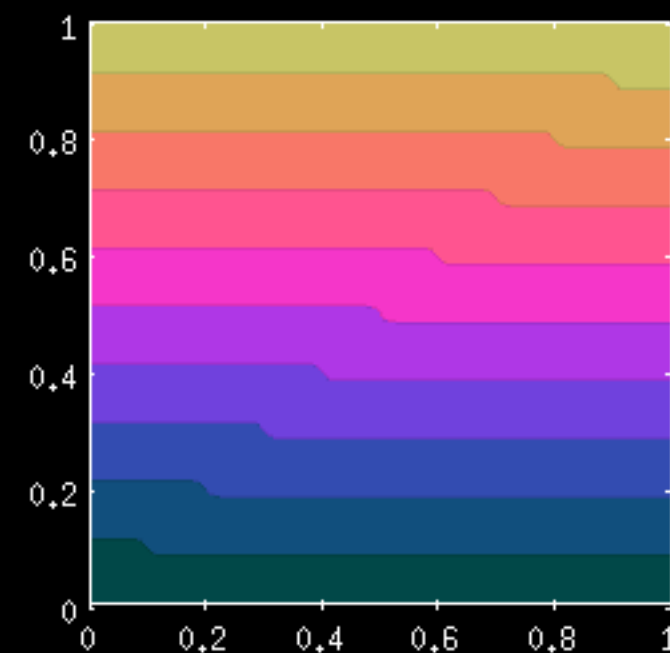
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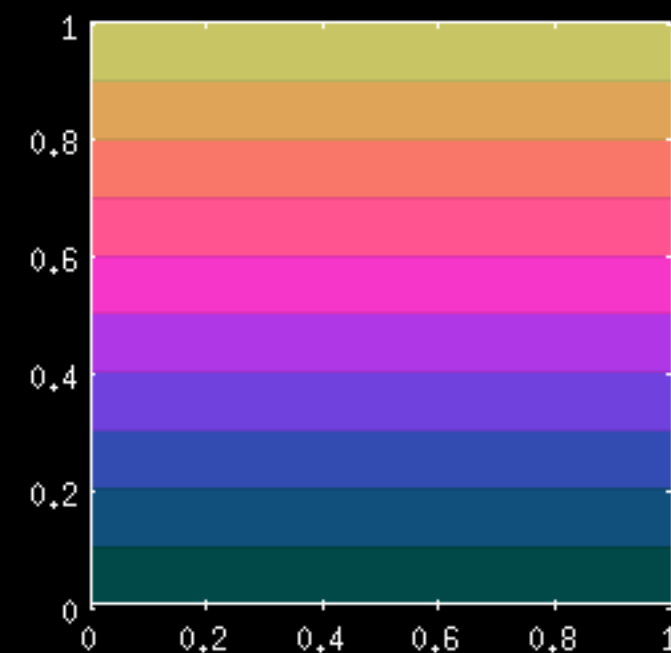
1.7143



1.8571



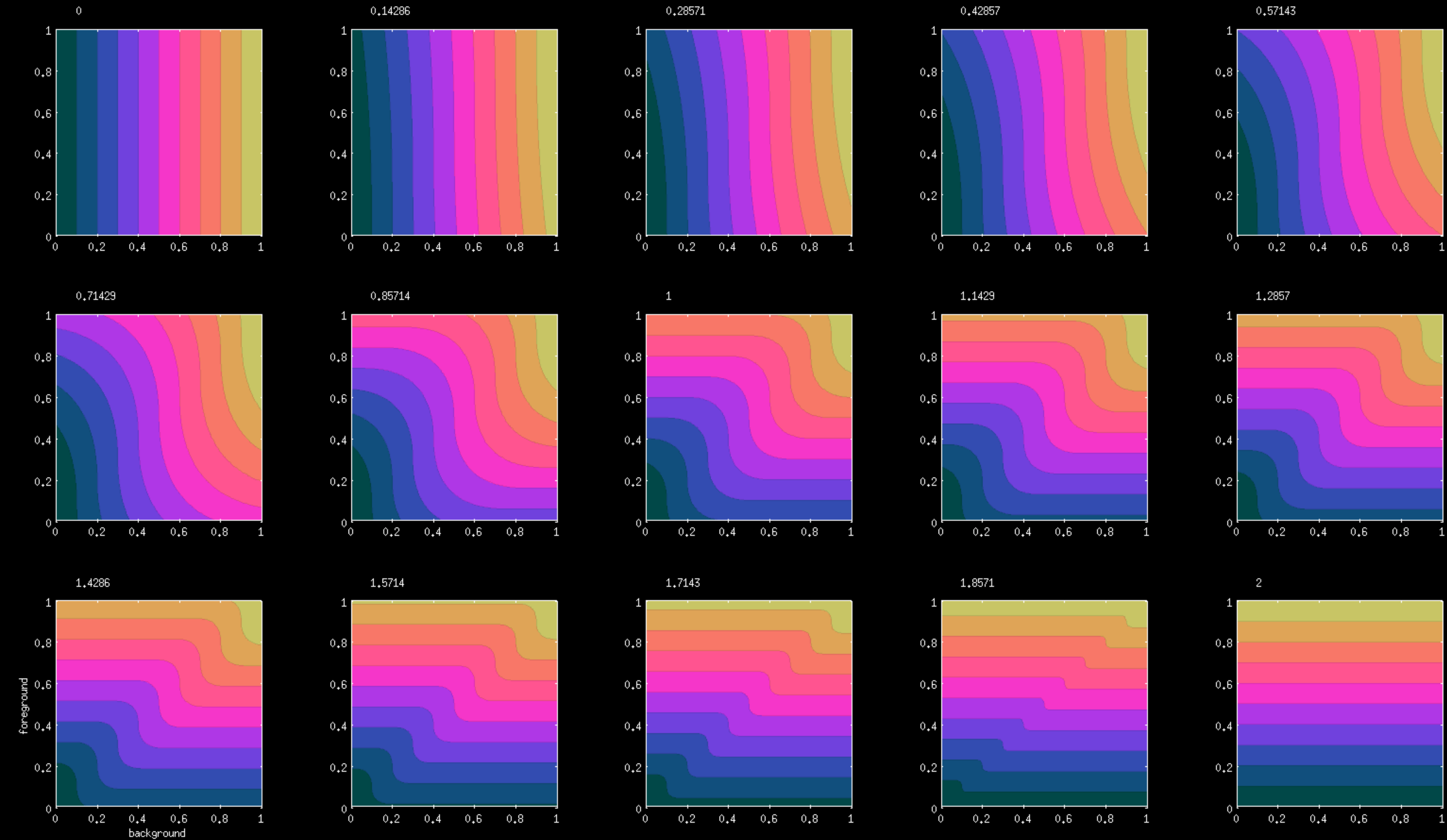
2



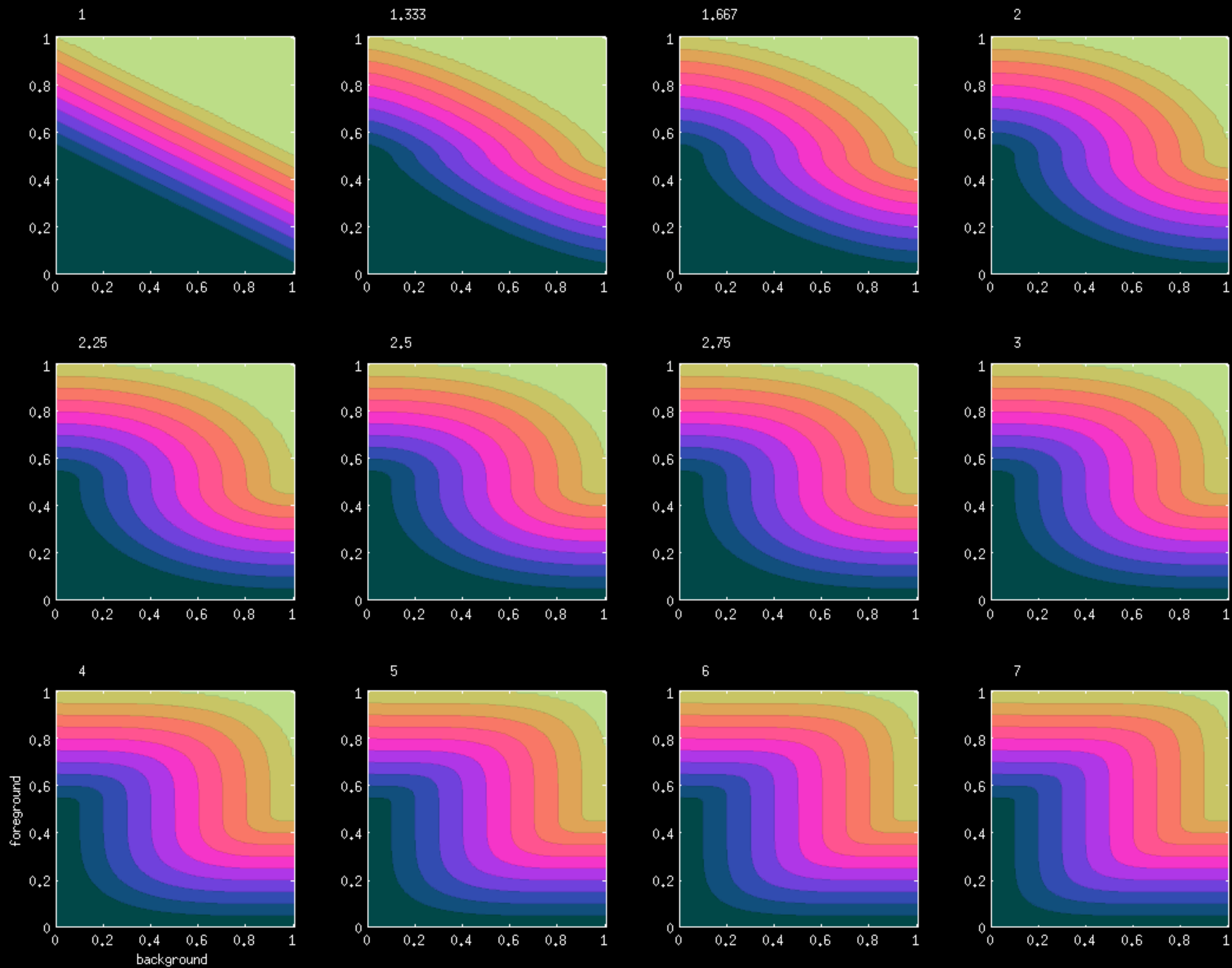
foreground

background

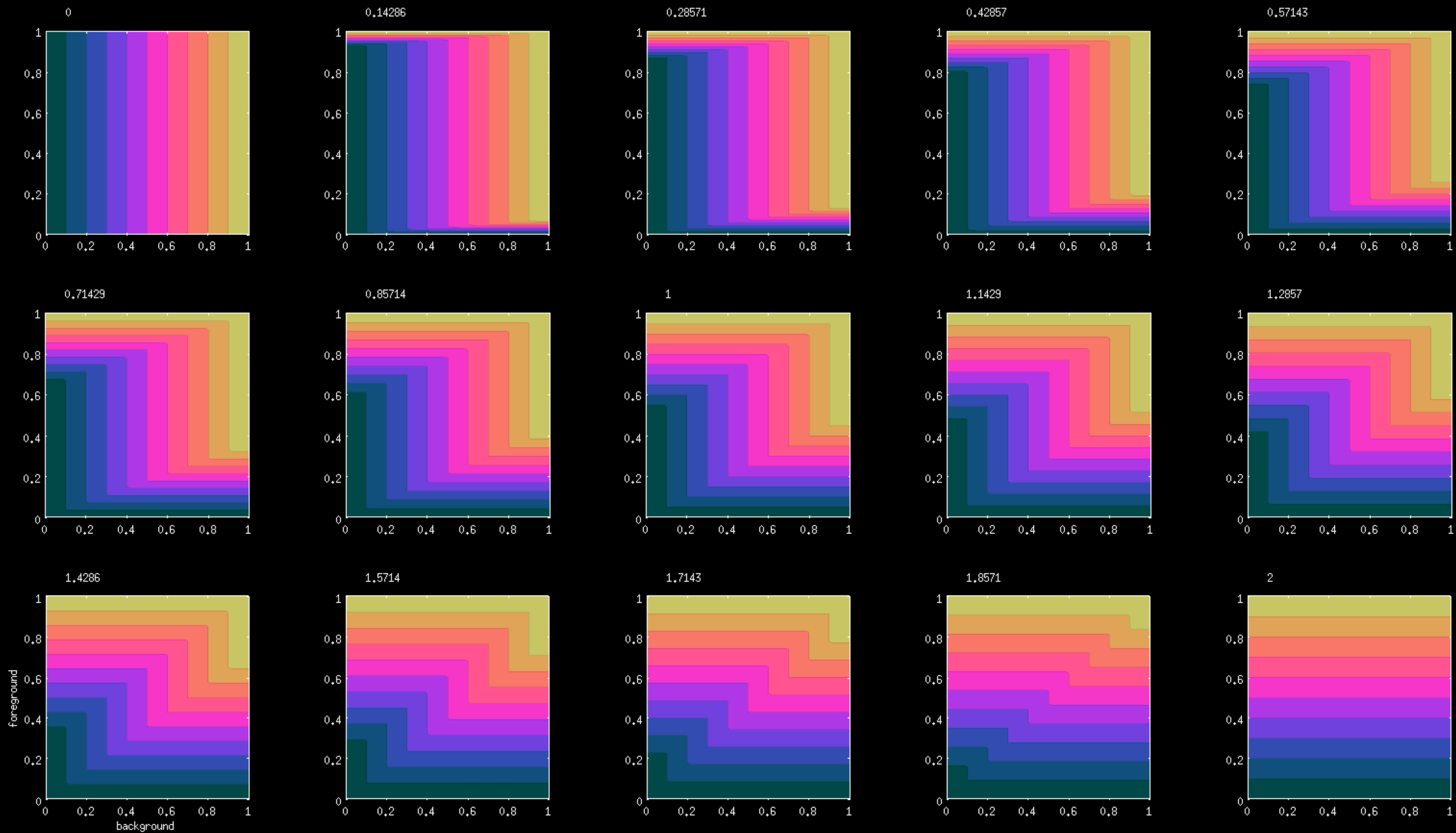
moonlight2



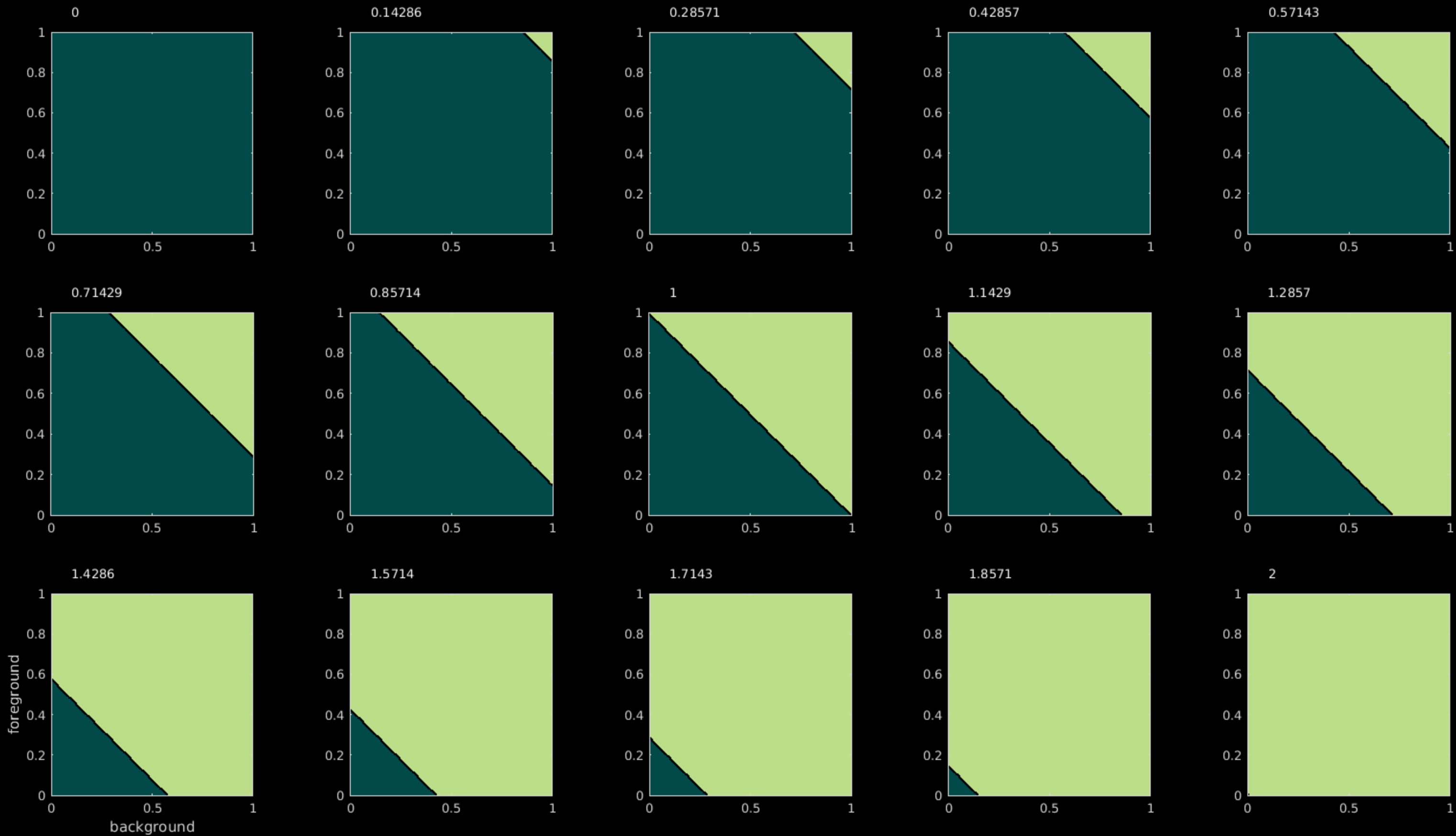
superlight



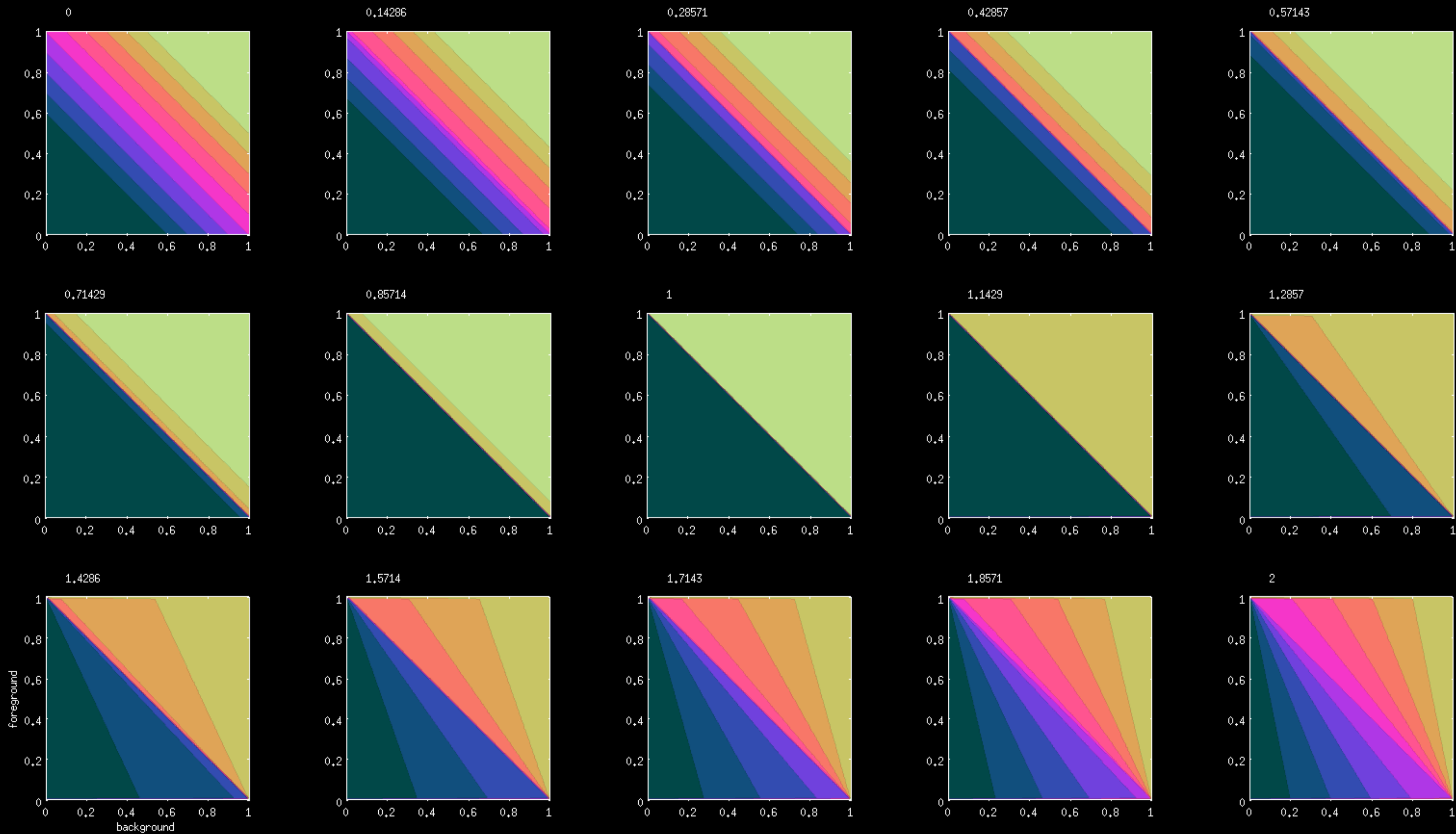
pinlight



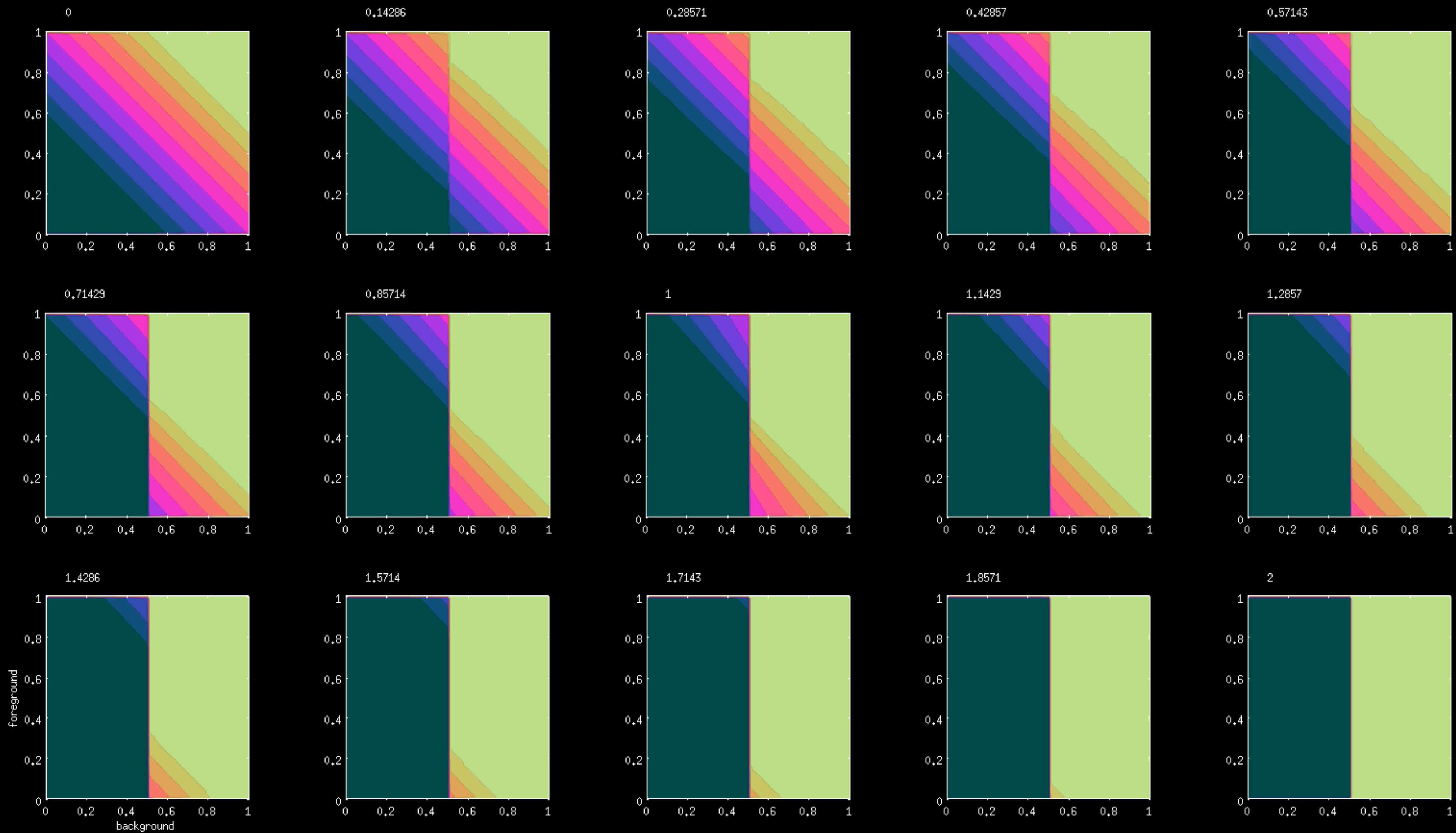
hardmixps



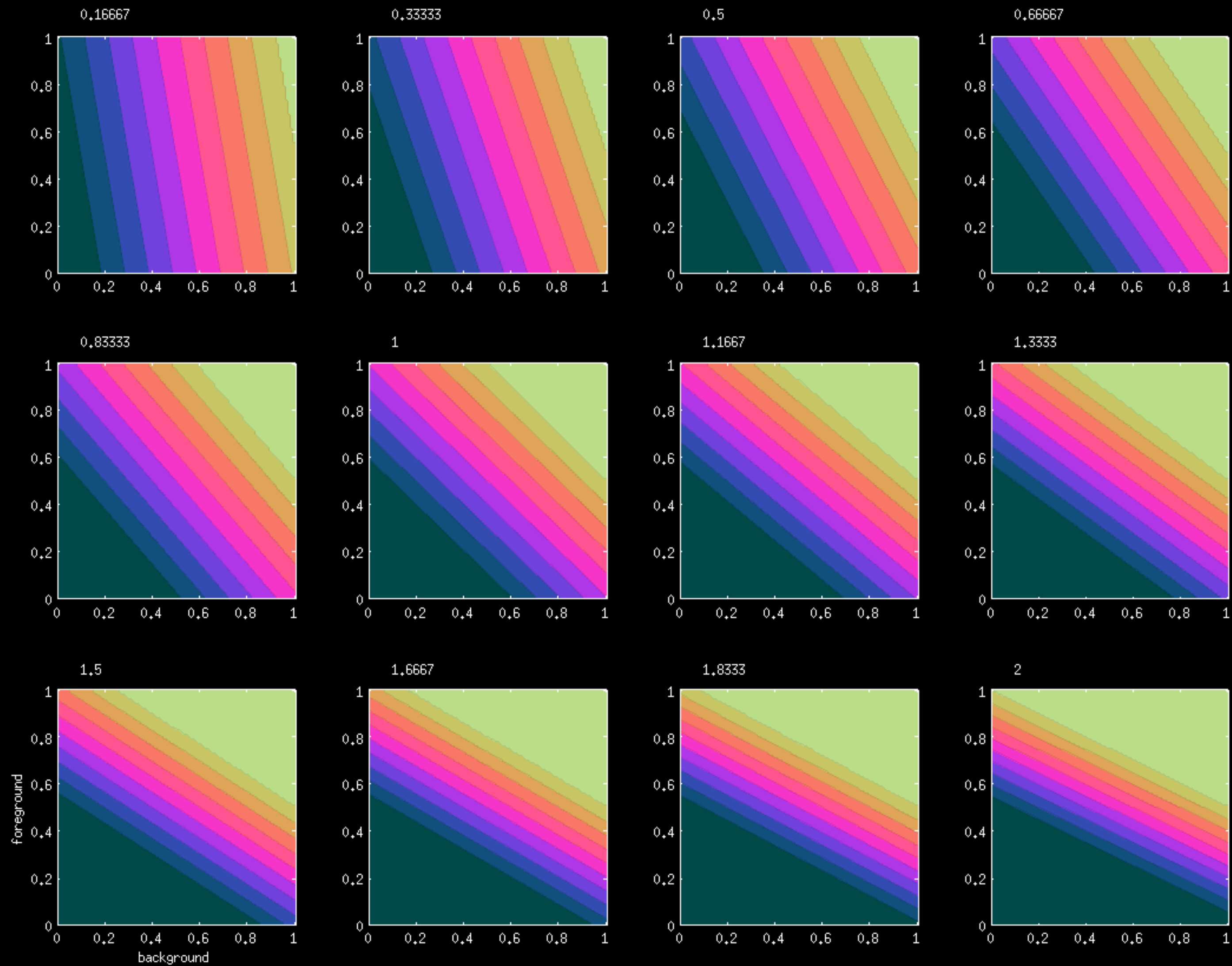
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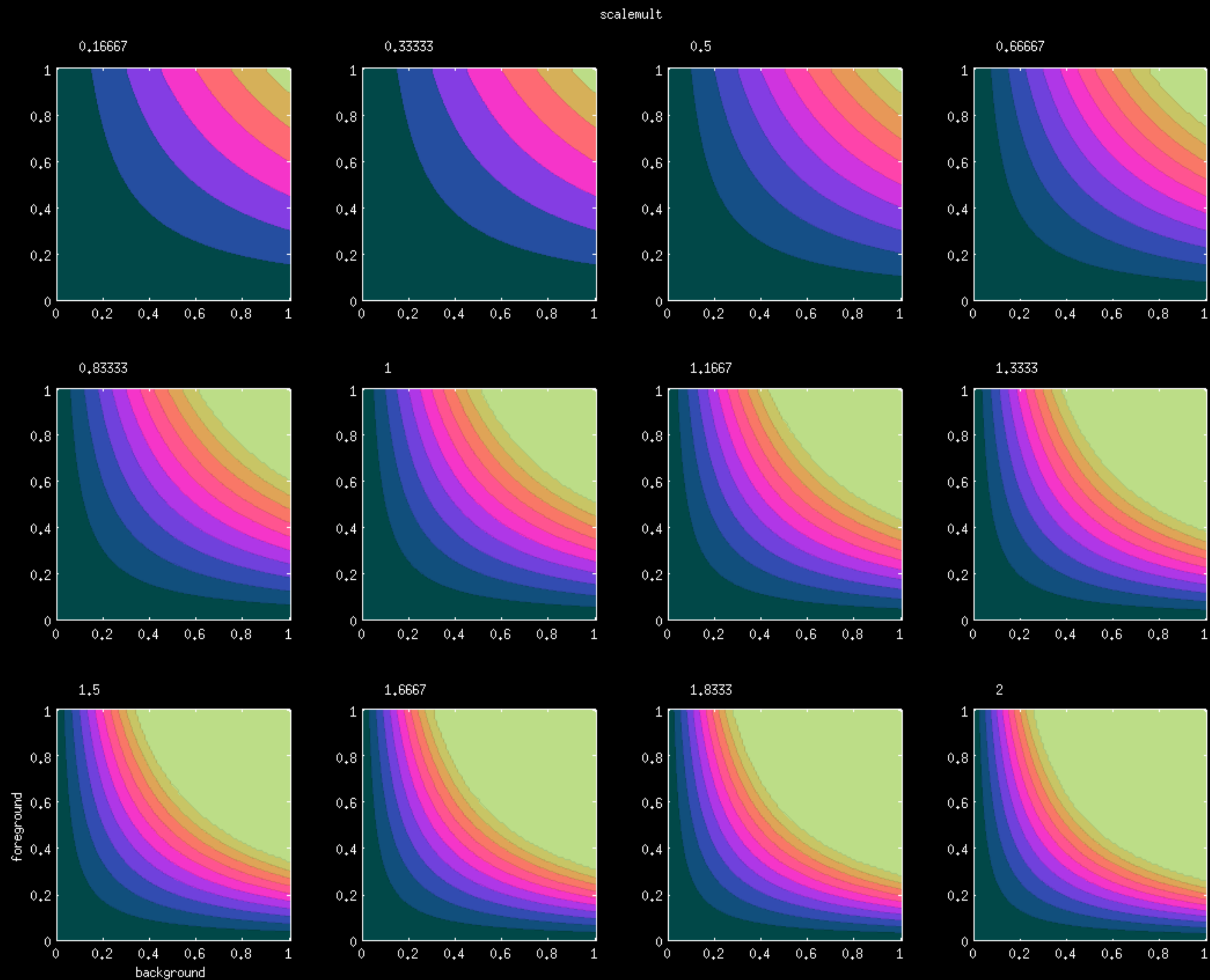


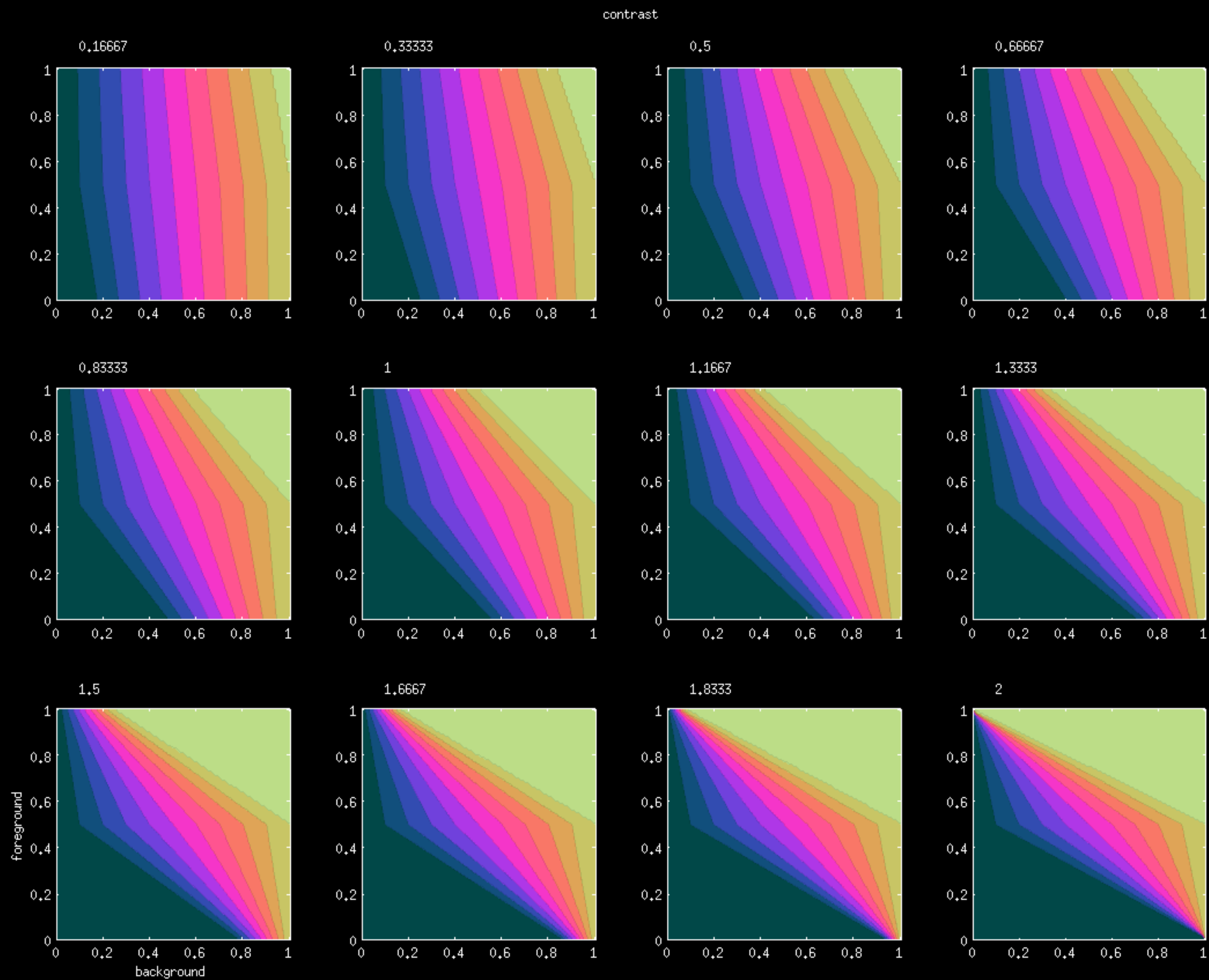
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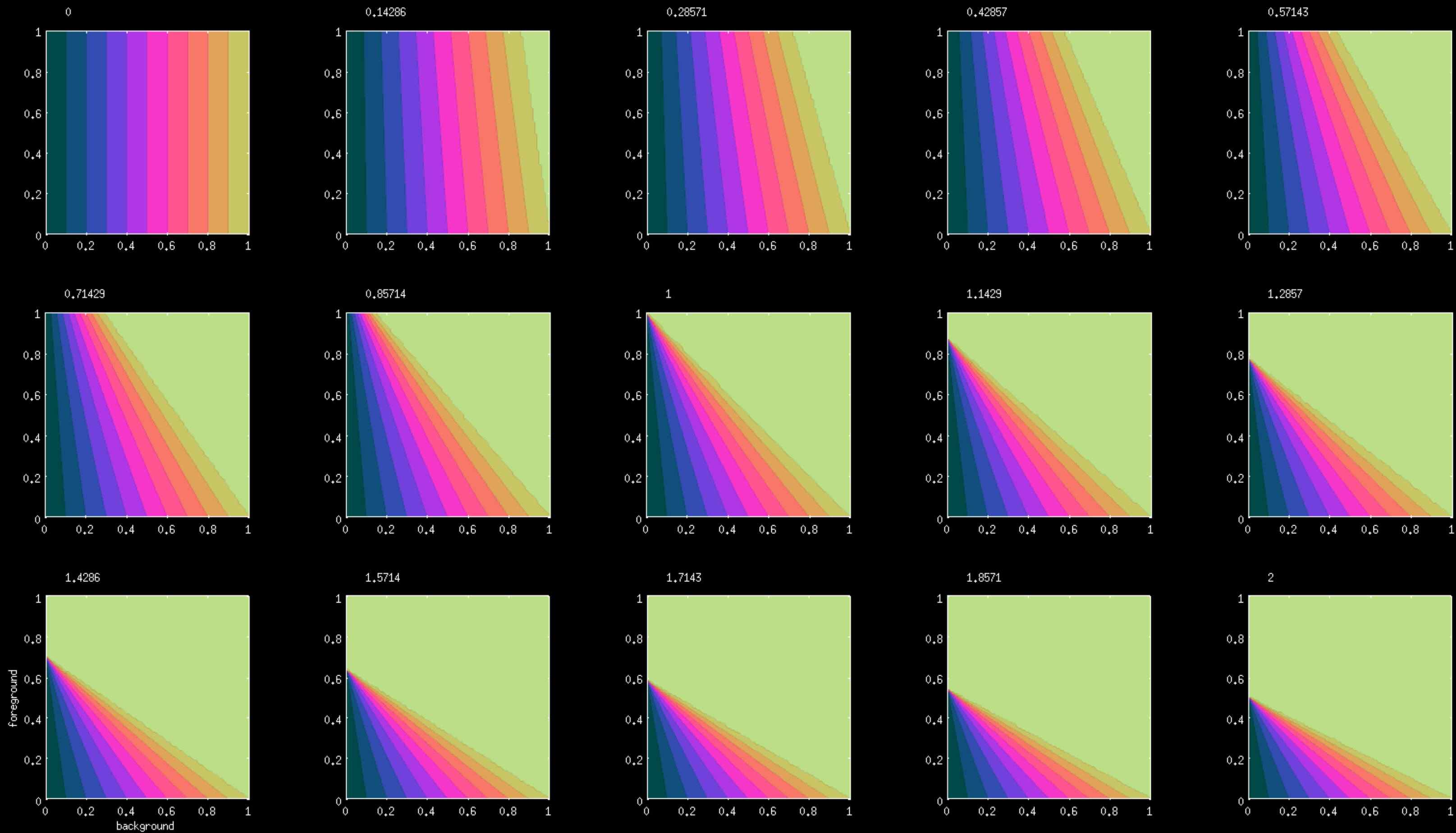
scaleadd



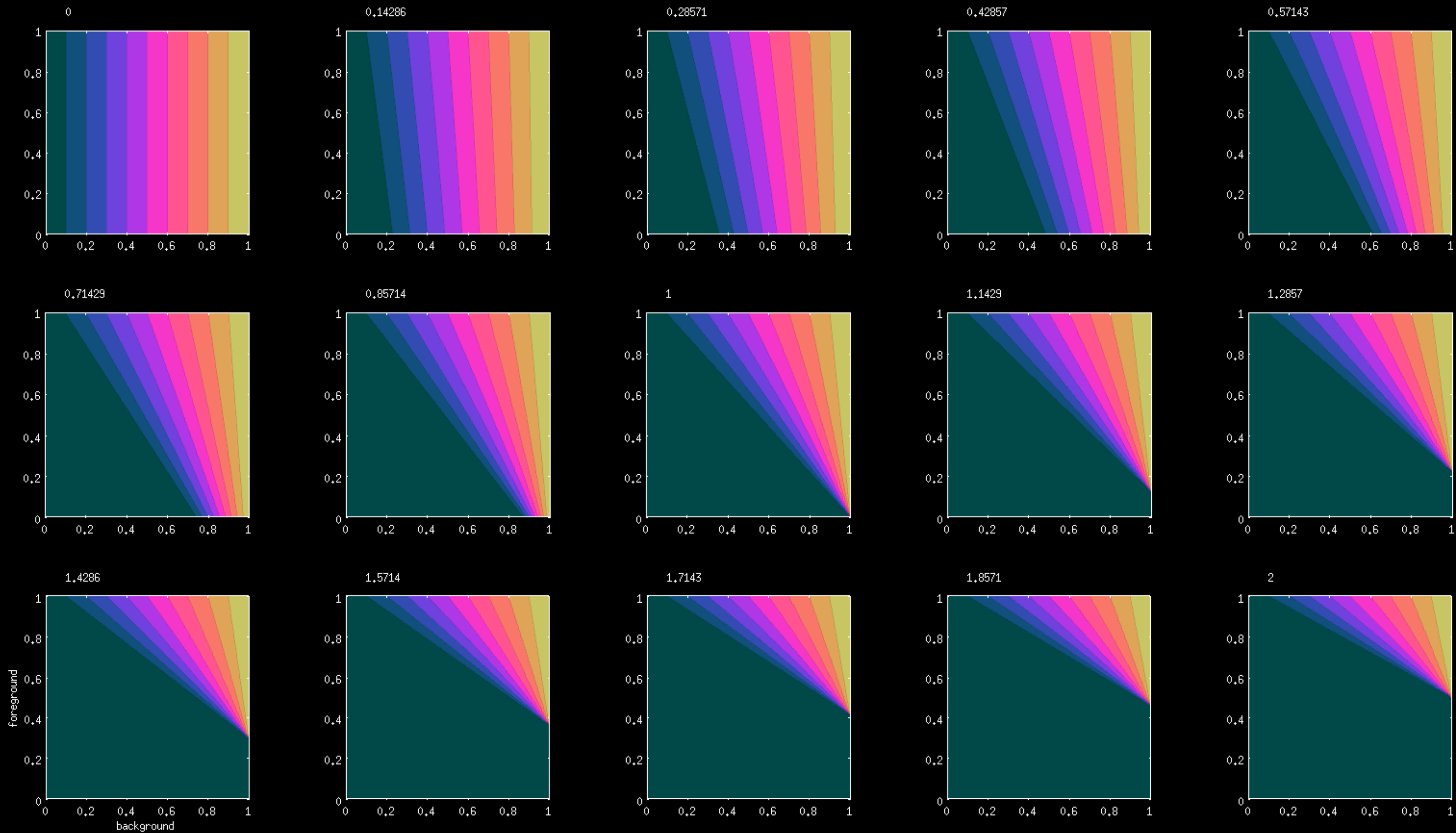




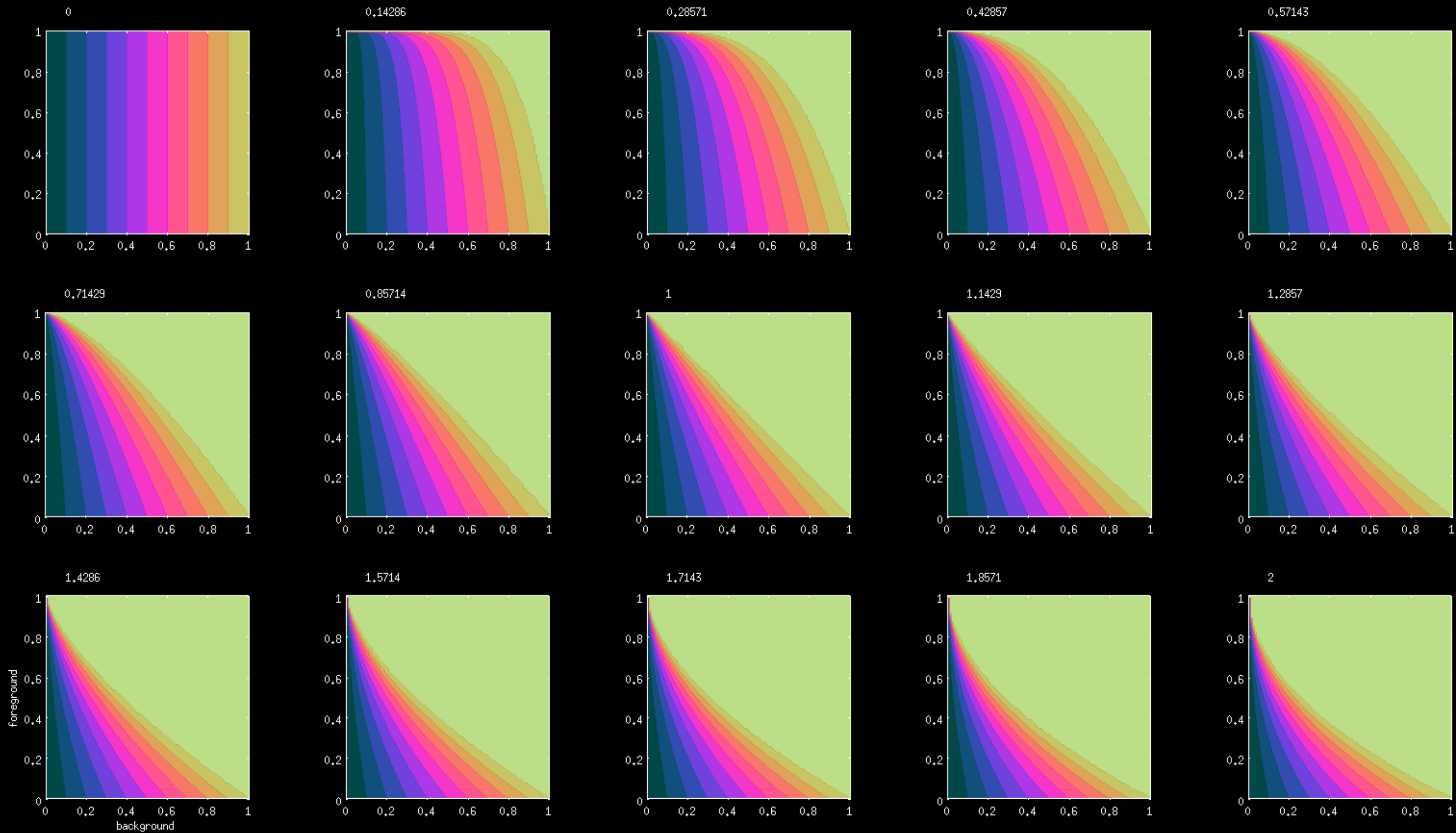
colordodge



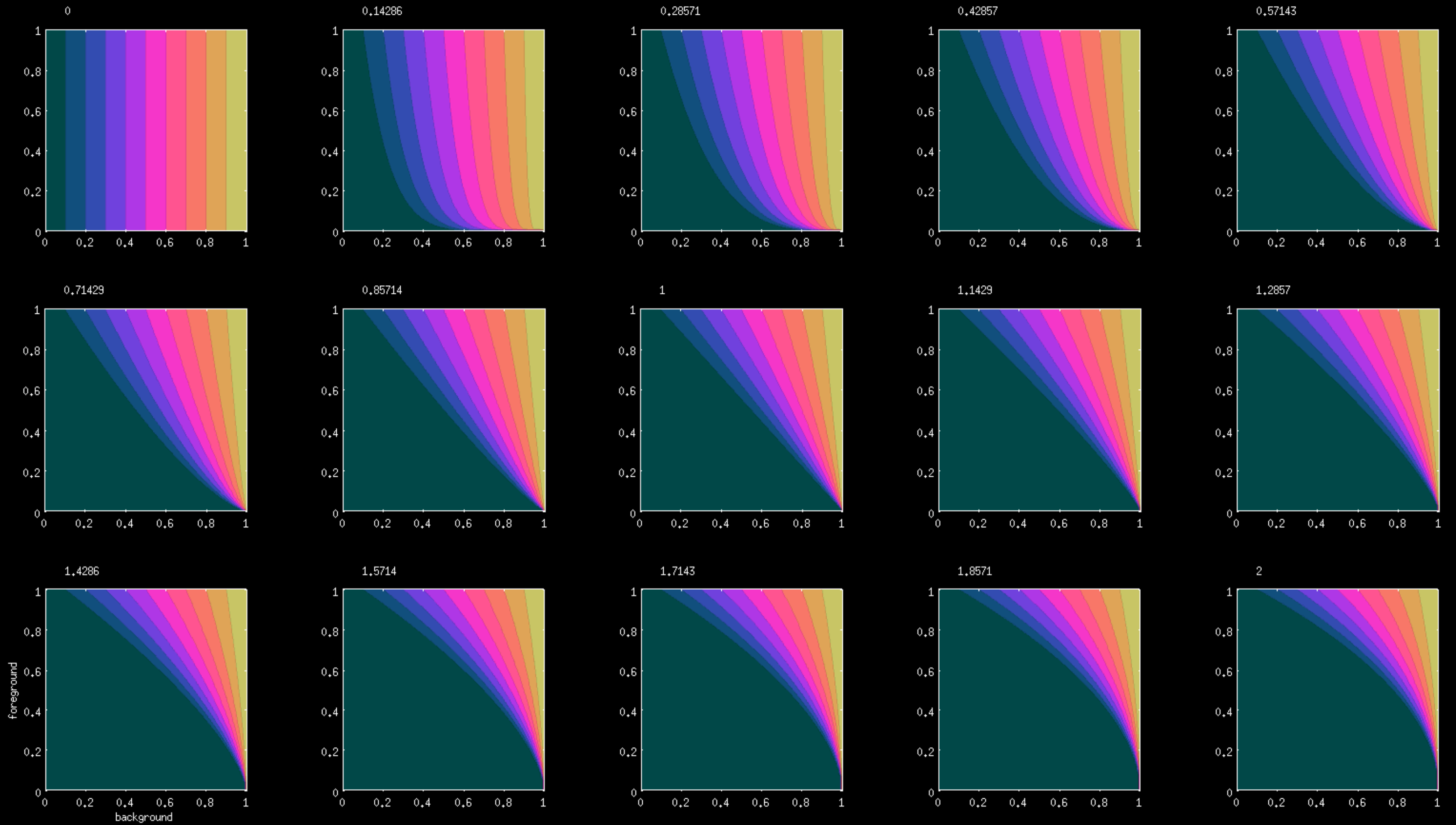
colorburn



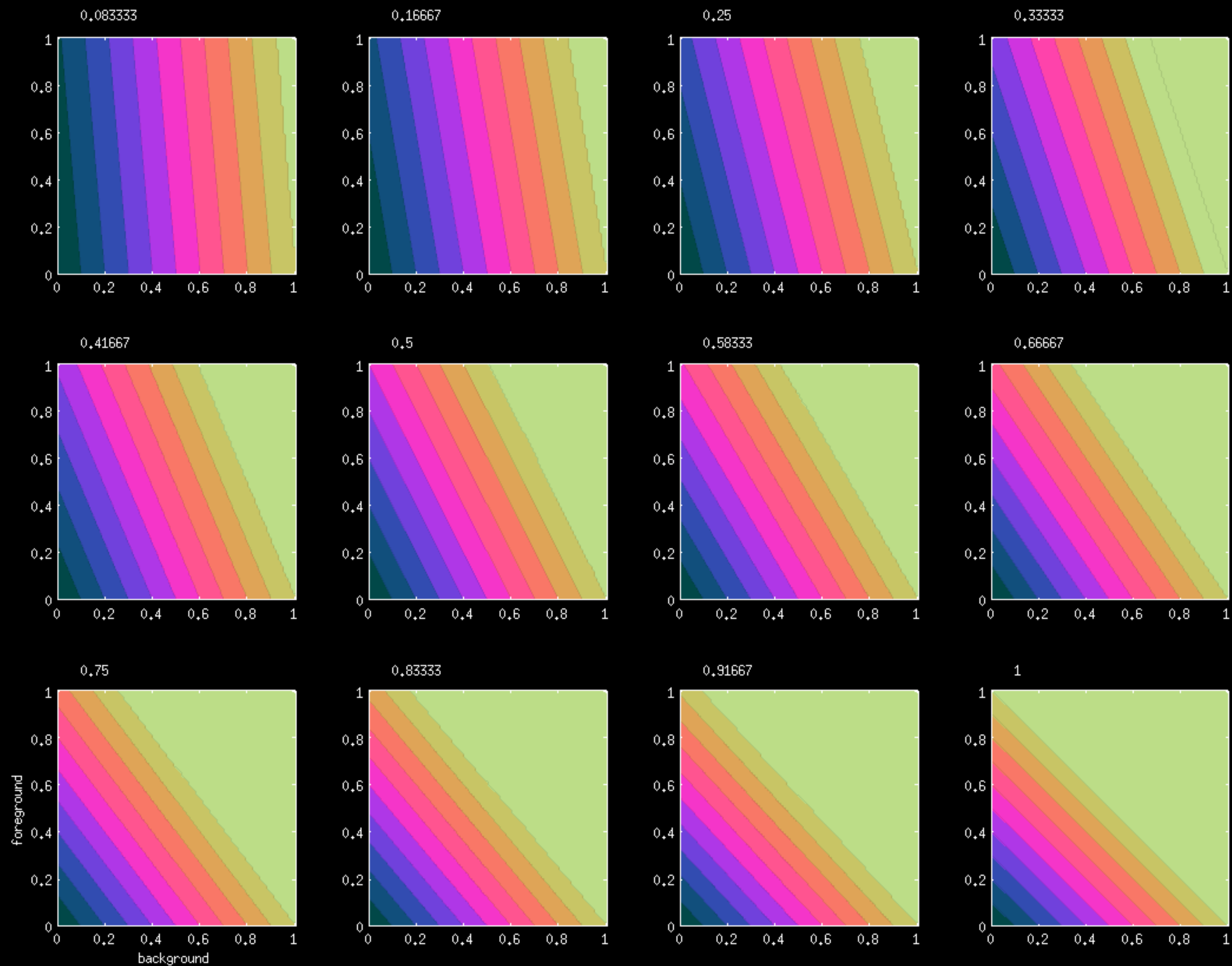
polydodge



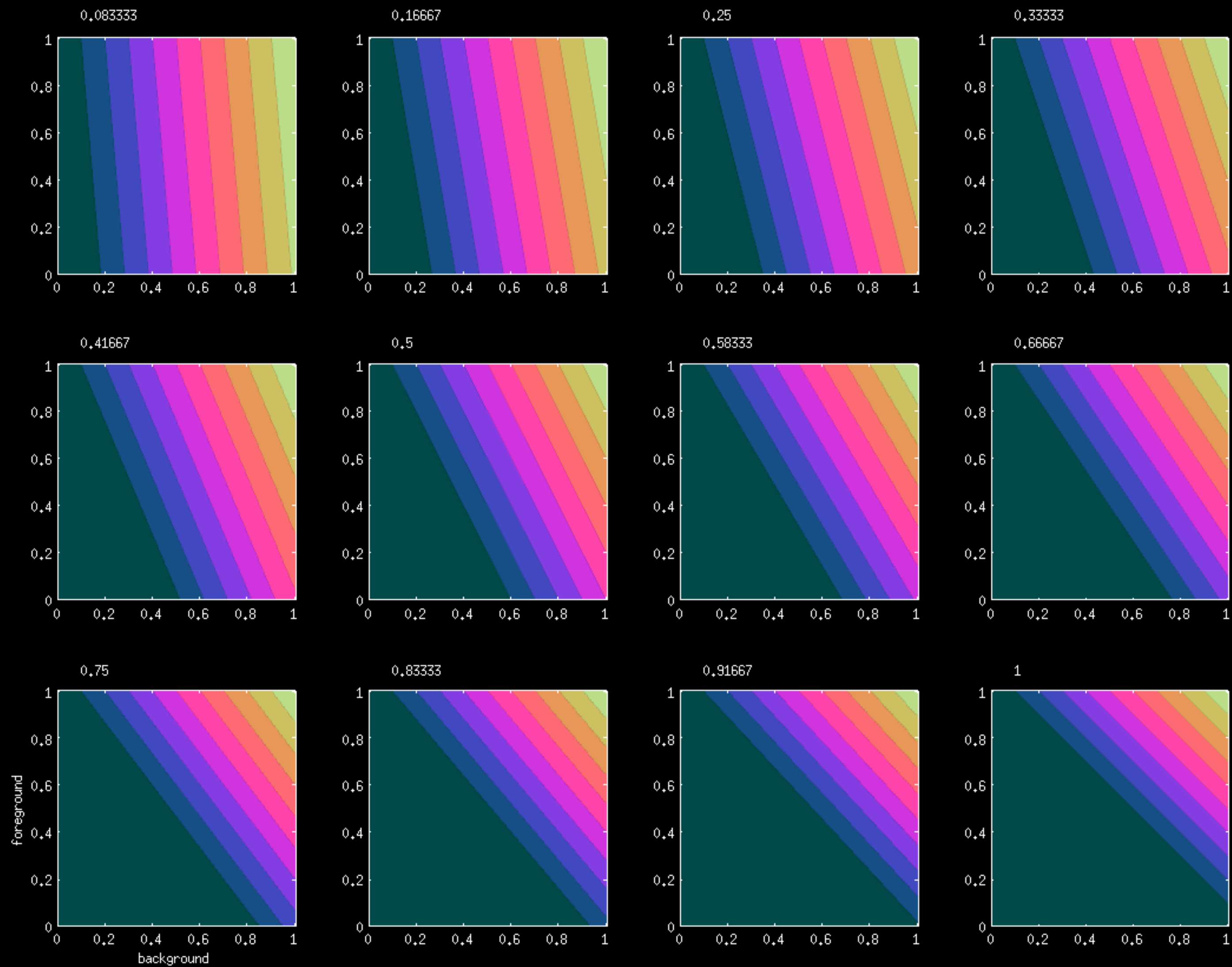
polyburn



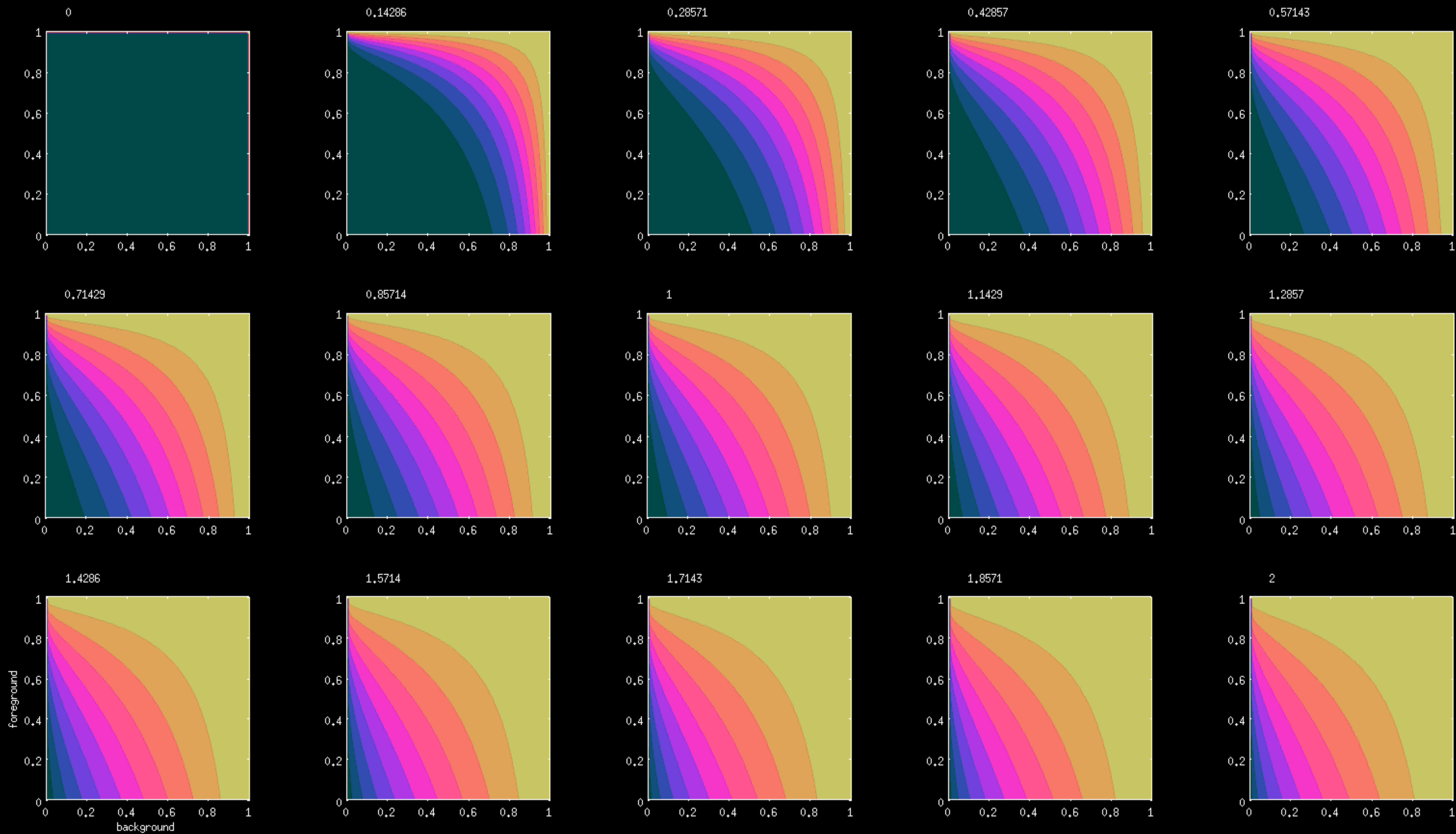
lineardodge



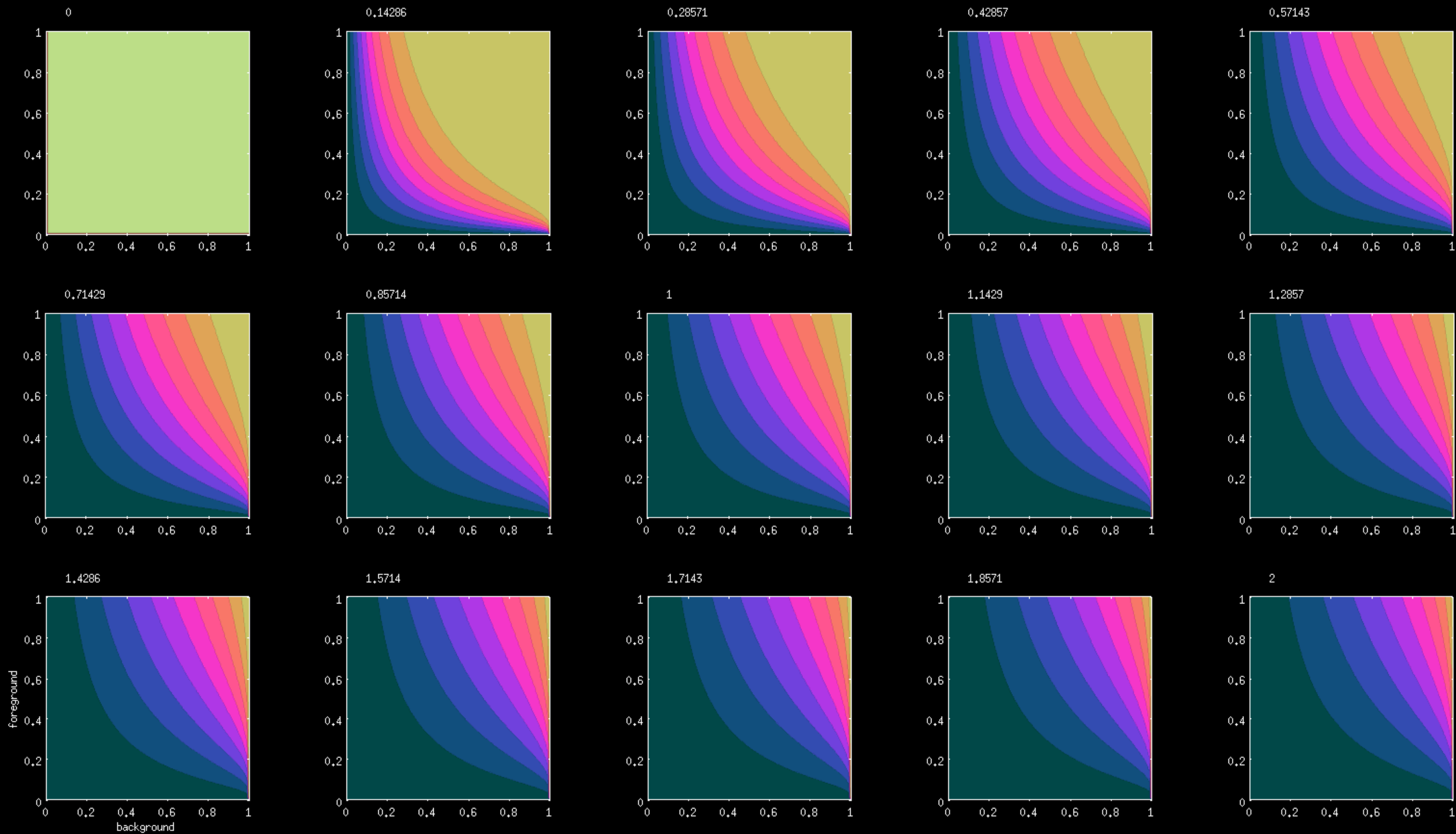
linearburn



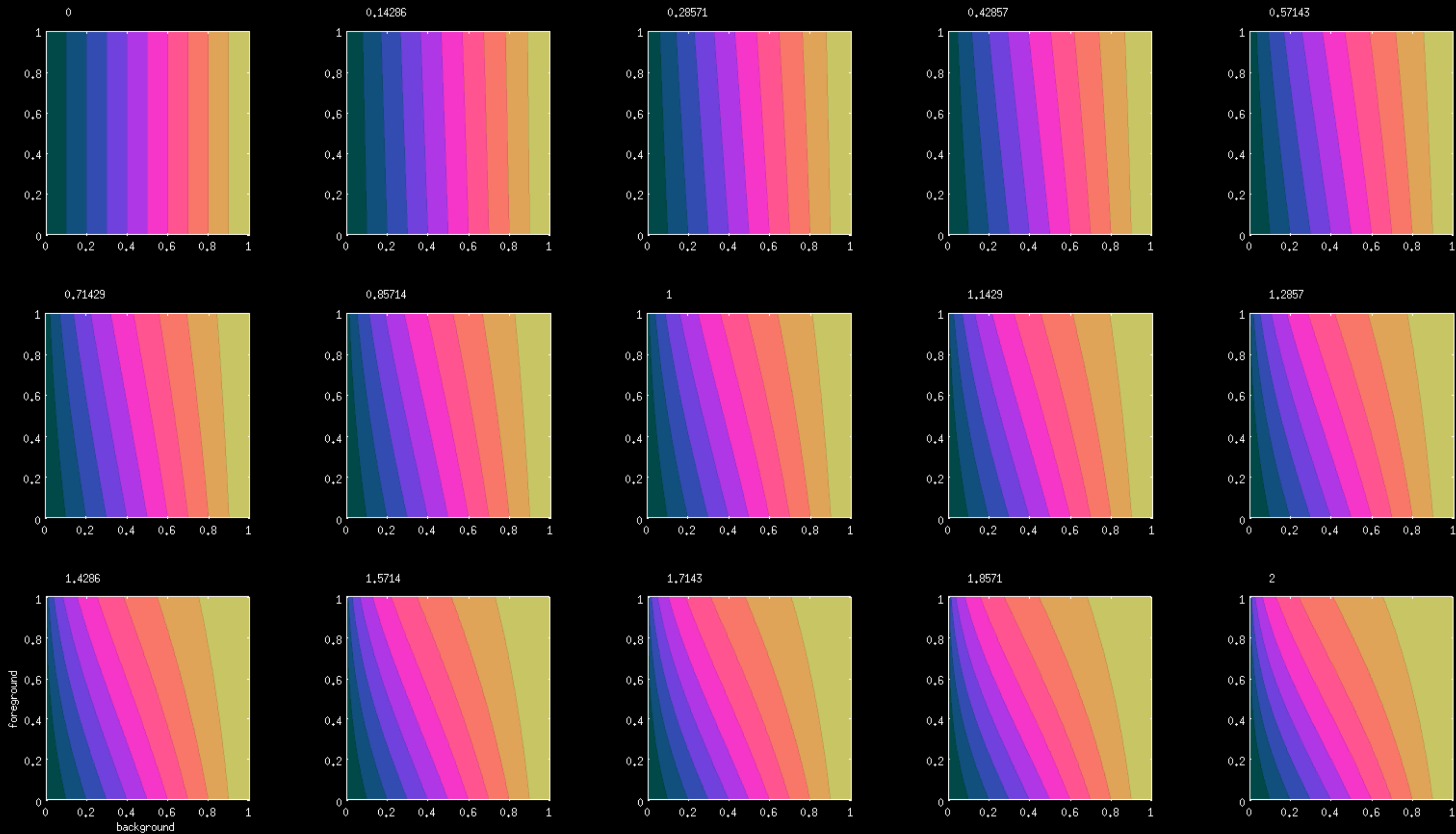
easydodge



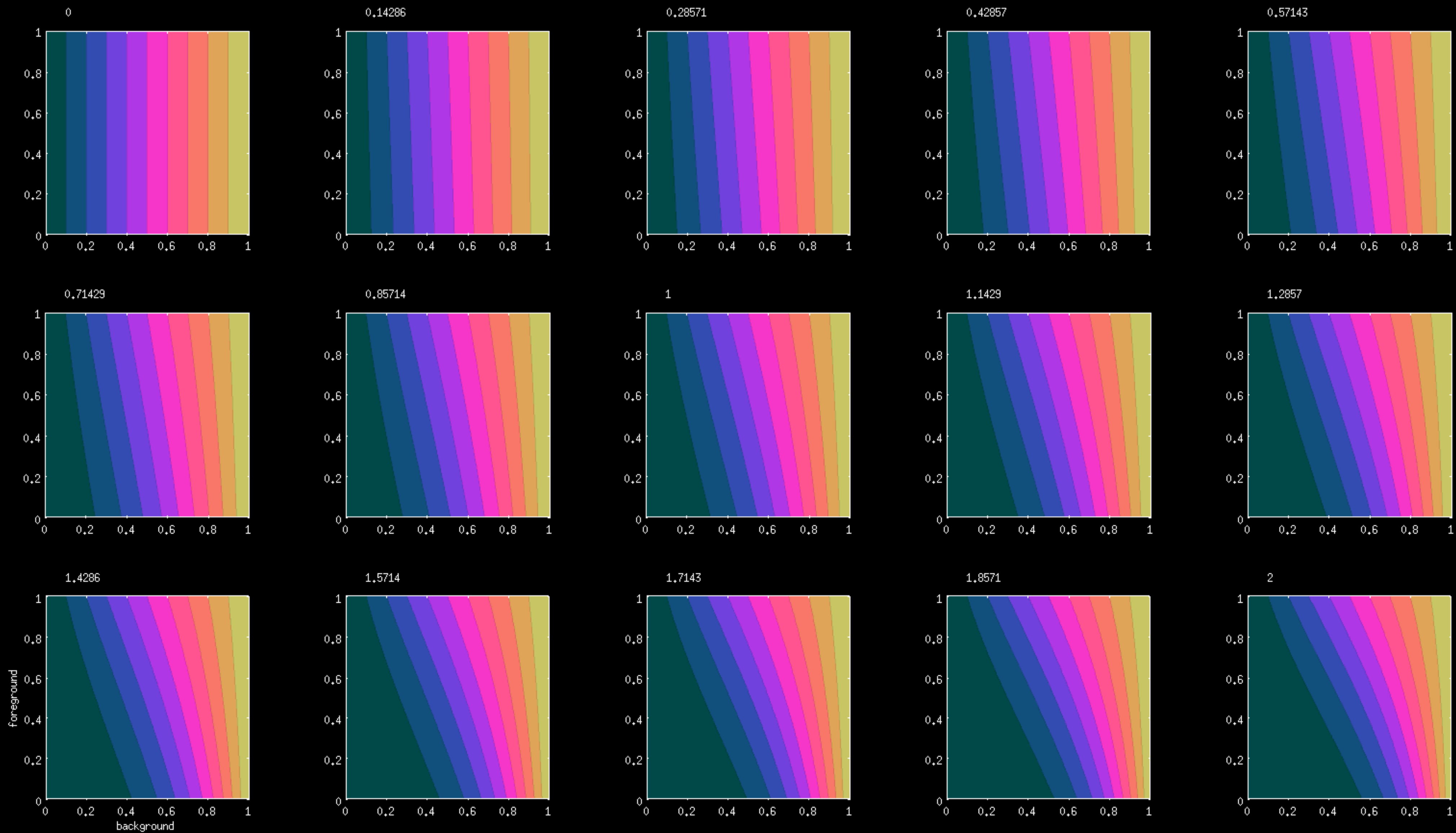
easyburn



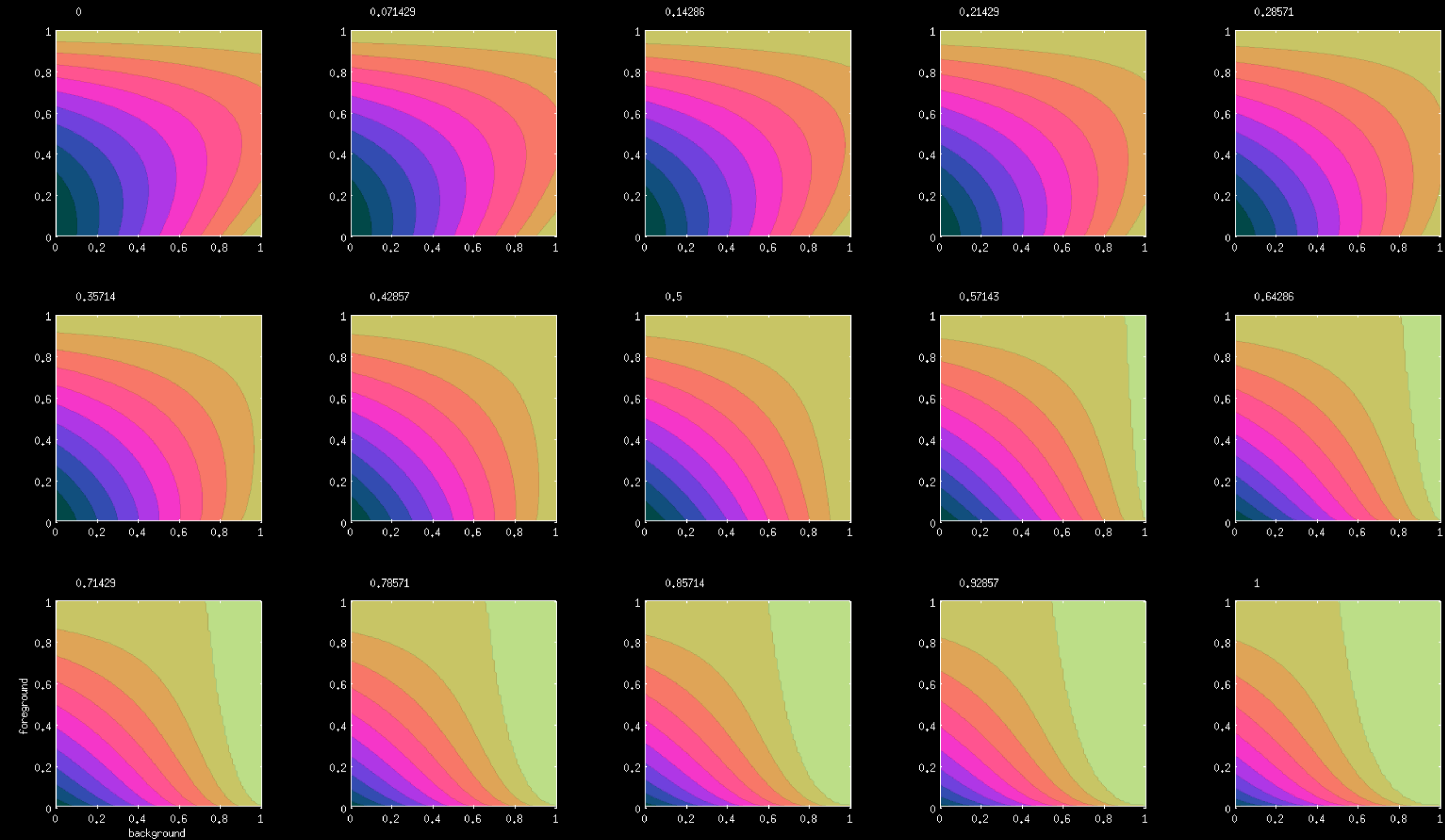
suaudodge



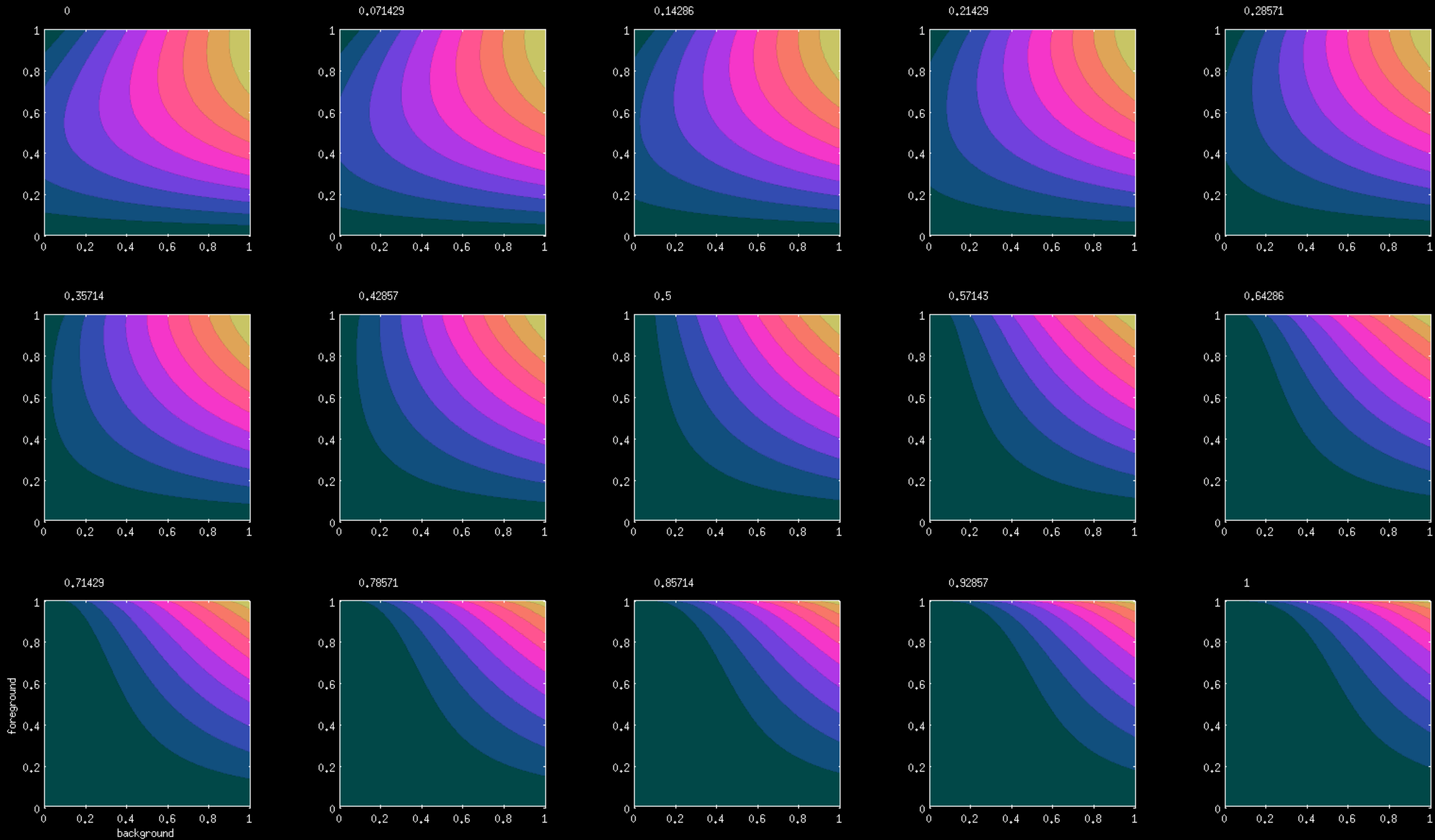
suauburn



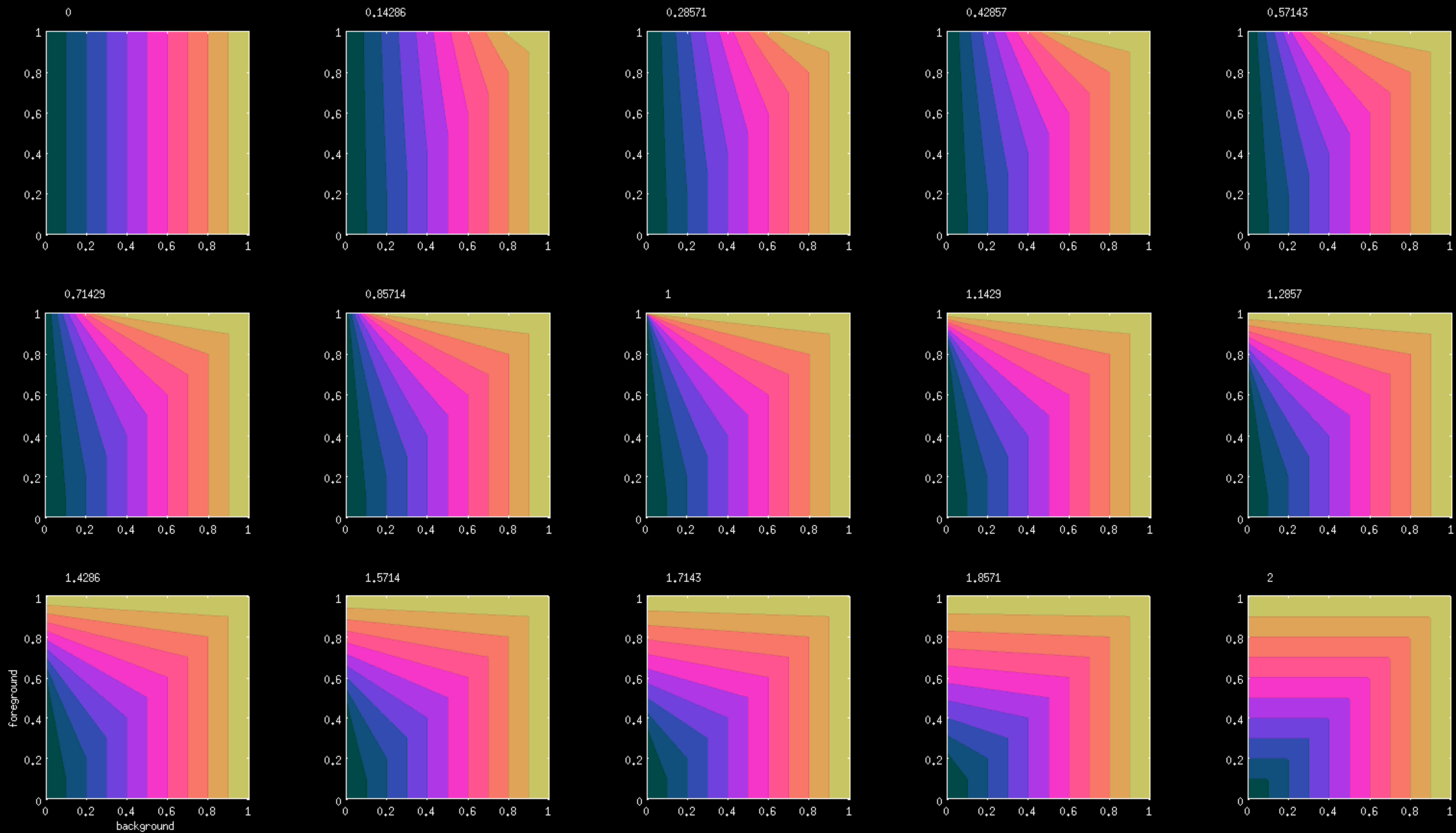
malekidodge



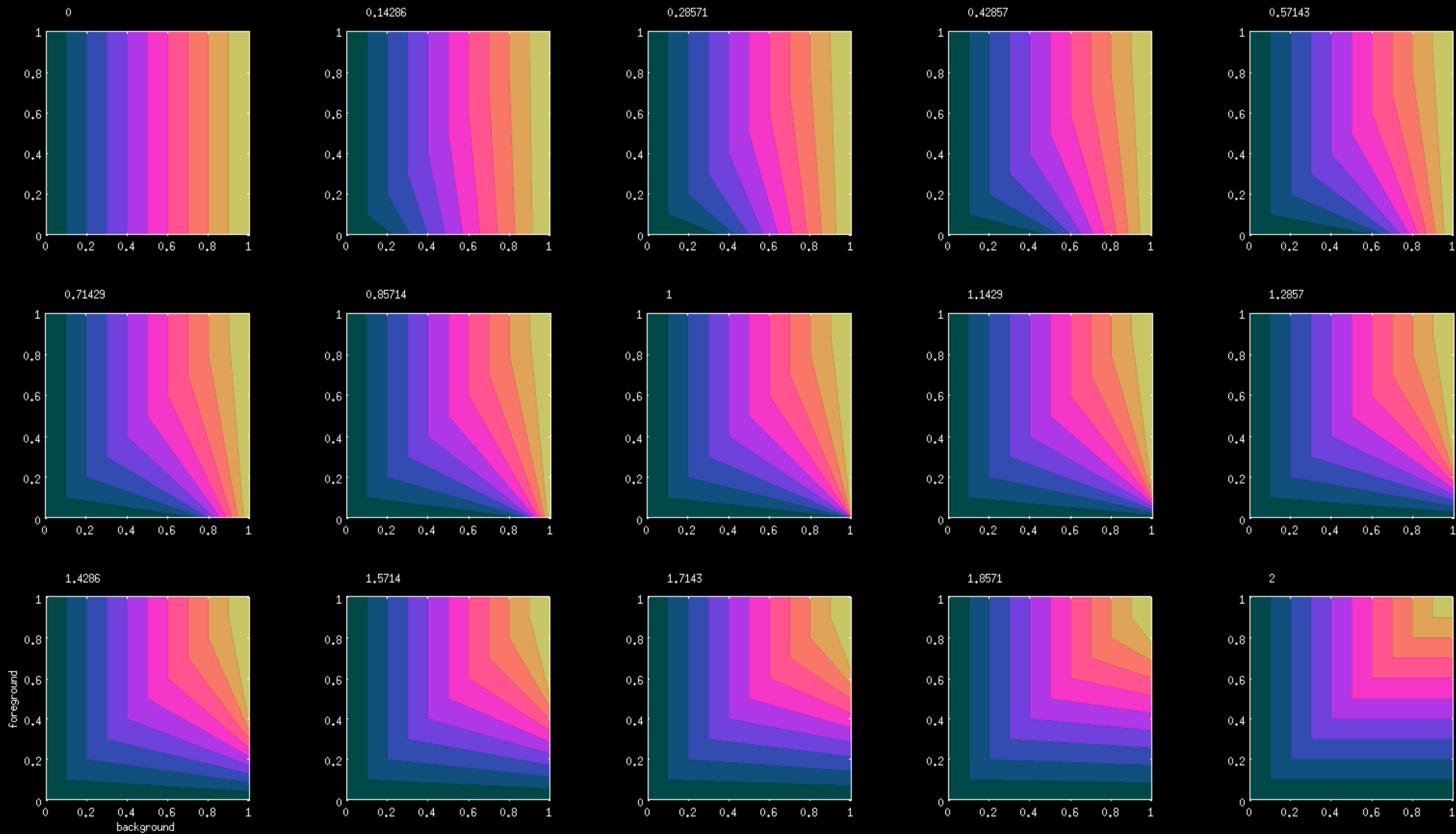
malekiburn



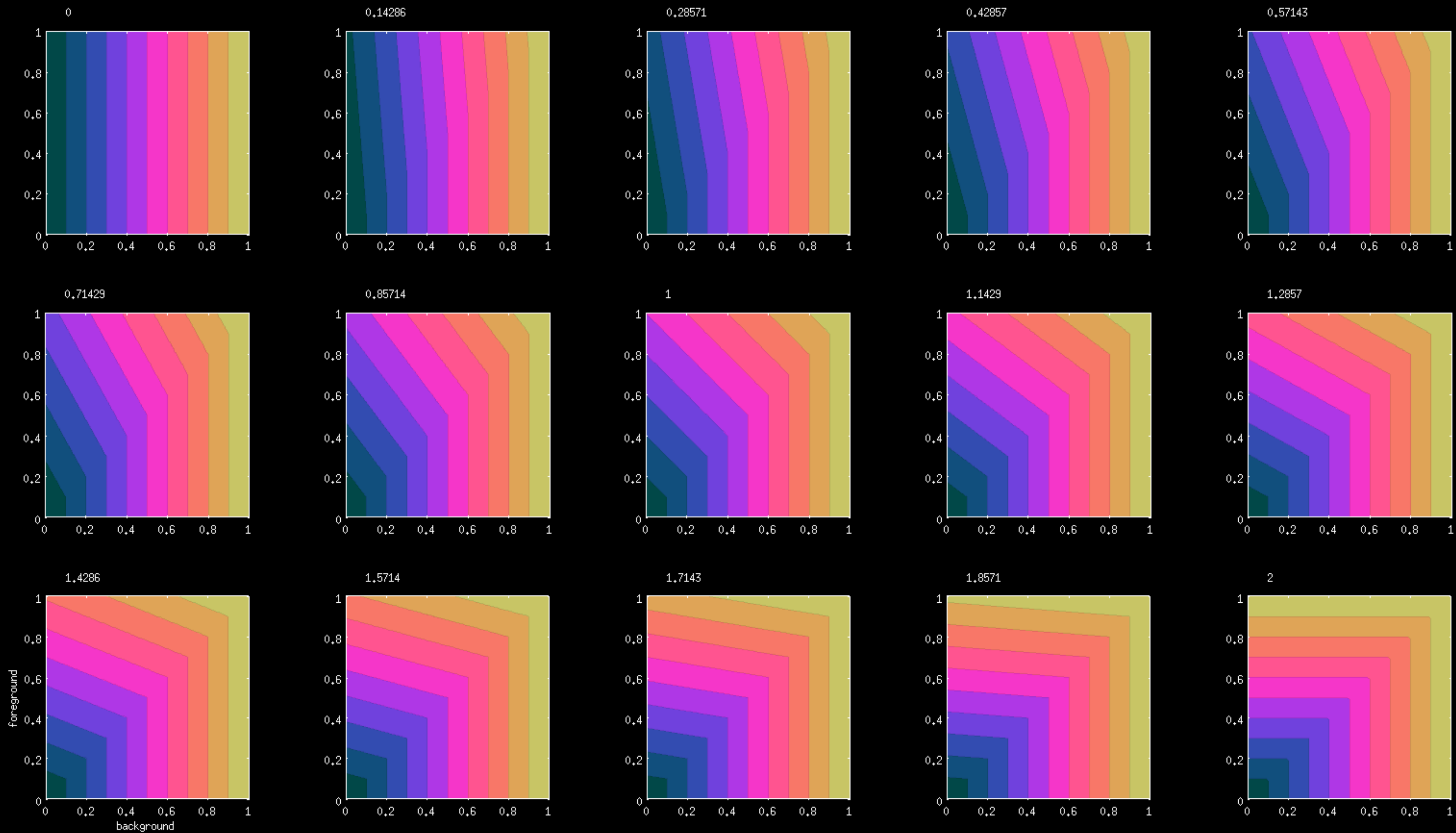
flatglow



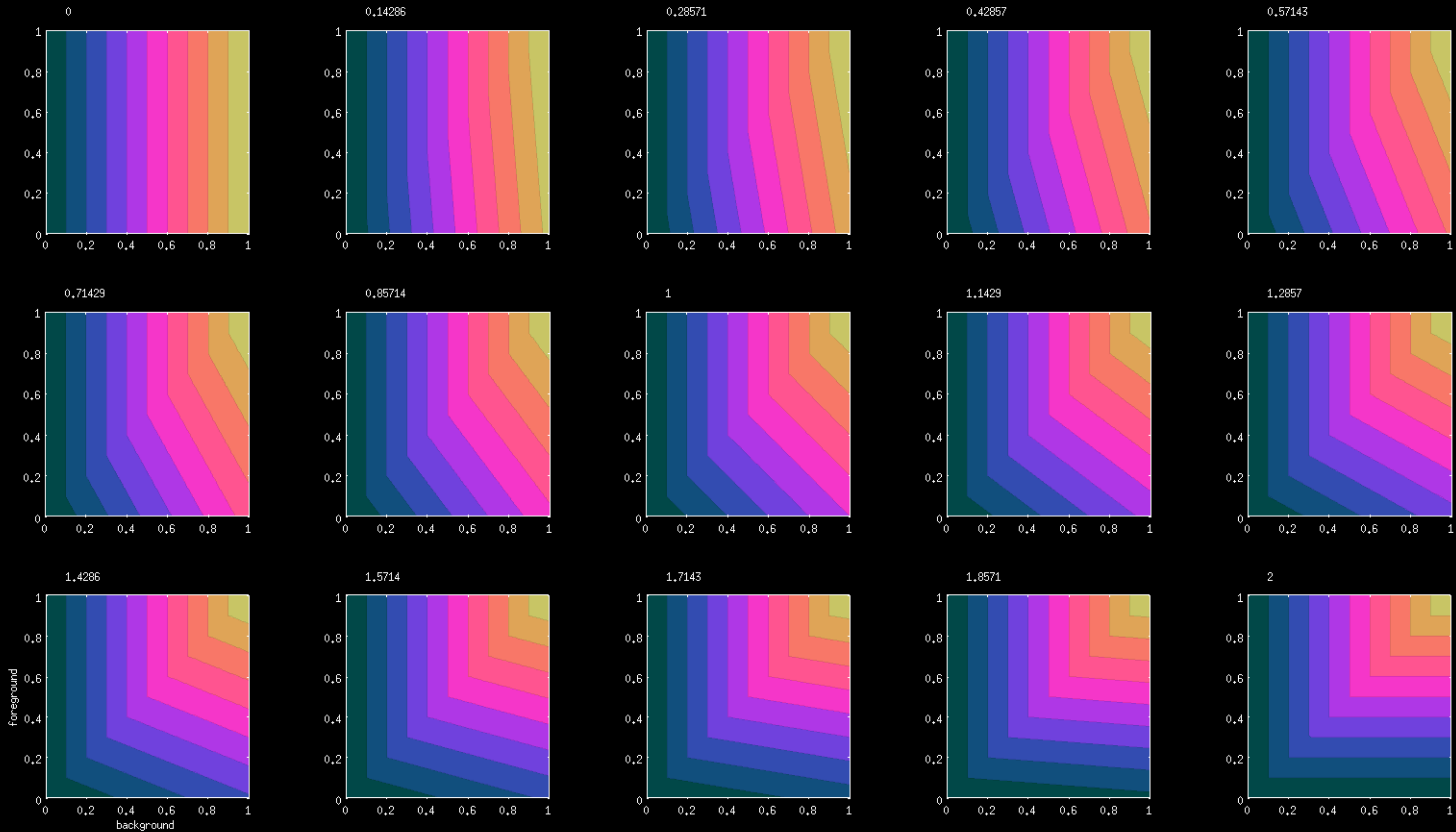
flatshadow



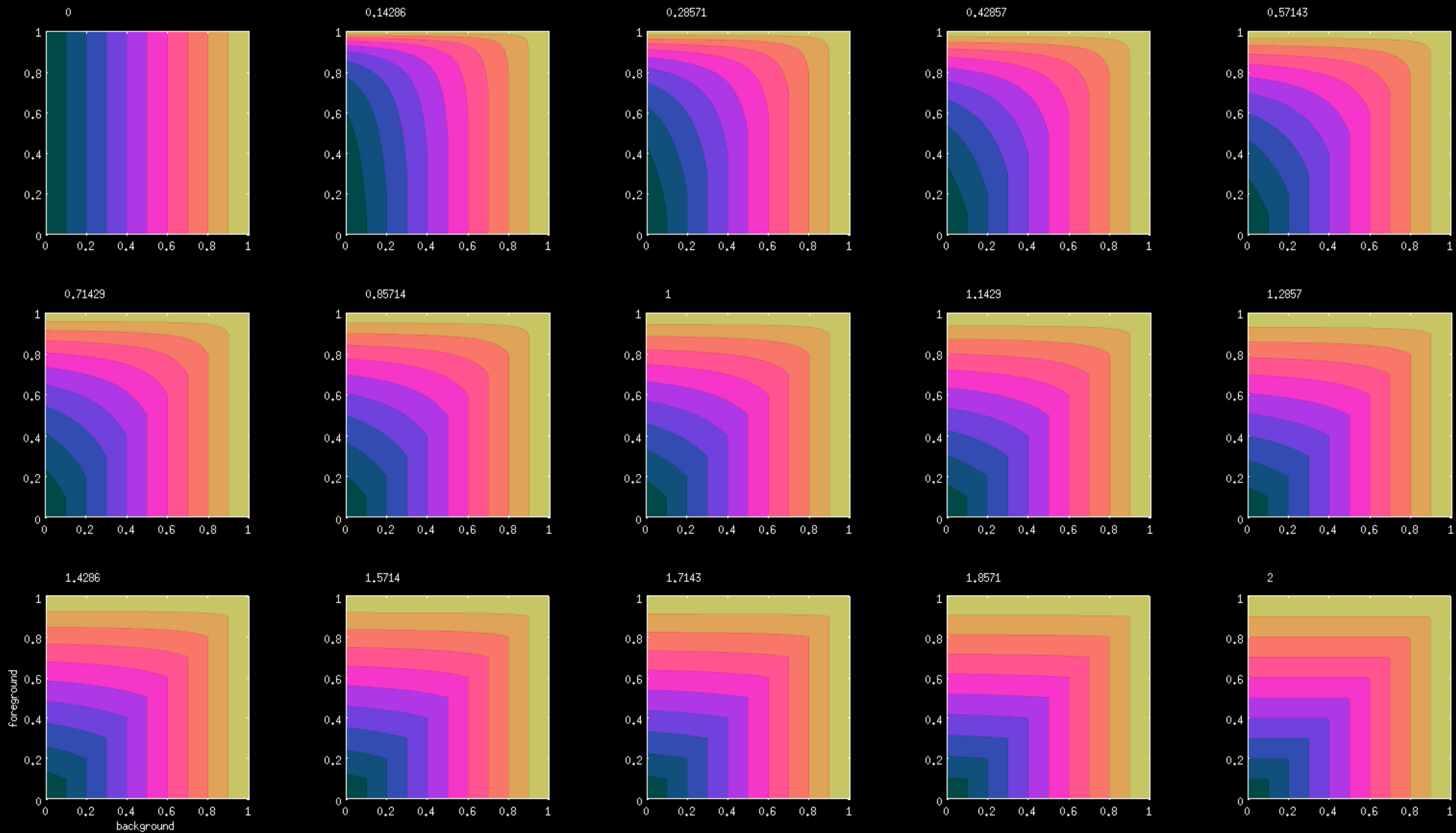
meanglow



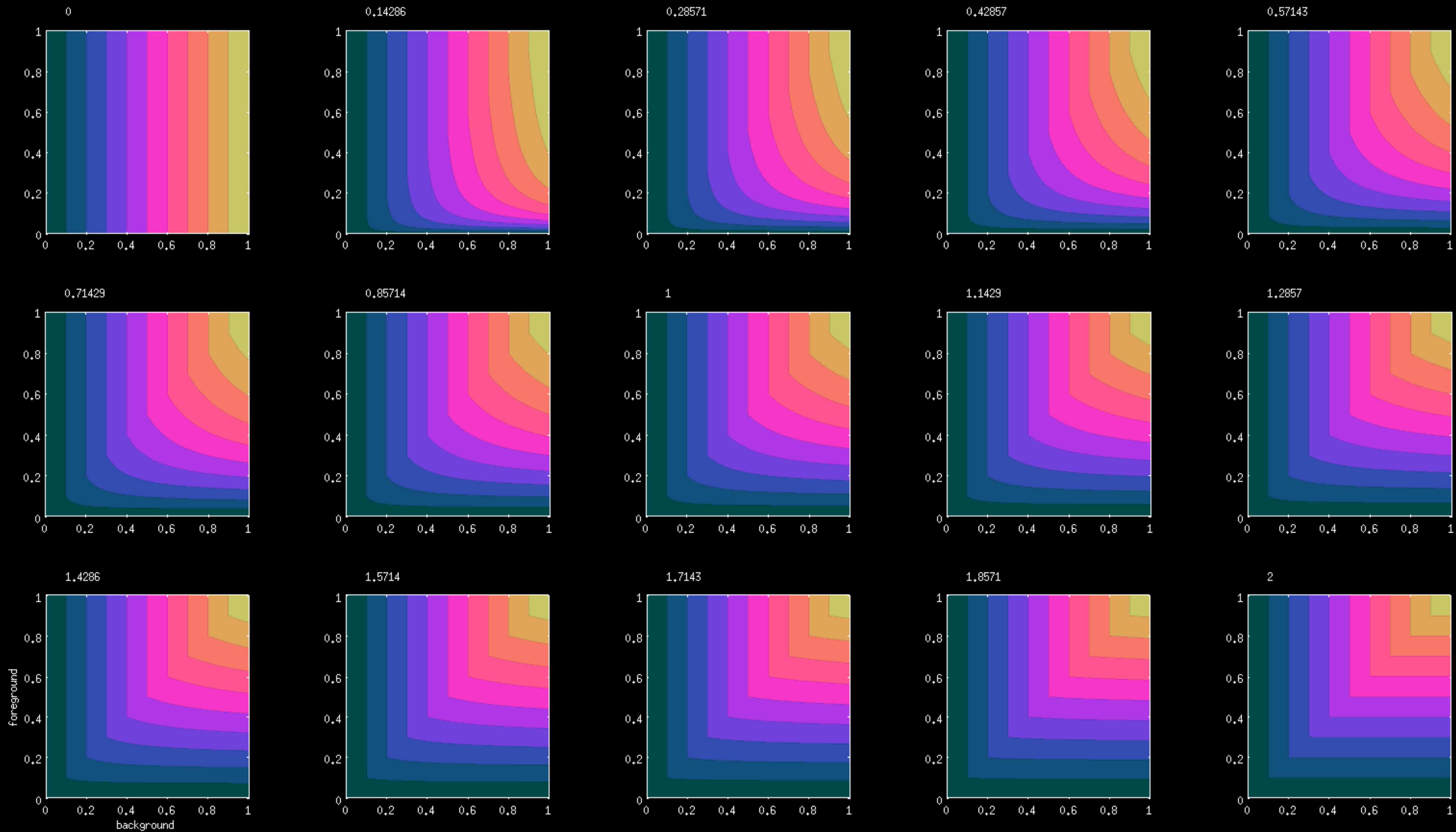
meanshadow



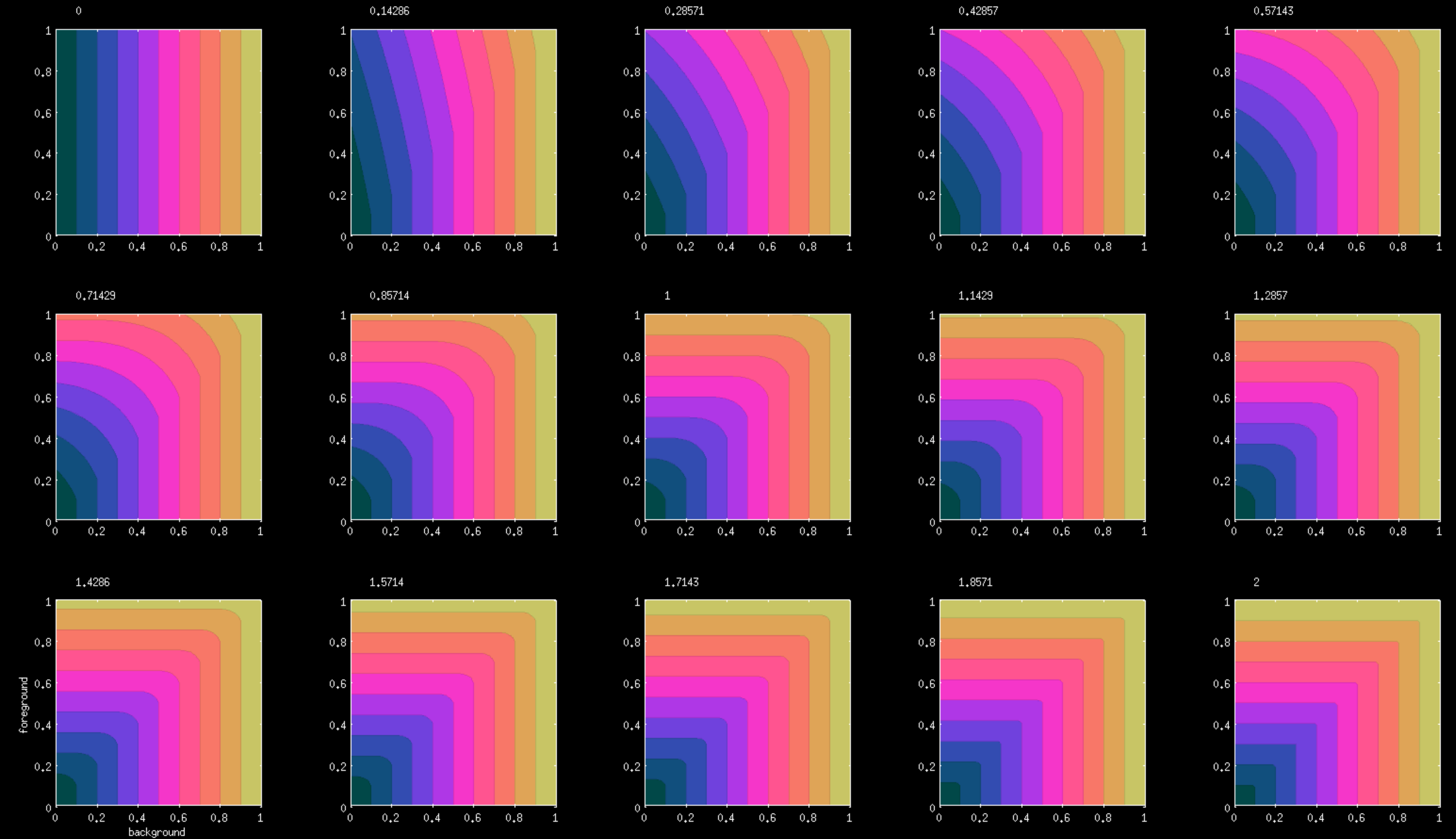
starglow



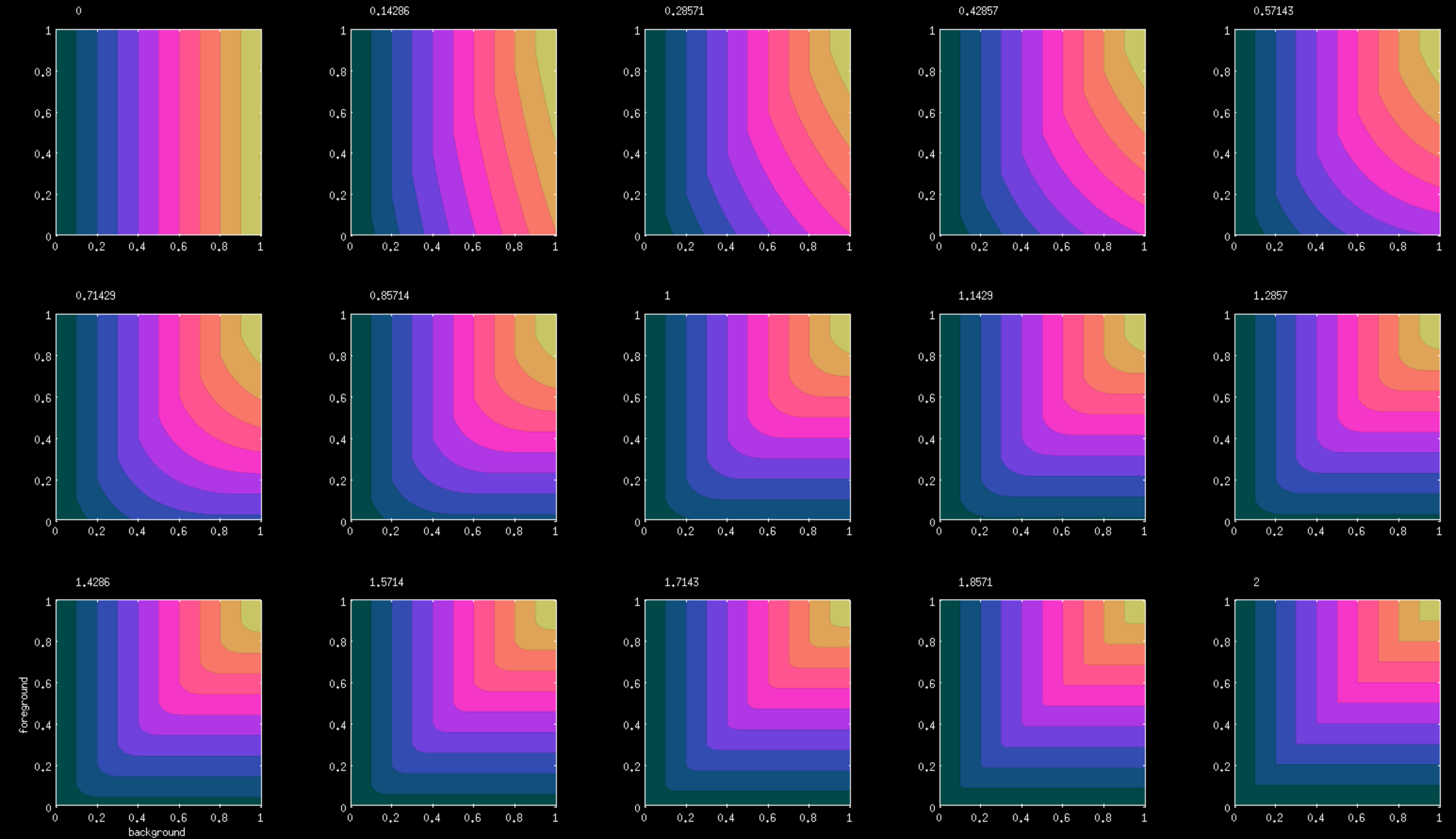
starshadow



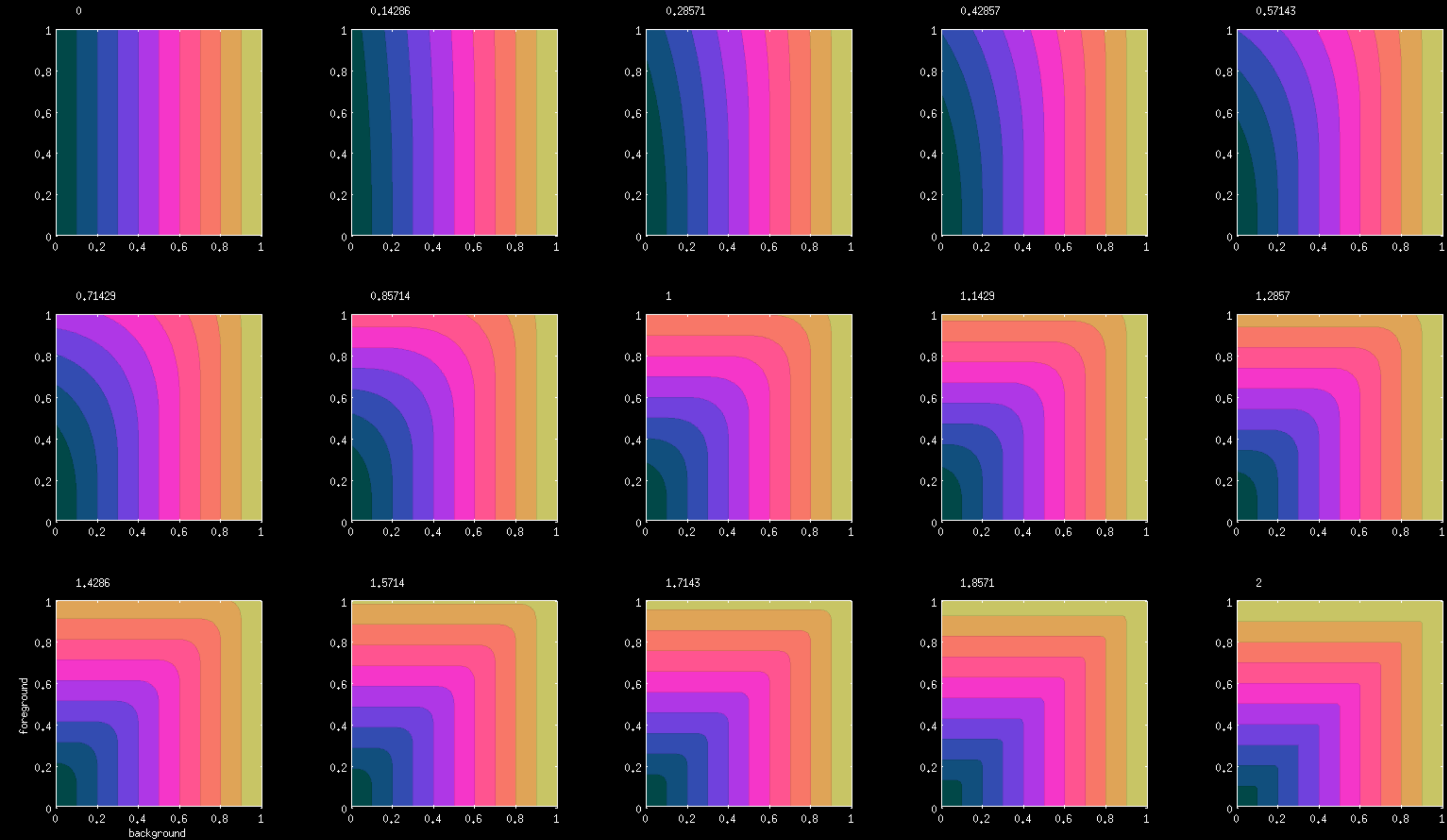
moonglow



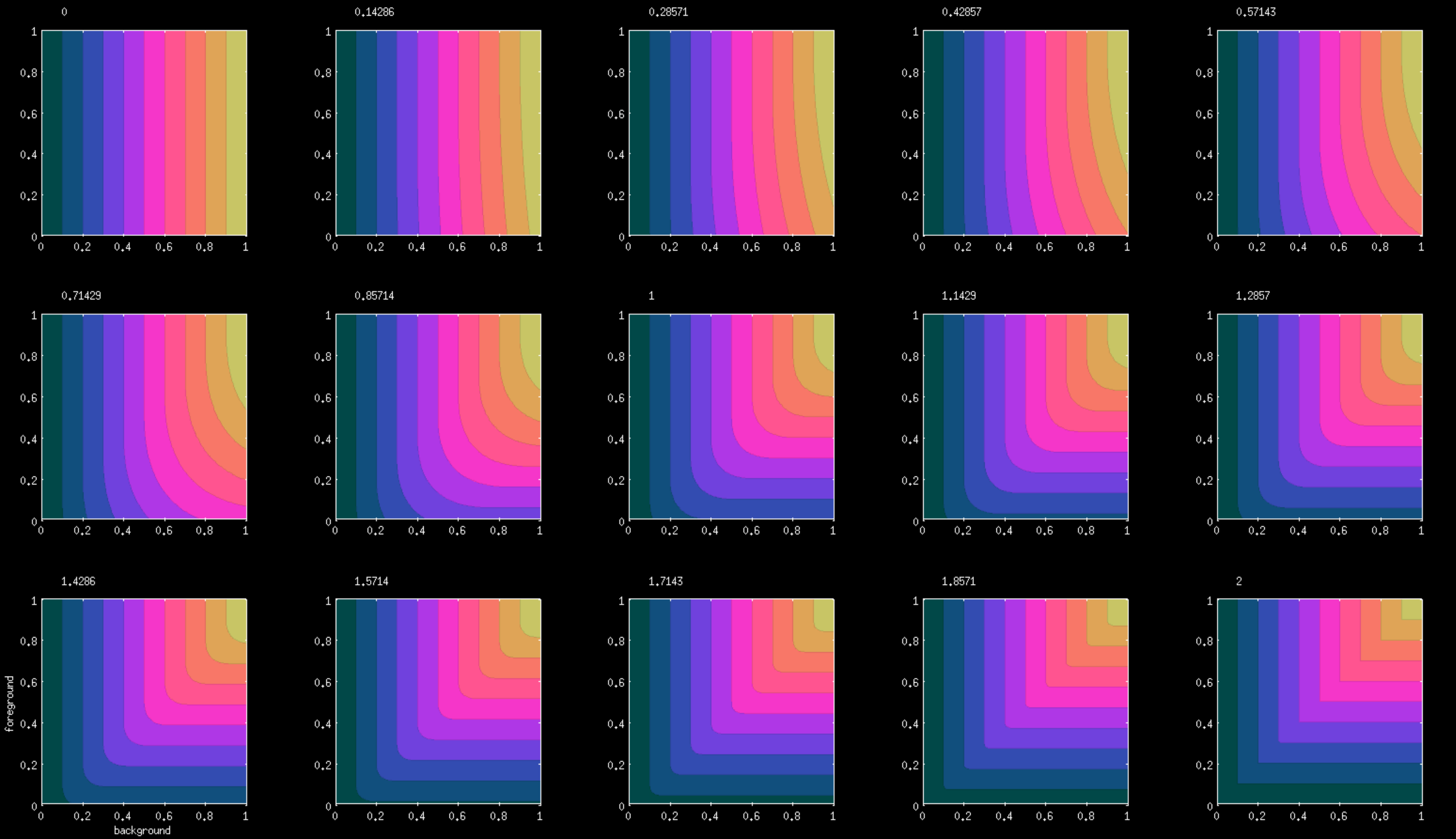
moonshadow



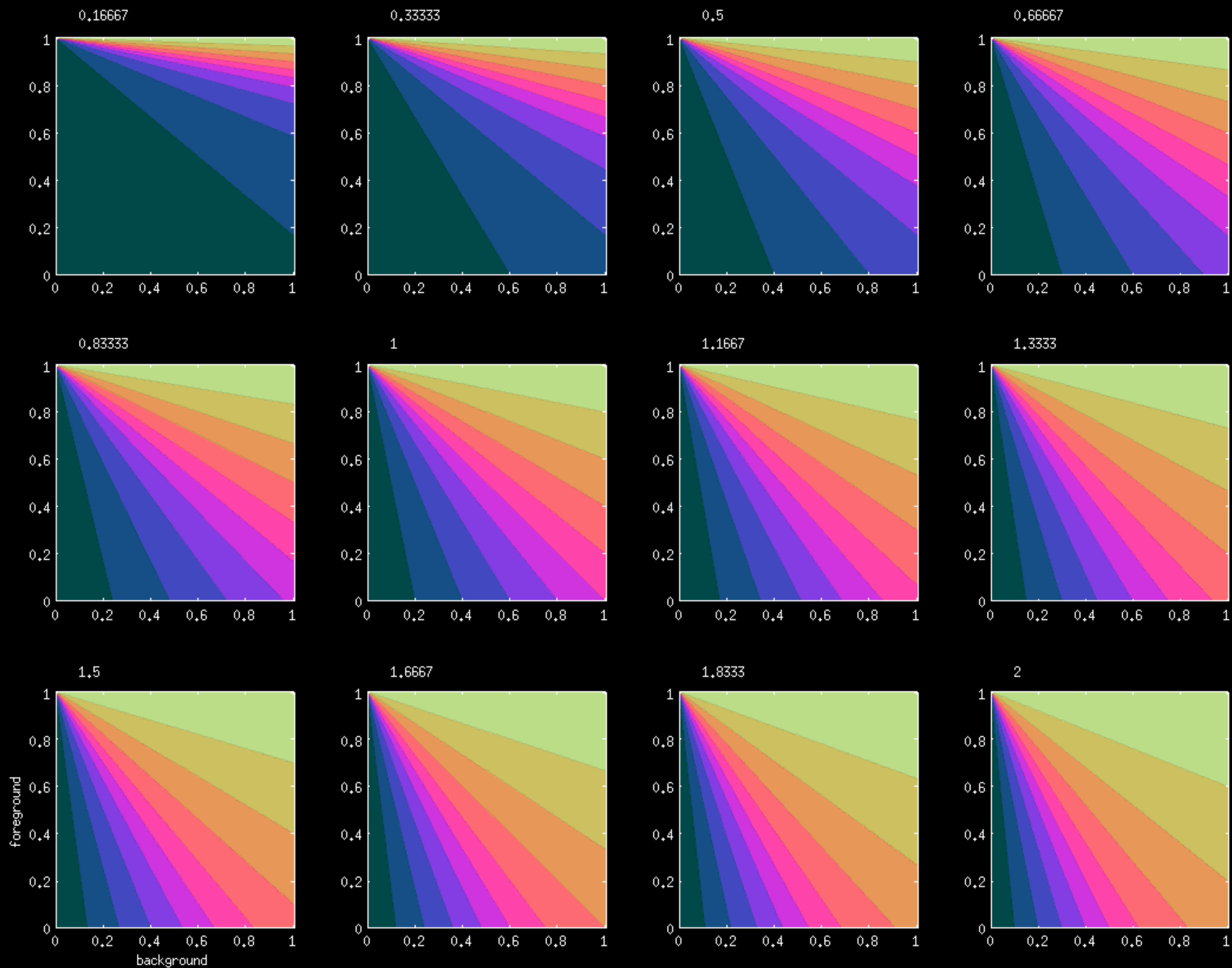
moonglow2



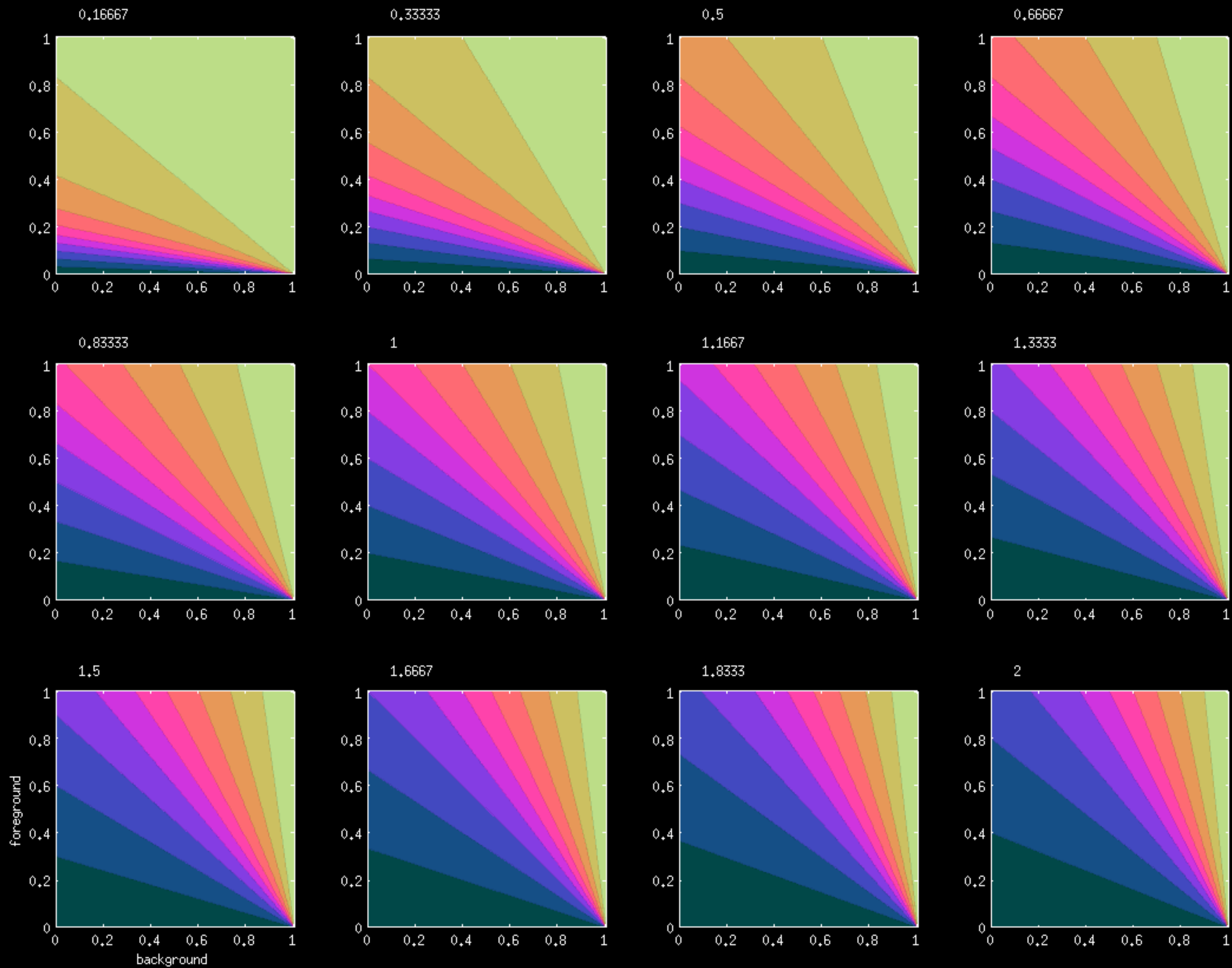
moonshadow2

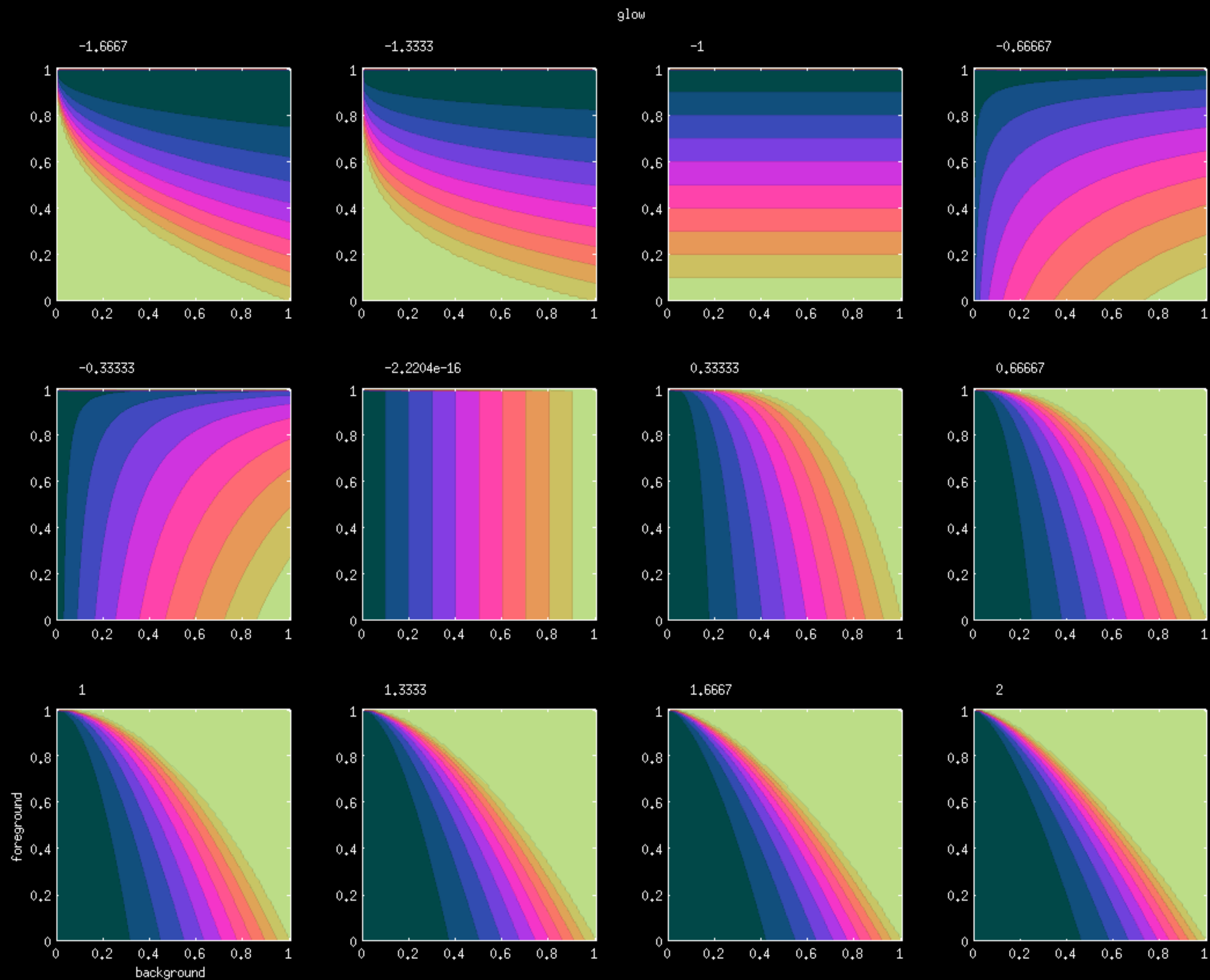


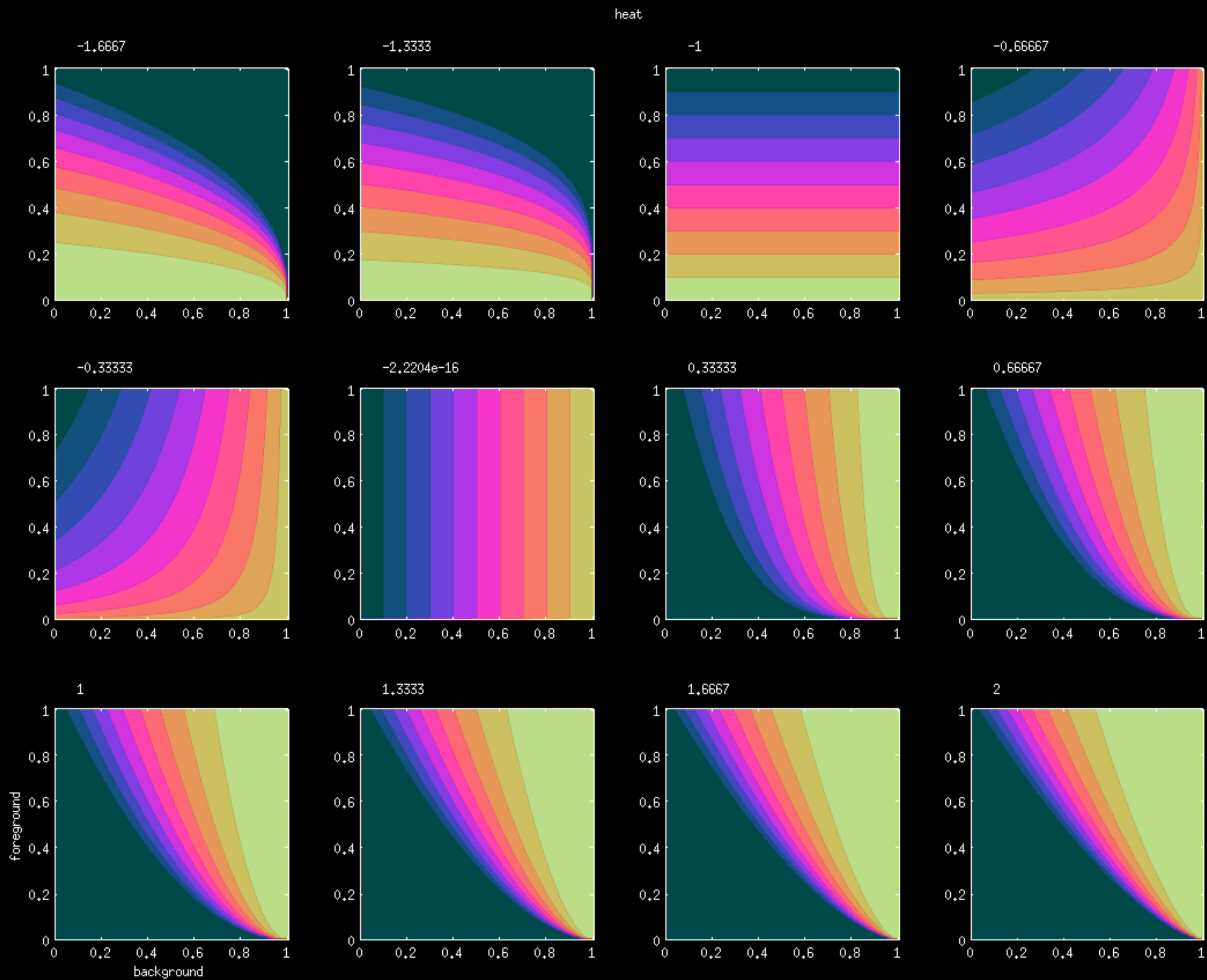
softdodge

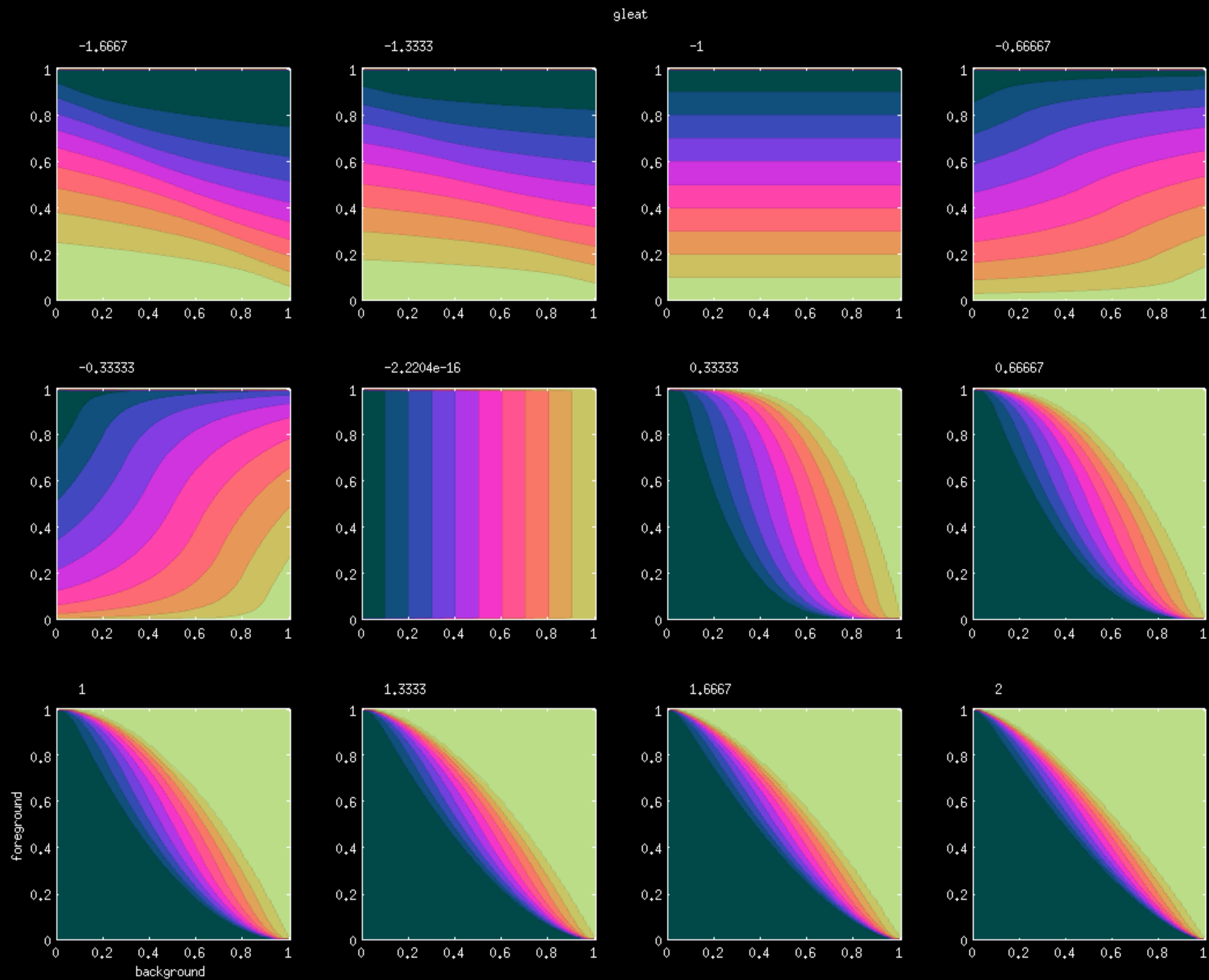


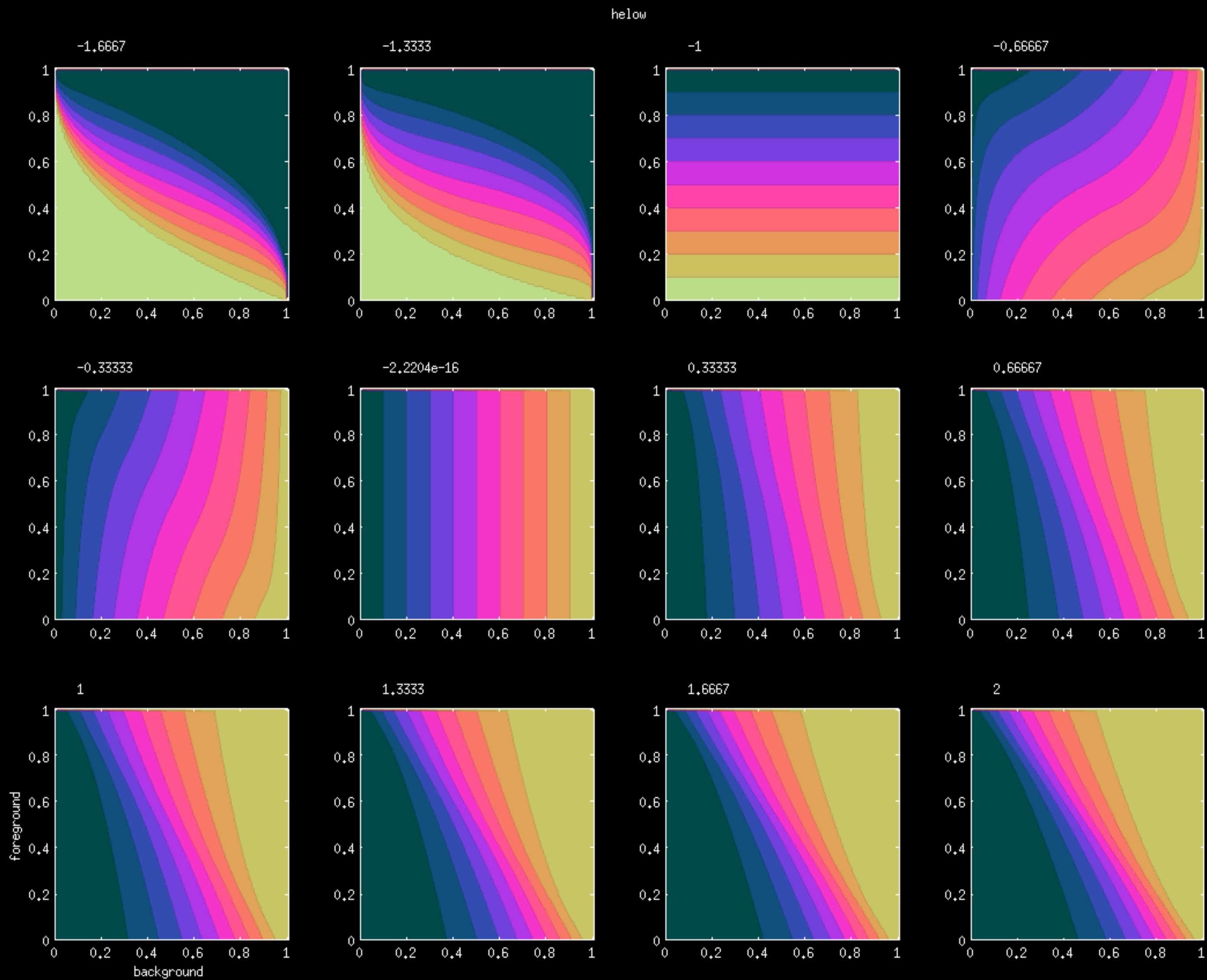
softburn

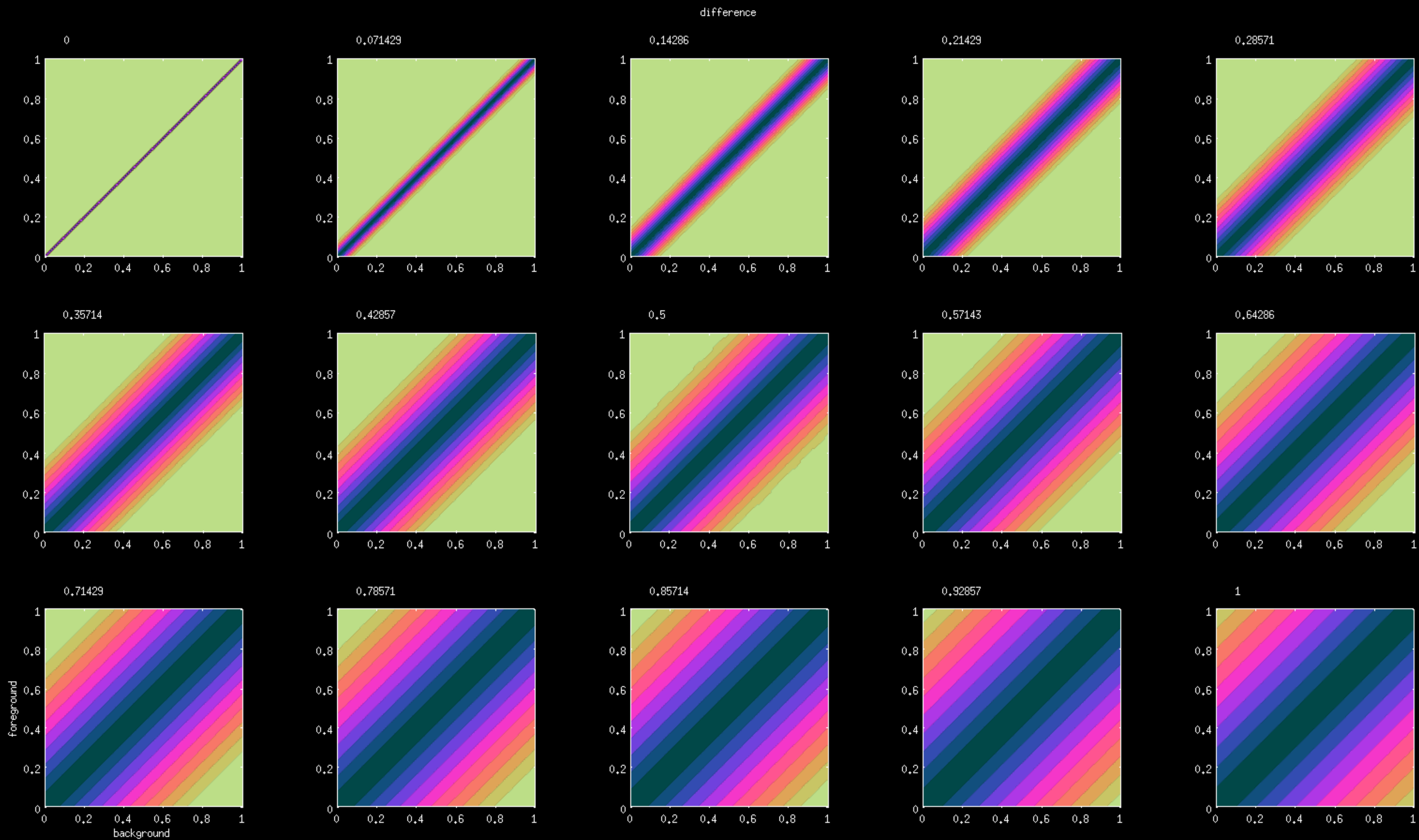




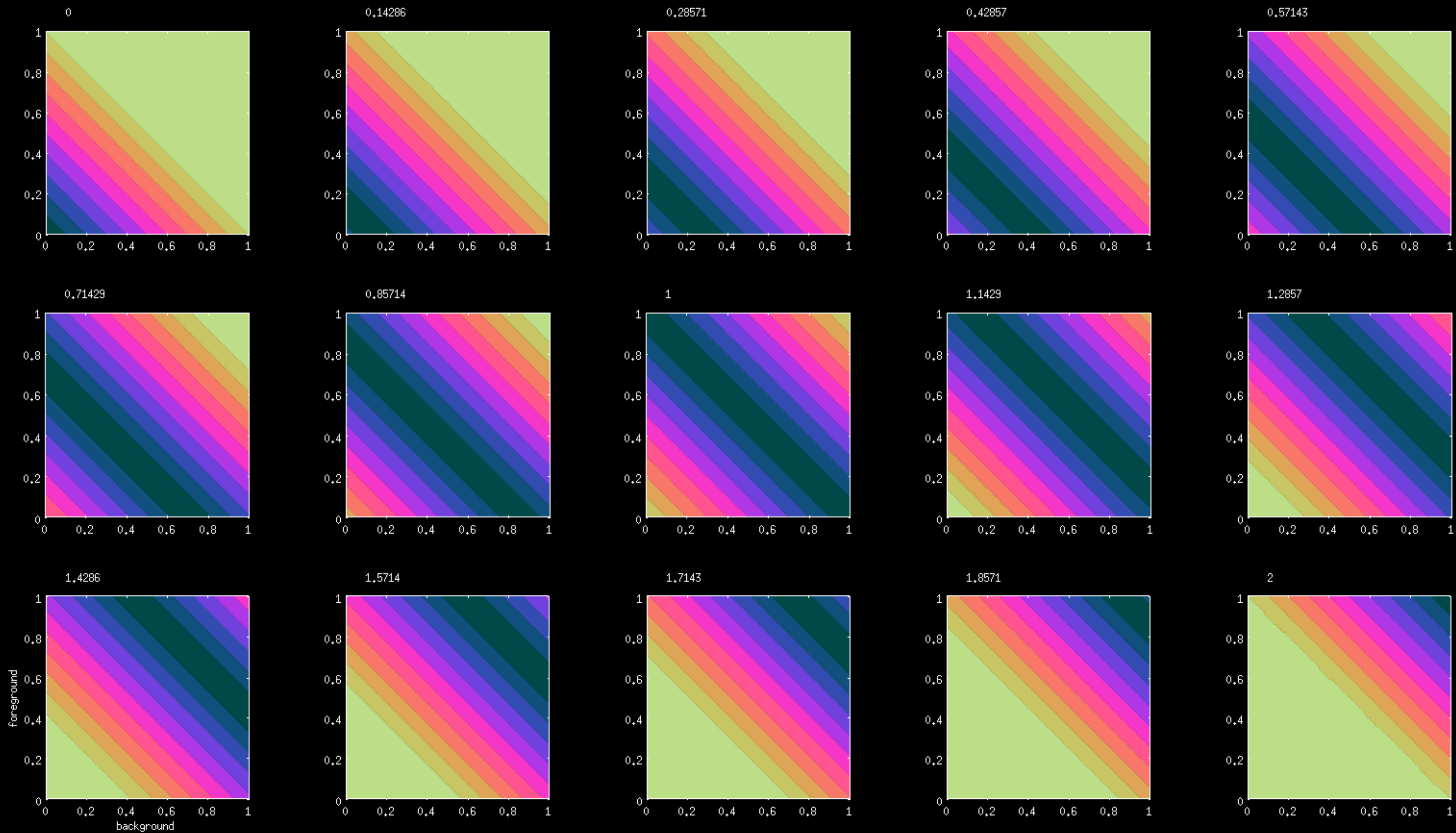




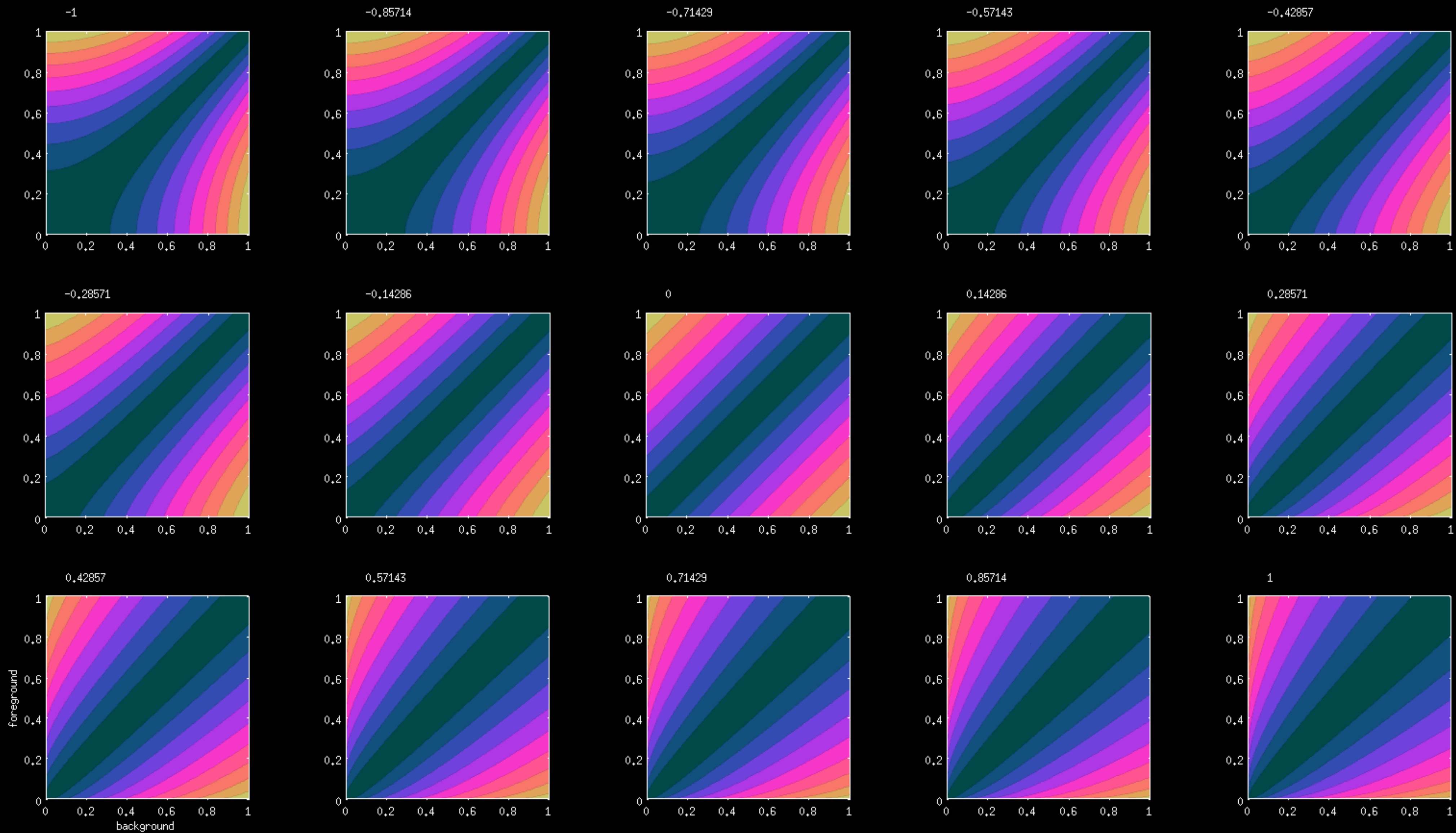




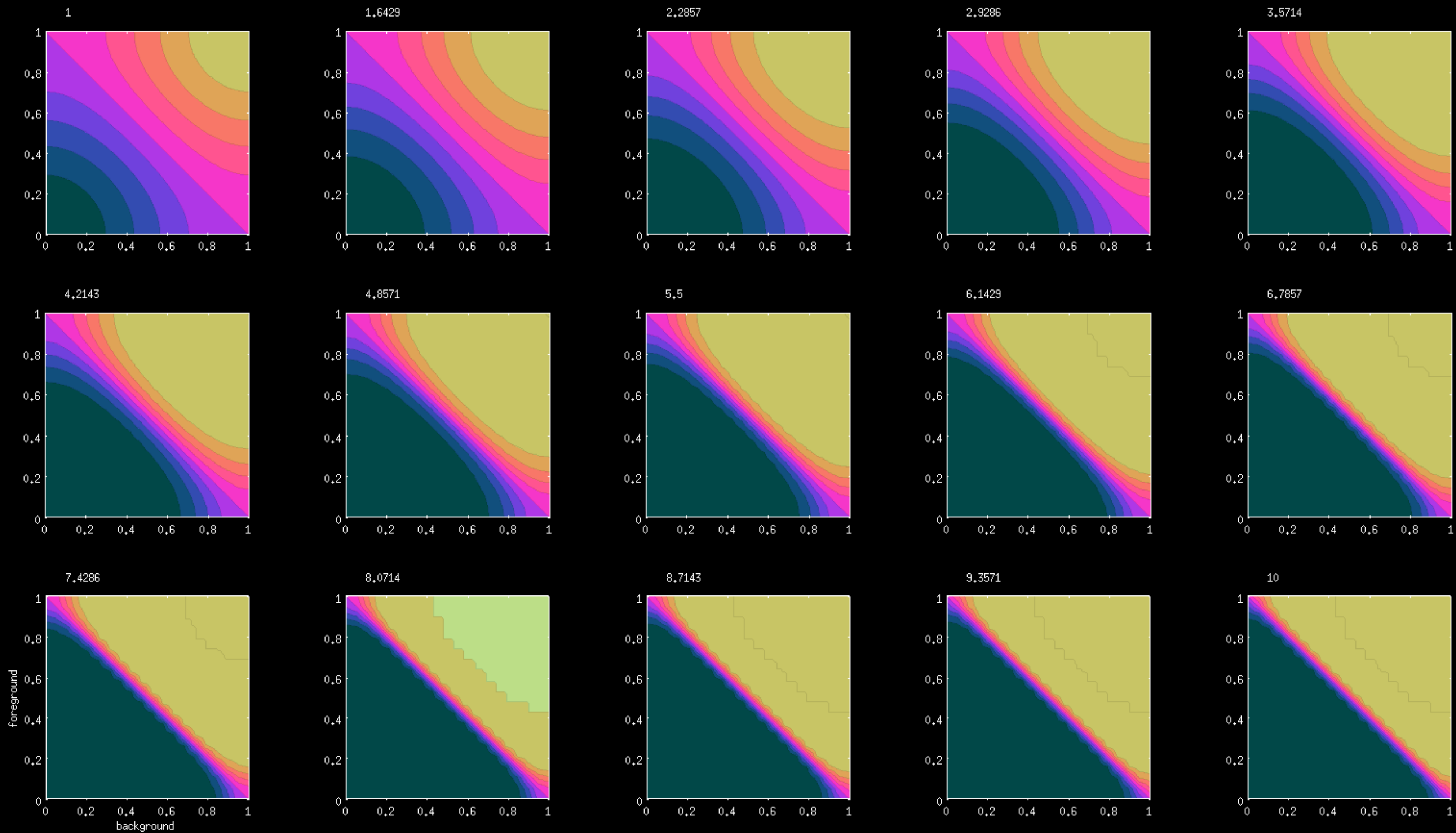
extremity

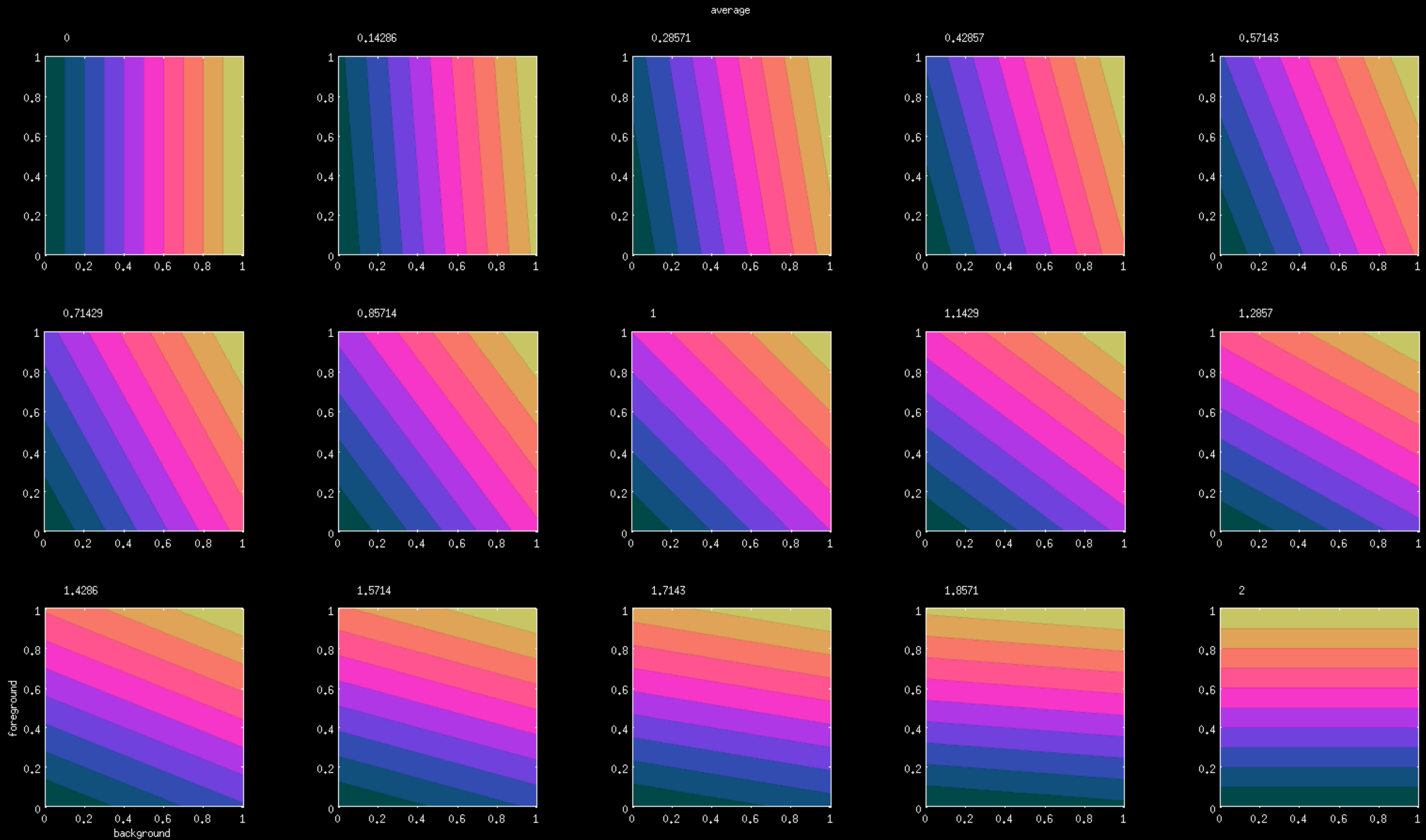


sqrtldiff

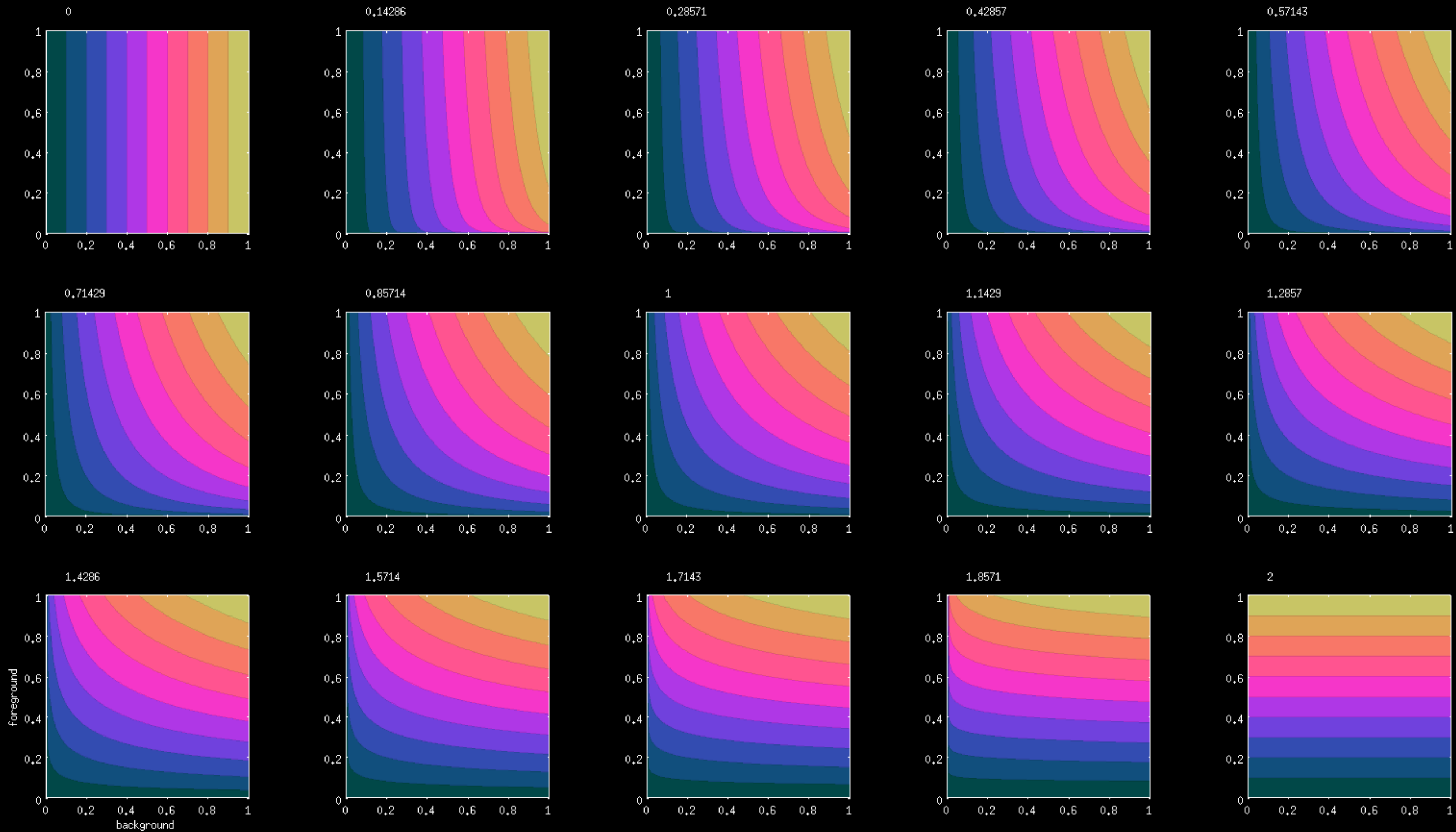


interpolate

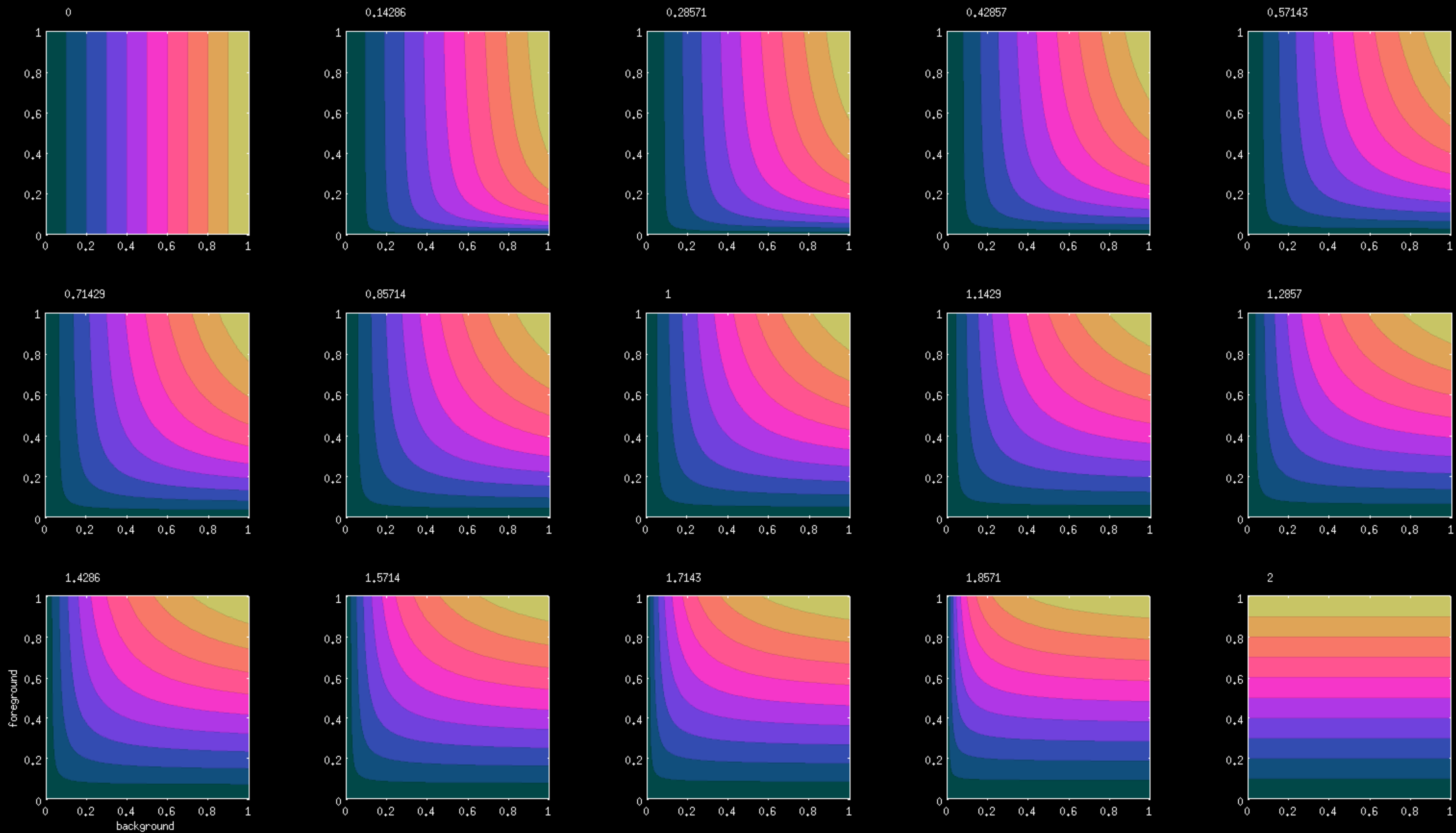


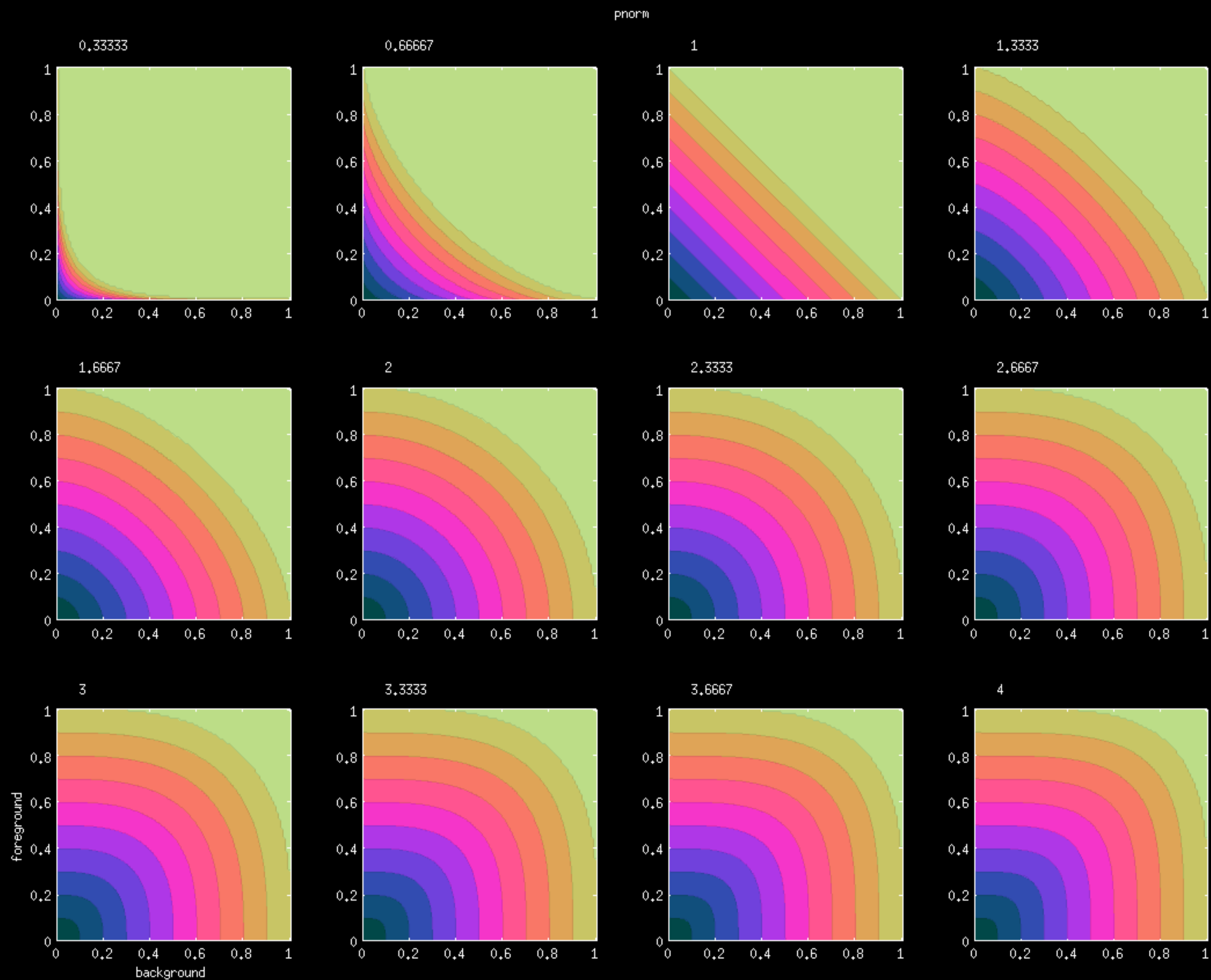


geometric

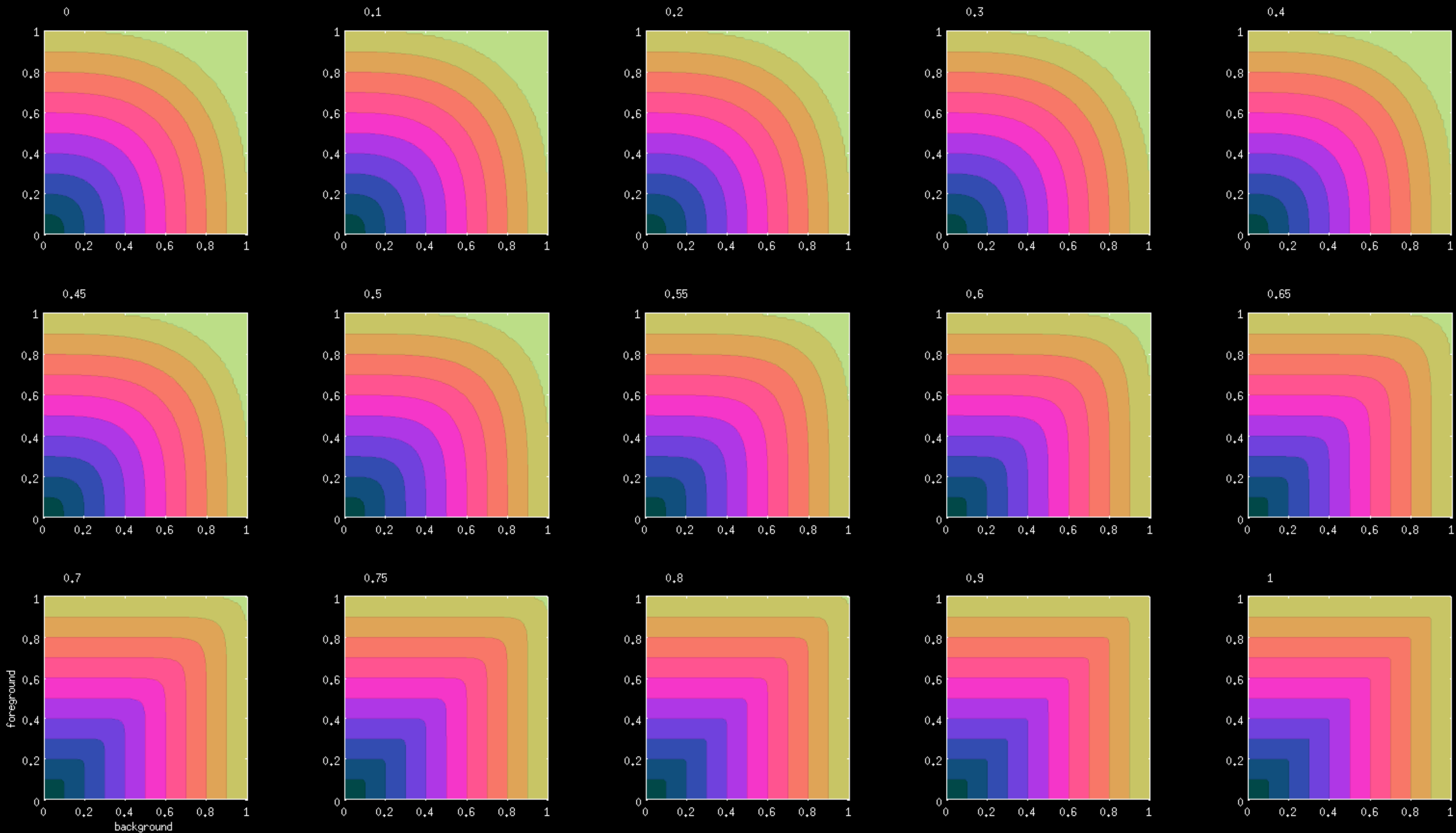


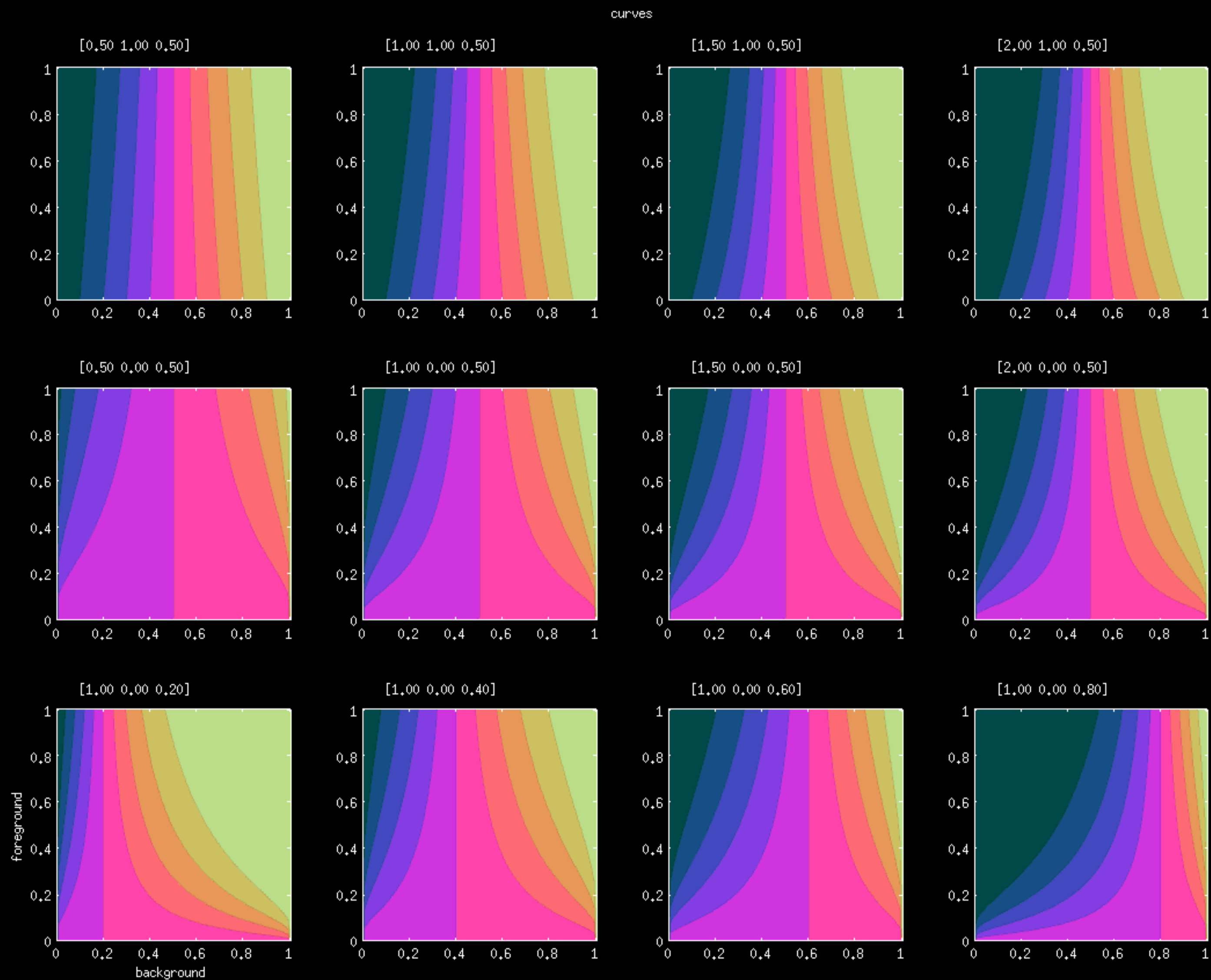
harmonic





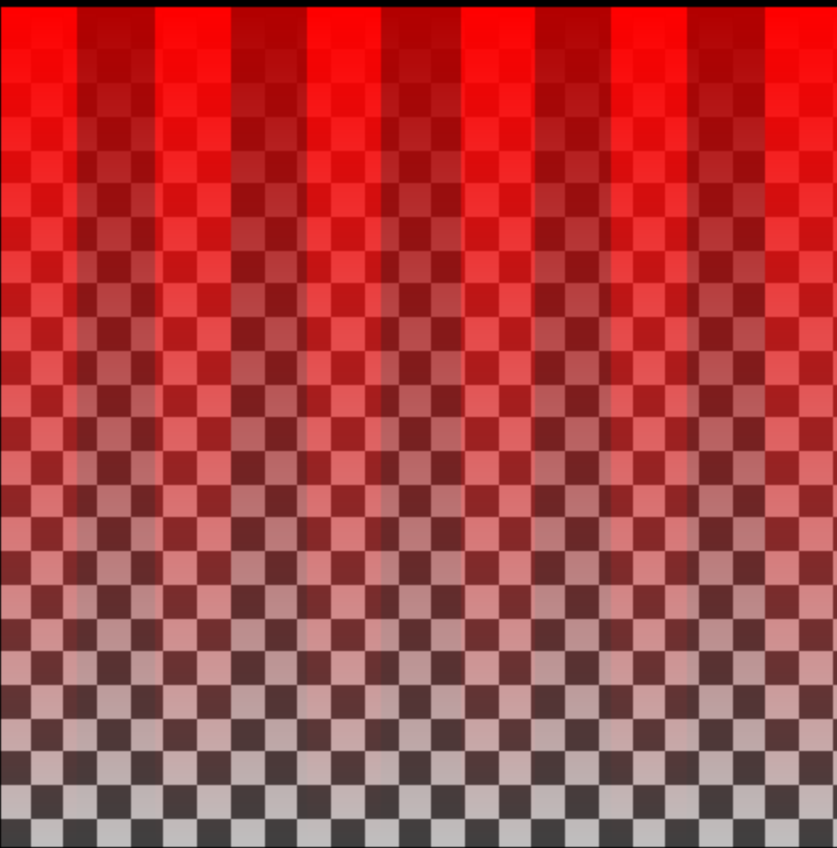
lightenrgb



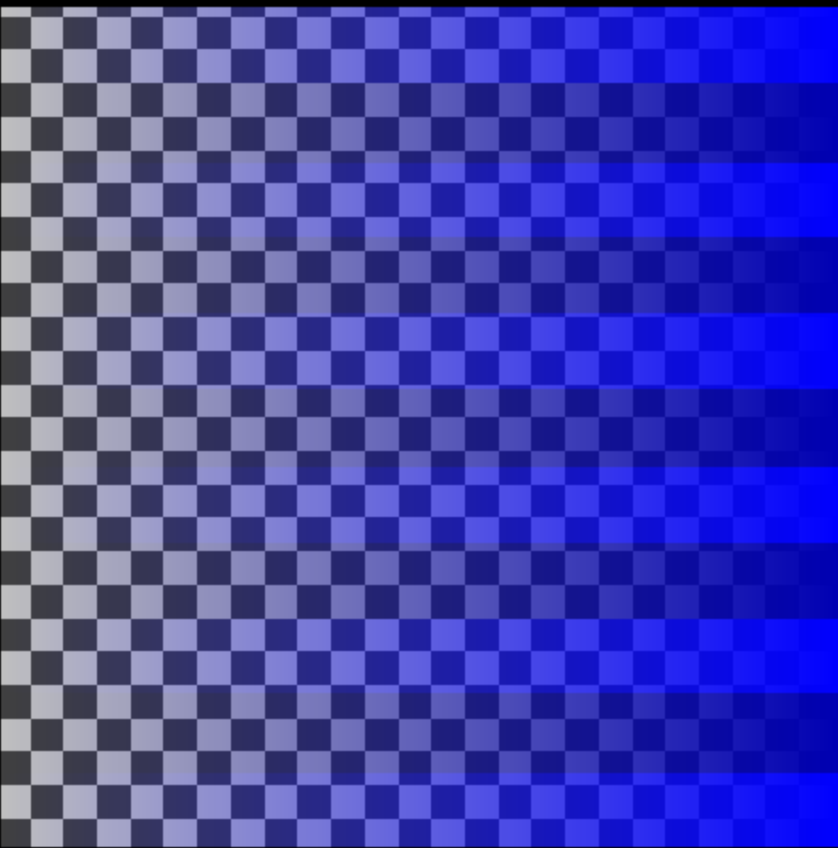


COMPOSITION MODE EXAMPLES ('normal' blend mode)

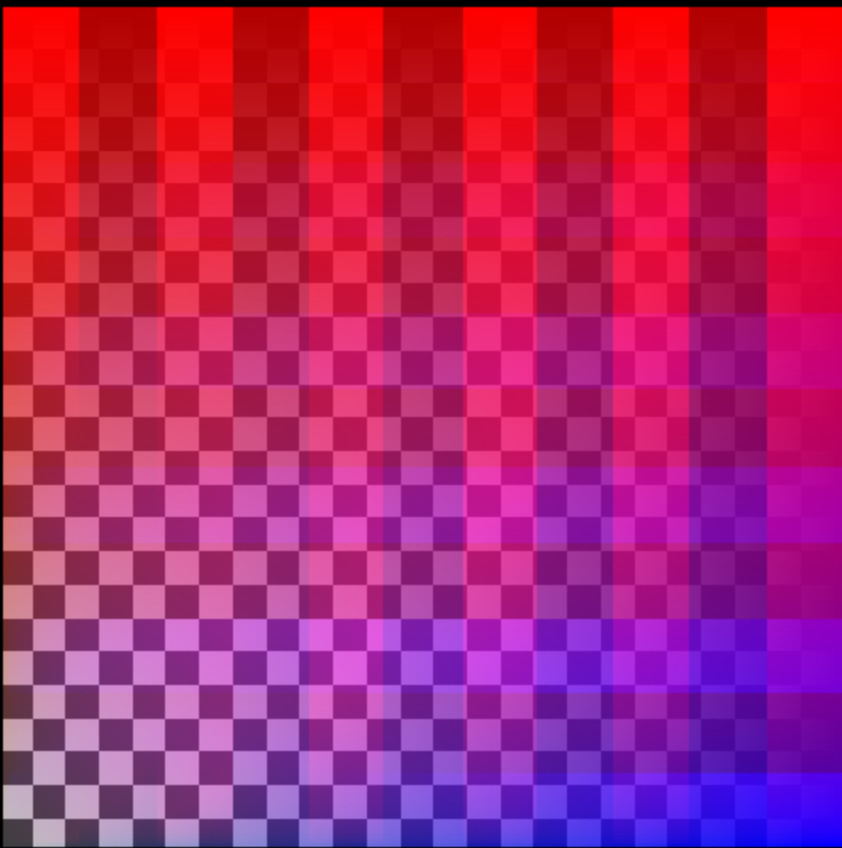
FG



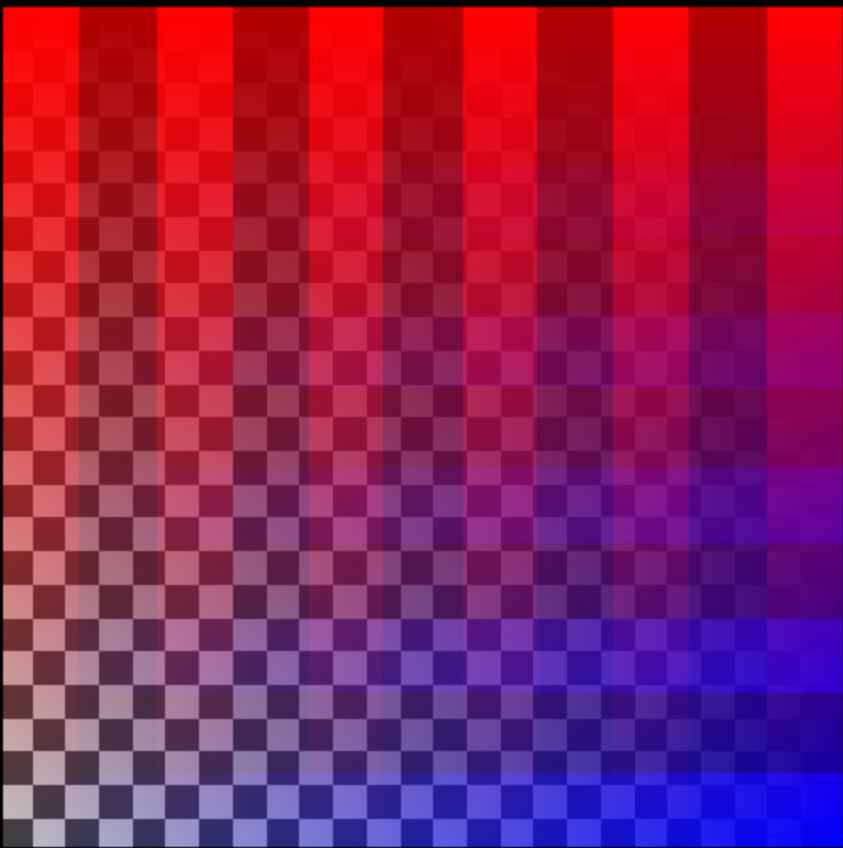
BG



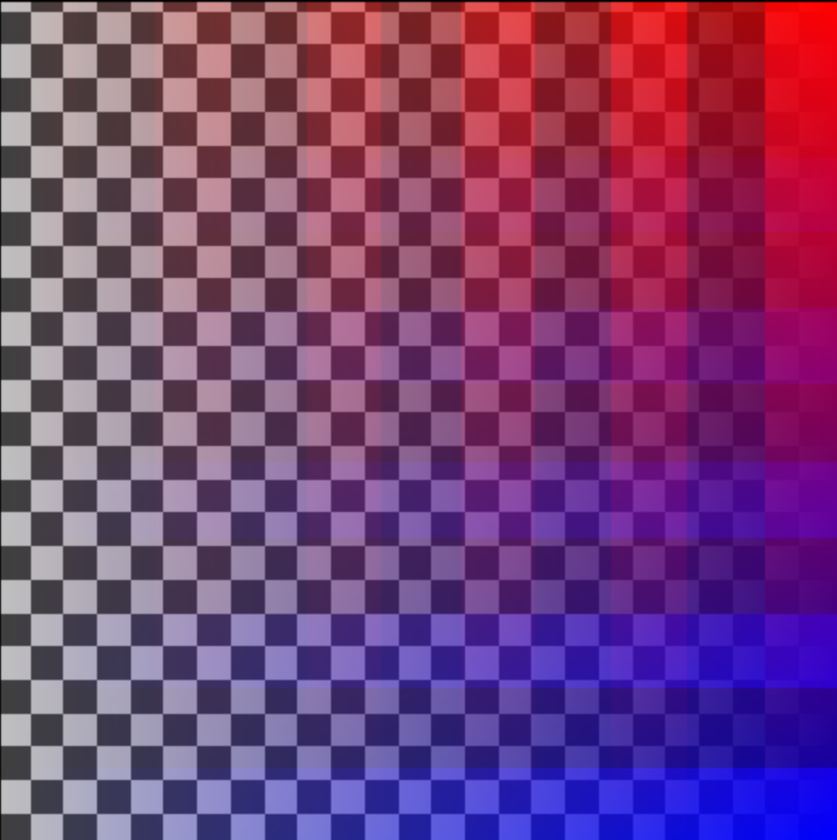
translucent



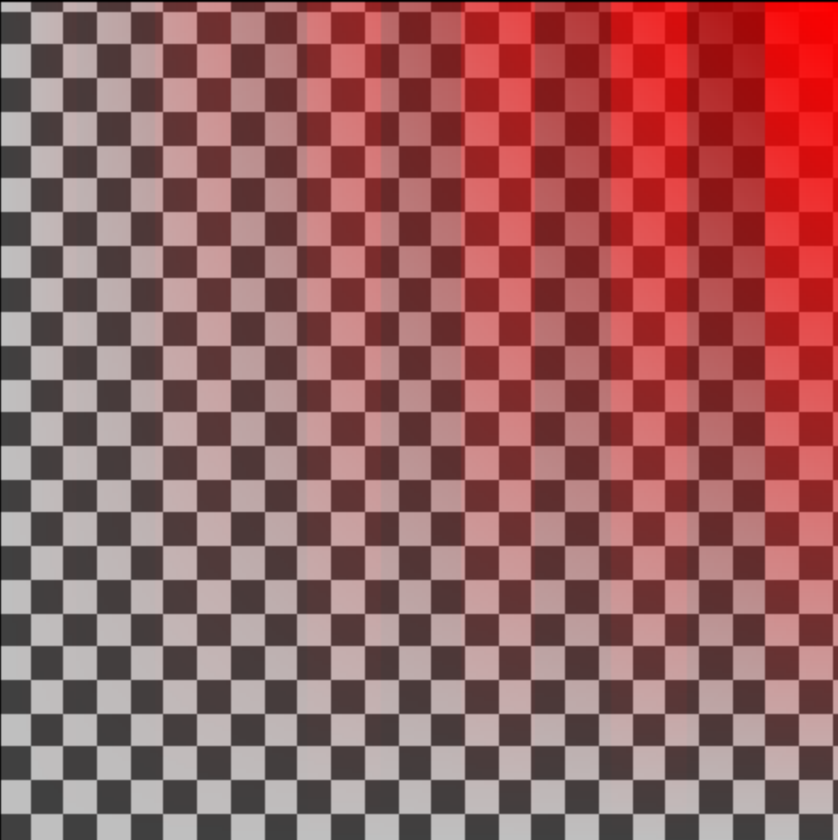
srcover



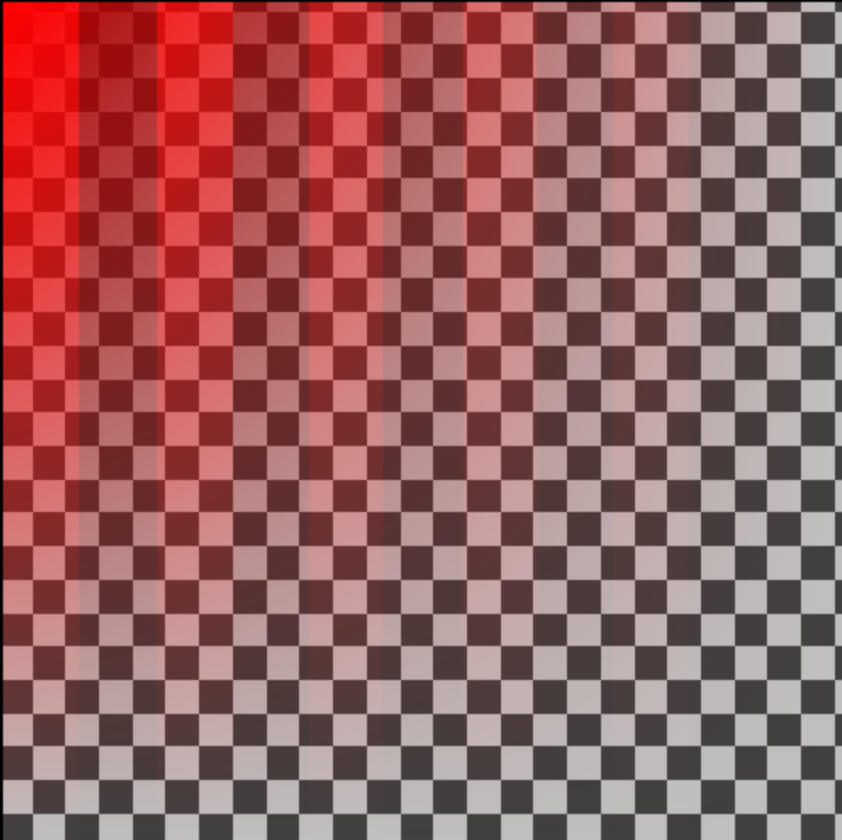
srcatop



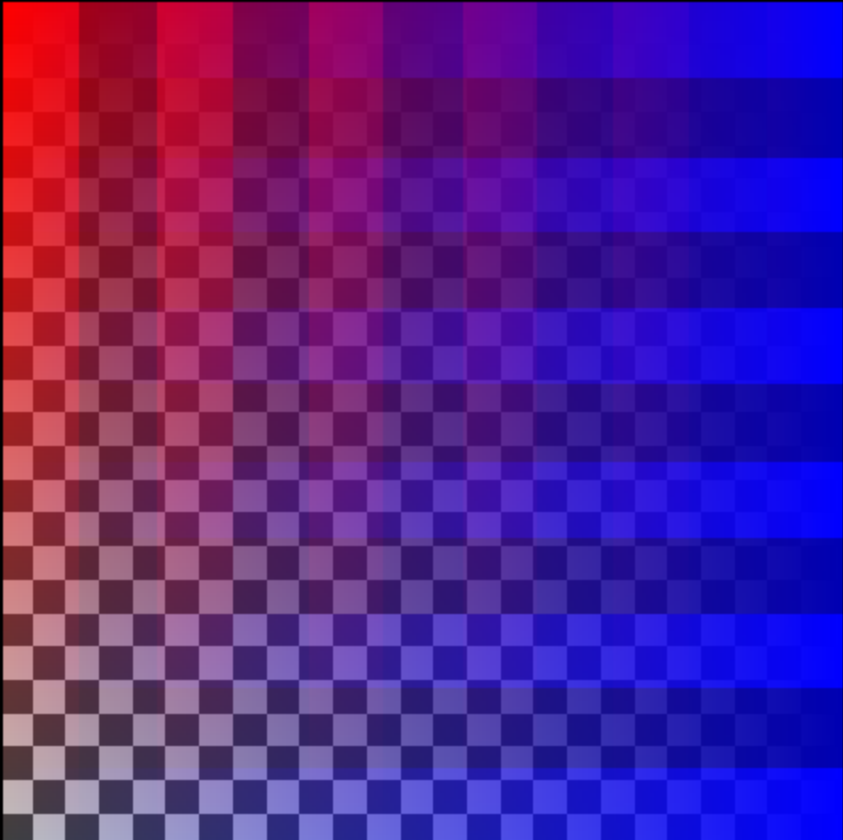
srcin



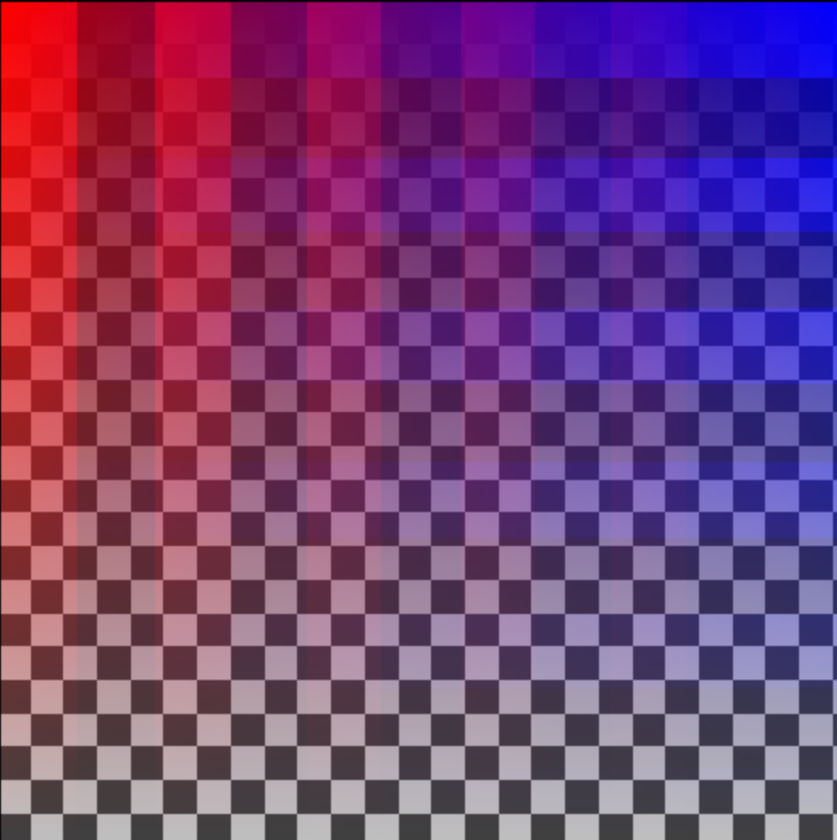
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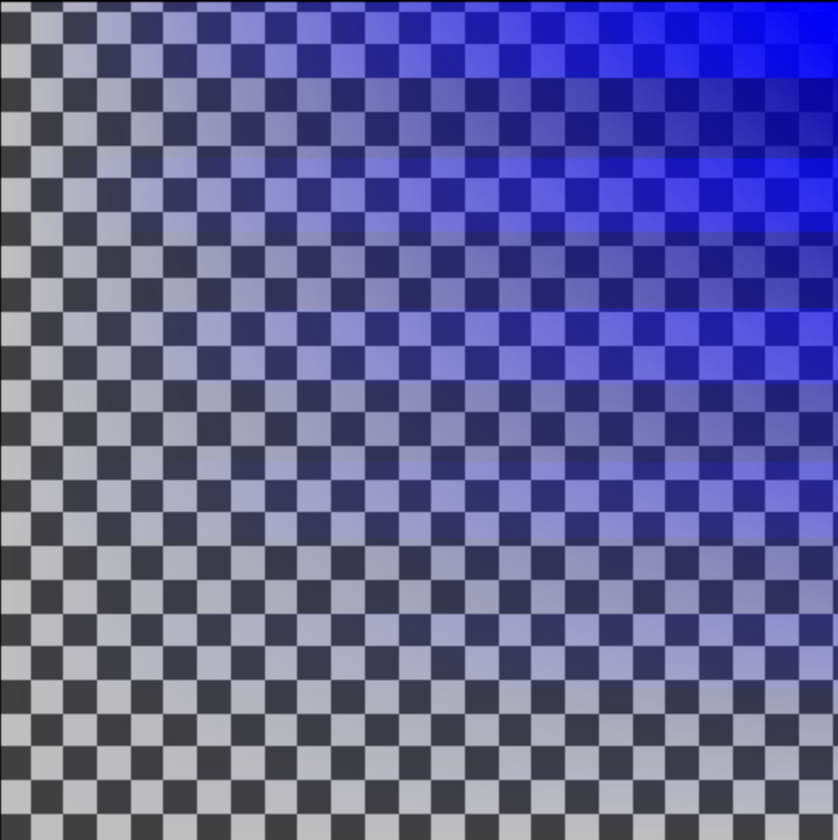
dstover



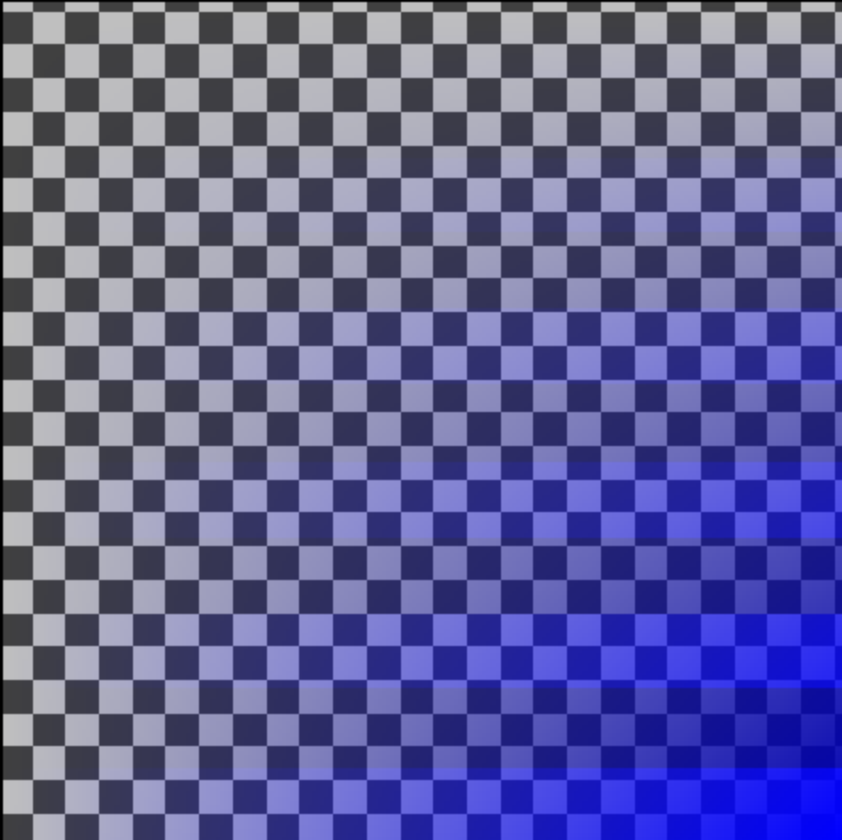
dstatop



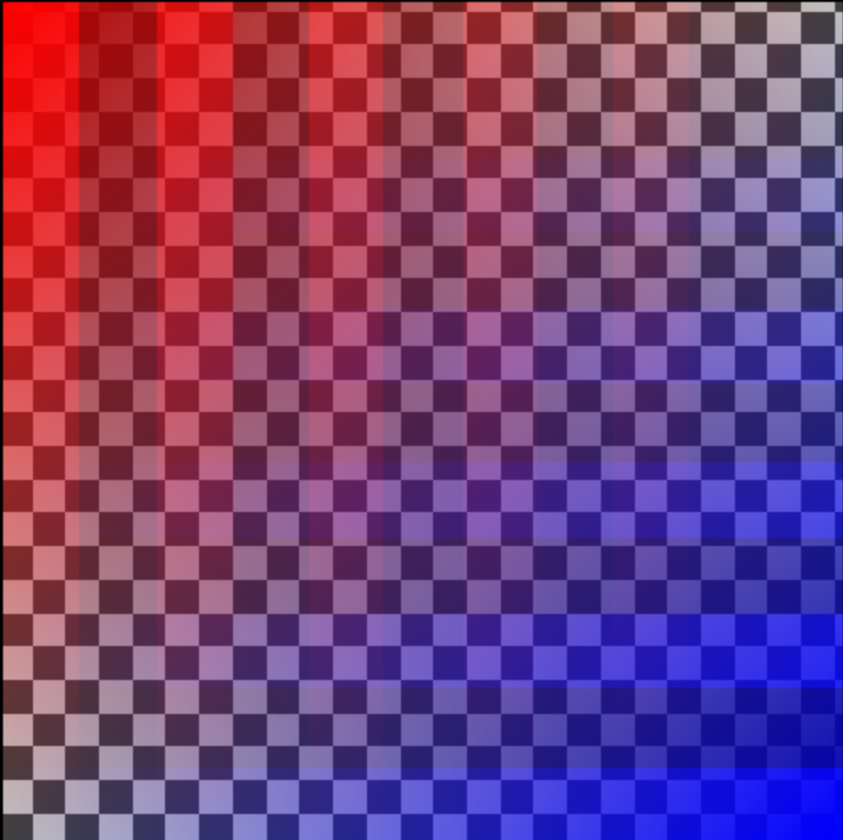
dstin



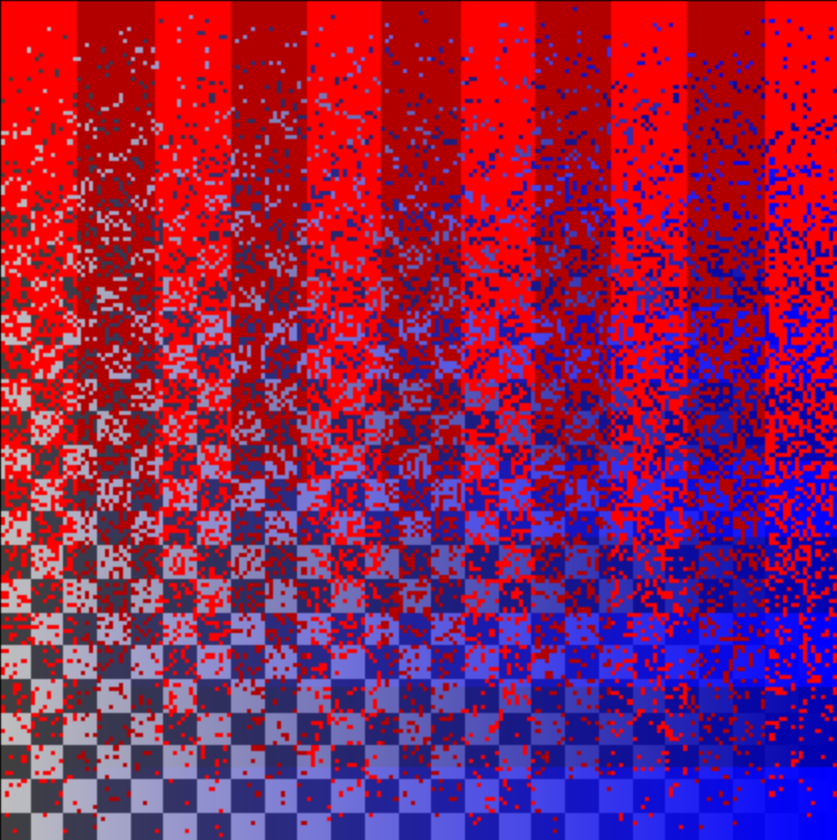
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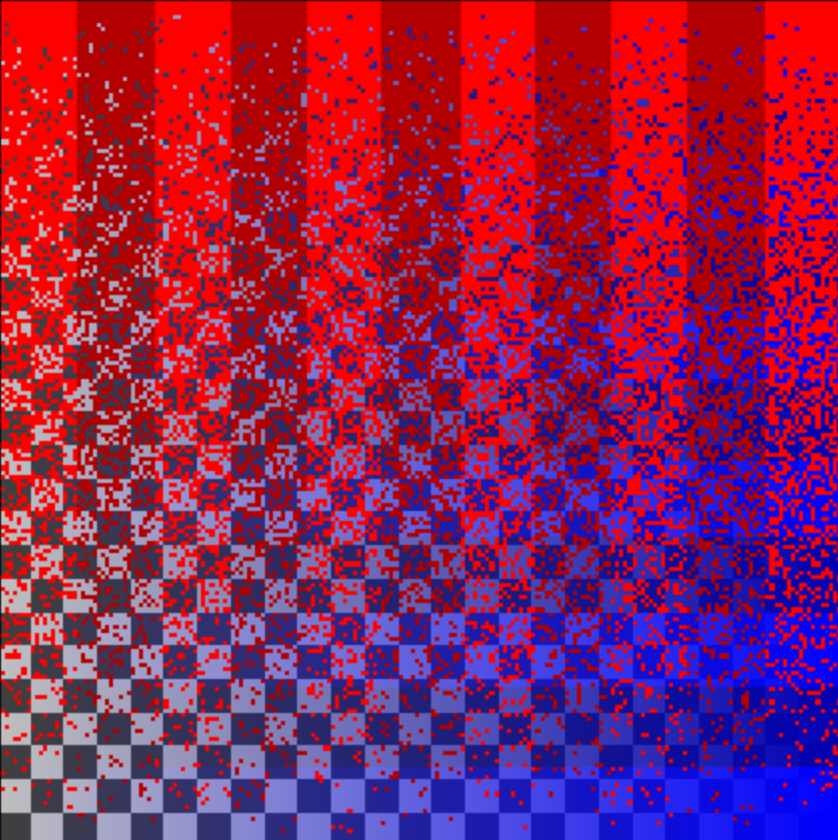
xor



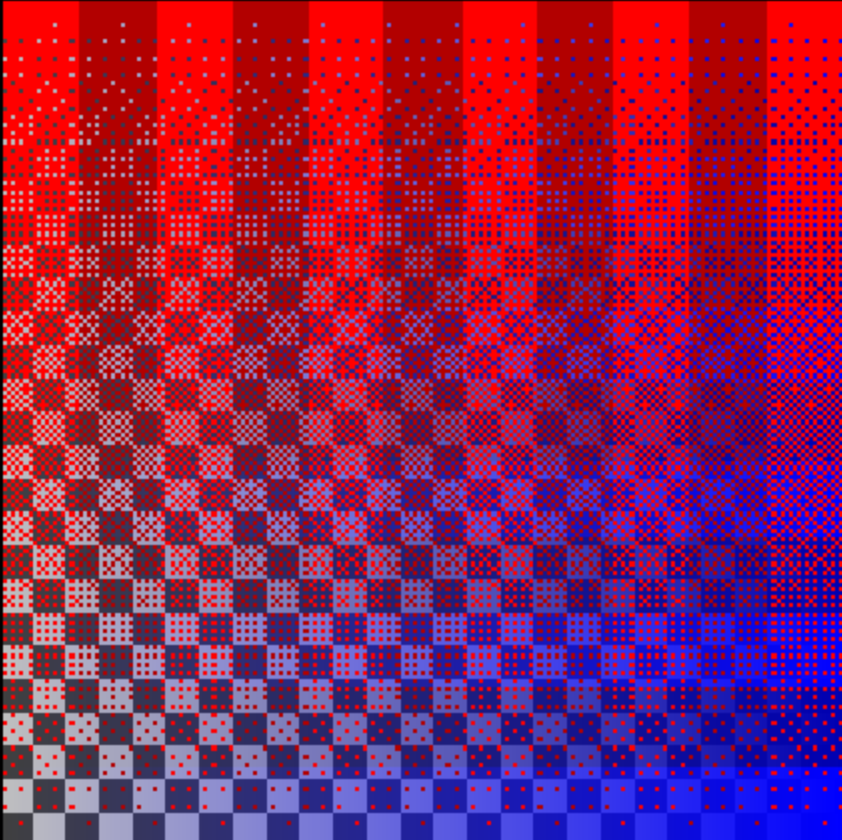
dissolve



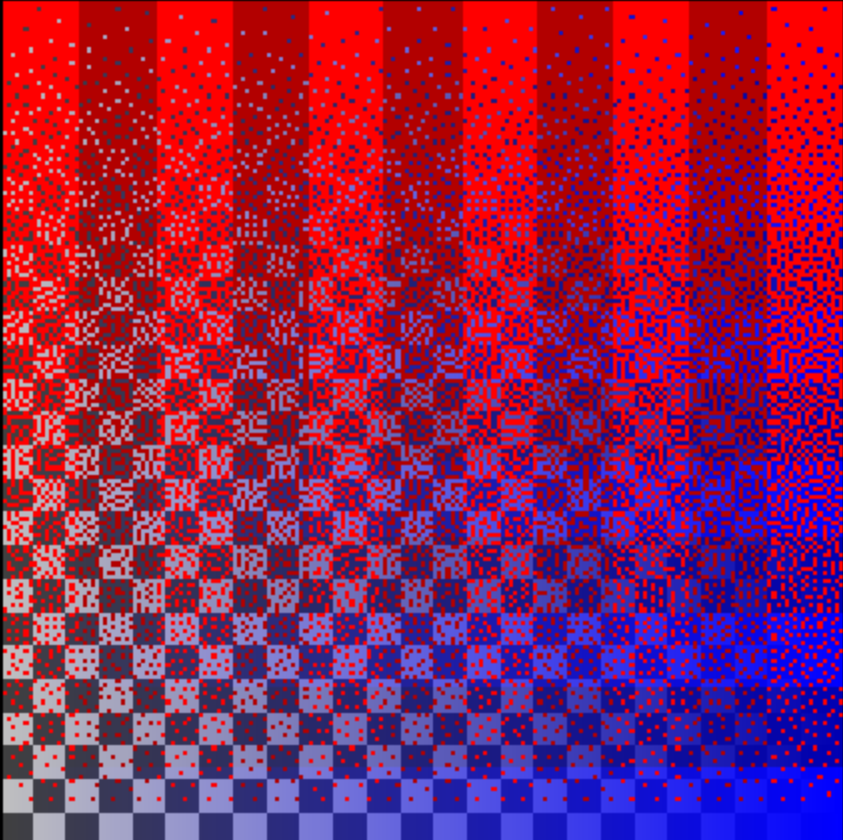
dissolvebn



dissolveord



dissolvezf



bg alpha

fg alpha